

Adaptively Secure Threshold Decryption from LWE with Polynomial Modulus

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Abstract. Threshold Decryption (TD) enables a set of decryptors, each holding a secret key share, to collaboratively decrypt ciphertexts. Lots of TD schemes consider only CPA security under static corruptions, but a stronger and more reasonable security notion in practice is *CCA security* under *adaptive corruptions*, which enhances the ability of adversaries to obtain partial decryptions of chosen ciphertexts and adaptively corrupt decryptors during the protocol. There are many works on TD from lattices, seeking for post-quantum security. However, none of these TD schemes achieves both adaptive security (i.e., security under adaptive corruptions) and polynomially-bounded modulus in lattices. Given the fact that polynomial modulus provides more post-quantum confidence than super-poly modulus, Devevey et al. [PKC 2021] left constructing an adaptively secure TD with polynomial modulus from lattices as an open problem.

In this paper, we resolve the above open problem by proposing three adaptively secure (t, N) -TD schemes based on the LWE assumption, all with polynomial modulus under appropriate settings.

- TD₀: an adaptively CPA-secure scheme in the asynchronous setting in the standard model, whose modulus is polynomial for small number of users N .
- TD₁: an adaptively CCA-secure scheme in the asynchronous setting in the standard model, whose modulus is polynomial for small N and bounded decryption queries.
- TD₂: an adaptively CCA-secure scheme in the synchronous setting in the random oracle (RO) model, whose modulus is polynomial for bounded decryption queries.

The main technical challenge is to limit the leakage of secret key shares arising from decryption queries, while keeping the modulus a polynomial. To overcome this barrier, we develop a refined *polynomial noise flooding technique* based on a detailed min-entropy analysis of secret shares conditioned on linear matrix hints, leveraging recent advances on Matrix-Hint LWE. Based on our new technique, we build TD₁ using the replicated secret sharing (RSS) scheme, hence supporting only small N . To enable larger N , we design TD₂ using the Shamir secret

sharing scheme, in which we further integrate our new technique with the zero-sum masking technique [Katsumata et al., CRYPTO 2024] to restrict the secret key leakage. To the best of our knowledge, our TD_1 and TD_2 are the *first* non-interactive lattice-based threshold decryption schemes achieving adaptive CCA security and polynomial modulus, simultaneously. Moreover, they achieve the strongest notion of adaptive CCA security among those compared in [Brzuska et al., PKC 2026]. We further establish robustness for both TD_0 and TD_1 via publicly verifiable partial decryptions, ensuring that the combination either outputs the correct plaintext or aborts.

1 Introduction

Threshold Decryption (TD) is a type of public-key encryption (PKE), whose decryption functionality is fulfilled by multiple decryptors. More precisely, with a (t, N) -TD scheme, the secret (decryption) key is pre-shared among N decryptors, each holding a secret key share. To decrypt a ciphertext, each decryptor computes a partial decryption with its own secret key share. Then t partial decryptions can be combined to recover the underlying plaintext from the ciphertext. TD supports distributed decryptions and hence is very friendly to distributed systems. Since its introduction by Desmedt and Frankel in [19], TD has been widely deployed as a useful cryptographic tool to support applications in blockchains [2, 12], multi-party computation [9, 54], e-voting [15], etc.

Distributed protocols are usually interactive ones and need multi-round interactions between users. But non-interactive ones are especially welcomed in the real deployments due to their communication round-optimality and low latency. In this paper, we focus on *non-interactive* TD, where the process of computing partial decryptions does not need interactions between the decryptors.

Besides, according to recent works [55, 33], TD schemes can be broadly categorized into asynchronous and synchronous settings. In the *asynchronous* setting, any set of t partial decryptions can be combined to recover the underlying plaintext, without knowing in advance which t decryptors will contribute. In the *synchronous* setting, the set of t participating decryptors is known at the outset of the decryption protocol. We study both types of TD schemes in this paper.

Robustness of TD. In the application of a (t, N) -TD, there might exist malicious decryptors who do not follow decryption procedure to contribute the correct partial decryptions for a target ciphertext. If the existence of one malicious decryptor can fail the whole decryption of the ciphertext, then a TD scheme might never fulfill its decryption function in the application. This is similar to DoS attack launched by some malicious decryptors. The *robustness* of (t, N) -TD is just a solution to this problem. In a robust (t, N) -TD, a verification algorithm is used to check the validity of partial decryptions. As long as there are at least t valid partial decryptions, these partial decryptions can be successfully combined to recover the right plaintext under the target ciphertext. It is desirable for TD schemes to have robustness, and some existing schemes are equipped with verification algorithms to accomplish robustness [20, 16, 53].

Security of TD. Like PKE, the security of TD is also formalized in an indistinguishable (IND) way: any PPT adversary is not able to distinguish a challenge ciphertext encrypting plaintext m_0 from a ciphertext encrypting plaintext m_1 . However, (t, N) -TD has to consider more than PKE, since there are N decryptors instead of one. The IND security of TD must allow the adversary to corrupt up to $t - 1$ decryptors, reflecting its resilience to $t - 1$ malicious decryptors.

- **Static corruptions vs. Adaptive corruptions.** If the adversary can designate a corruption set of (at most $t - 1$) decryptors right after it sees the public key of (t, N) -TD, then the security is under static corruptions and hence called static security. If the adversary can adaptively obtain the secret key shares of any (up to $t - 1$) corrupted decryptor during the IND game, the attack is under adaptive corruptions and hence called adaptive security. Obviously adaptive corruptions is more powerful than static corruptions, and achieving adaptive security is much more challenging than static security.
- **CPA security vs. CCA security.** Similar to PKE, the IND security can be classified into IND-CPA and IND-CCA security (CPA and CCA security for short), depending on whether adversary $\mathcal{A}$ has access to a decryption oracle, which decrypts any ciphertext of $\mathcal{A}$'s choice, other than the challenge ciphertext. Boneh et al. [3] also proposed an enhanced CPA/CCA security for TD. The enhanced CPA/CCA security further allows $\mathcal{A}$ to access an encryption oracle, which not only encrypts the message submitted by $\mathcal{A}$ and returns the corresponding ciphertext ct to $\mathcal{A}$, but also decrypts ct with each secret key share and returns all the N partial decryptions to $\mathcal{A}$.

Most existing constructions for threshold decryption [3, 6, 42, 14, 8, 33, 46, 47, 55] consider only static corruptions. Only few works achieve CCA security under adaptive corruptions. Examples include the TD scheme from pairings in [37] and the DCR/LWE-based TD scheme in [20]. Recent works [16, 53] constructed TD schemes with enhanced CCA from the DDH assumption under adaptive corruptions in the RO model. Recently, Lapiha and Prest [35] proposed a lattice-based 3-round threshold KEM in the synchronous setting and proved its CCA security under static corruptions. Subsequently, Boudgous et al. [5] observed that the scheme in [35] can also achieve adaptive-CCA security, however, it lacks a formal proof of adaptive-CCA security and still requires 3-round interactions.

In this paper, we focus on non-interactive (t, N) -TD and its enhanced CPA/CCA security under adaptive corruptions, which are simply called CPA/CCA security throughout our paper.

Post-Quantum Secure TD. In the past a few years, NIST (the National Institute of Standards and Technology) has been calling for and working on post-quantum transition. There are many efforts on post-quantum secure non-interactive threshold decryption schemes, most of which are constructed from lattices. See Table 1 for their features and securities.

It is not easy to achieve CCA security under adaptive corruptions. Recently, Brzuska et al. [10] proposed a generic CPA-to-CCA transformation. However, to the best of our knowledge, no lattice-based adaptively CPA-secure TD scheme

is known to satisfy the requirements for the transformation in [10]. As shown in Table 1, most of the existing works [3, 6, 42, 14, 8, 33, 46, 47, 55] obtain only CPA-security under static corruptions, and the only exception is the TD scheme in [20]. Devevey et al. [20] proposed the first non-interactive threshold decryption with CCA security under adaptive corruptions based on the LWE assumption. However, to support decryption services for adversaries, they used a so-called noise flooding technique, which introduces a super-polynomial modulus q to the LWE problem. It is known that LWE is at least as hard as quantumly approximating certain “short vector” problems on n -dimensional lattices within $\tilde{O}(q\sqrt{n})$ factors. The super-polynomial size of the modulus q weakens the hardness of LWE assumption [48]. Therefore, Devevey et al. [20] left the following open question: *“It remains an open problem to prove adaptive security under a more common LWE assumption with polynomial approximation factor.”*

Addressing this open question is also identified by [27] as a particularly challenging task.

Later, many works tried to reduce the modulus size to polynomial so that the hardness of the underlying LWE assumption guaranteed by polynomial approximation factor. Boudgoust and Scholl [6] proposed a (t, N) -TD scheme with modulus $q = O(\sqrt{\ell} \cdot \text{poly}(\lambda))$, where ℓ indicates the number of encryption queries. However, their TD scheme only achieves CPA security under static corruptions in the random oracle model. Recent progress by Micciancio and Suhl [42] made the modulus size in LWE independent of the number of encryption queries ℓ , and their scheme achieves CPA security in the standard model. However, their TD scheme supports only $t = N$ (i.e., (N, N) -TD scheme) and their security is proven under static corruptions. Cini et al. [14] proposed a CPA secure TD scheme under static corruptions with modulus $q = O(N^t \cdot \sqrt{\ell} \cdot \text{poly}(\lambda))$.

In summary, the existing post-quantum TD schemes from lattices either have adaptive security with super-polynomial modulus, or use a polynomial modulus to achieve only static security. Therefore, the problem left by Devevey et al. is still open up to now.

Our Contributions. We resolve the aforementioned open problem by proposing three non-interactive (t, N) -TD schemes, which have polynomial modulus under appropriate settings.

- We propose an adaptively CPA-secure asynchronous (t, N) -TD scheme TD_0 in the standard model with modulus $q \leq N^t \cdot \text{poly}(\lambda)$. The modulus is independent of the number of the adversary’s queries ℓ , and is polynomial for settings of small user number N (like $N \leq 16$). It also achieves robustness.
- We propose an adaptively CCA-secure asynchronous (t, N) -TD scheme TD_1 in the standard model with modulus $q \leq N^t \cdot \sqrt{\ell} \cdot \text{poly}(\lambda)$, where ℓ denotes the number of encryption and partial decryption queries. The modulus is polynomial for settings of small user number N and bounded query number ℓ (e.g., $\ell \leq 2^{64}$ as suggested by NIST [45, 4.A.2]). It also achieves robustness.
- We propose an adaptively CCA-secure synchronous (t, N) -TD scheme TD_2 in the RO model with modulus $q \leq \sqrt{\ell} \cdot \text{poly}(\lambda)$. The modulus is polynomial

for settings of bounded decryption queries like TD_1 . In contrast to the above two schemes, TD_2 supports any polynomial number of N of users.

To our best knowledge, our TD_1 and TD_2 are the *first* non-interactive TD schemes achieving CCA-security against adaptive corruptions and polynomial modulus¹ from lattices, simultaneously. Moreover, they achieve the strongest notion of adaptive CCA security among those compared in [10, Fig. 1]. In Table 1, We compare our TD_0 , TD_1 and TD_2 with existing non-interactive TD schemes from lattices.

In Appendix G, we provide an asymptotic efficiency comparison between our CCA-secure TD_1 and the lattice-based TD scheme of [20], as well as concrete parameters and efficiency estimates for our TD_2 at the 128-bit security level.

Concurrent Work [27]. We are aware of a concurrent work [27], which also constructs non-interactive CCA-secure threshold Regev encryption with a polynomial modulus for a small number of users. More precisely, they first establish a threshold Regev scheme with static simulation CPA security in the asynchronous setting, and then obtain simulation CCA security by applying the generic CPA-to-CCA transformation in [10] with their novel adaptive hint-MLWE assumption. Their work is primarily efficiency-oriented and provides optimized concrete parameters for up to $N = 32$ users. However, they only considered static corruptions and pointed out that CCA security under adaptive corruptions “appears particularly challenging to achieve with a polynomial moduli” [27, p. 13].

In contrast, we accomplish this challenging task by proposing the first adaptively CCA-secure threshold decryption schemes with polynomial moduli in both standard and random oracle model. Moreover, robustness against maliciously generated partial decryptions is not considered in [27], whereas our asynchronous construction TD_1 explicitly addresses this property.

Technical Overview. Our threshold decryption schemes stem from the dual Regev public-key encryption (PKE) scheme [25]. With the basic CPA-secure dual Regev scheme, one can use public key $\text{pk} = \mathbf{A}\mathbf{k}$ to encrypt a message μ and get a ciphertext consisting of $(\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top, d = \mathbf{s}^\top \text{pk} + e' + \text{Encode}(\mu) \in \mathbb{Z}_q)$, where $\mathbf{c}$ is an LWE instance, e' is an LWE error, and $\text{Encode}(\mu)$ helps to remove noises in the decryption. To leverage it to a CCA-secure PKE, we can resort to a tag-based non-interactive zero-knowledge (NIZK) proof with one-time simulation soundness (OTSS) [51, 17] to prove that $\mathbf{c}$ is an LWE instance with witness $(\mathbf{s}, \mathbf{e})$. Accordingly, the CCA-secure PKE works as follows.

- The secret key is a small-norm vector $\text{sk} = \mathbf{k}$ and the public key is $\text{pk} = \mathbf{A}\mathbf{k}$.
- The ciphertext for message μ is defined by

$$\text{ct} := (\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top, d = \mathbf{s}^\top \text{pk} + e' + \text{Encode}(\mu), \pi \leftarrow \text{Prove}(\tau, \mathbf{c}, (\mathbf{s}, \mathbf{e}))),$$

where $\mathbf{s}$ is uniform modulo q and $\mathbf{e}, e'$ are of small norm, and $\tau := H(\mathbf{c}, d)$ with H a collision-resistant hash function.

¹ Though q depends on ℓ , we take $q = O(\sqrt{\ell})$ as a polynomial modulus, following the convention of prior works [6, 14]. Furthermore, many recent works [18, 30, 49] consider $q = O(\sqrt{\ell})$ as an acceptable modulus for $\ell \leq 2^{64}$.

Table 1. Comparison with existing *non-interactive* threshold decryption schemes from lattices (a) in the asynchronous setting and (b) in the synchronous setting. Here the column “**#Shares**” shows the number of shares per user, which depends on the underlying secret sharing scheme: schemes with $O(1)$ shares can efficiently support large number of users N , while the remaining schemes are practical only for small N . “ $O(1)^*$ ” indicates that HKM+26 [27] does have $O(1)$ secret shares, but is practical for small N, t . “**Security**” indicates whether the TD scheme has CPA or CCA security, where the symbol CPA*/CCA* indicates that the CPA/CCA security is a simulation-based one. “**Adaptive Corruptions**” (resp., “**Standard Model**”) indicates whether the TD scheme is proven to be secure under adaptive corruptions (resp., in the standard model). “**Modulus**” shows the asymptotic growth of the modulus q , where ℓ denotes the number of encryption and/or partial decryption queries. “**Robustness**” indicates whether the TD scheme satisfies robustness. “**Assumption**” specifies the underlying hardness assumptions. If the underlying assumption can be reduced to a more standard assumption, we show the latter. For example, the security of MS24 [42] is based on the known-norm LWE assumption, which can be reduced to standard LWE as proved in [42], so we write LWE for their scheme. We do not include [43, 44] in the table, since Colin de Verdière et al. [55] demonstrated key-recovery attacks against these schemes in the IND-CPA security model.

(a) Non-interactive TD schemes from lattices in the *asynchronous* setting.

TD Schemes	#Shares	Security	Adaptive Corruptions	Modulus	Robustness	Assumption	Standard Model
BGG+18 [3]	$O(1)$	CPA*	×	$\text{superpoly}(\lambda)$	×	LWE	✓
BS23 [6]	$O(N^{4.3})$	CPA	×	$\sqrt{\ell} \cdot \text{poly}(\lambda)$	✓	MLWE	×
MS24 [42]	$O(1)$	CPA*	×	$\text{poly}(\lambda)$	×	LWE	✓
CLW25 [14]	$O(t \log N)$	CPA*	×	$N^t \cdot \sqrt{\ell} \cdot \text{poly}(\lambda)$	×	LWE	✓
BFM+25 [8]	$O(1)$	CPA*	×	$\text{superpoly}(\lambda)$	✓	RLWE	×
KLL+26 [33]	$O(1)$	CPA*	×	$\text{superpoly}(\lambda)$	×	RLWE	✓
HKM+26 [27]	$O(1)^*$	CCA*	×	$N^{O(t)} \cdot \sqrt{\ell} \cdot \text{poly}(\lambda)$	×	MLWE	×
DLN+21 [20]	$O(N^{4.3})$	CCA	✓	$\text{superpoly}(\lambda)$	✓	LWE	✓
Our TD₀	$O(N^t)$	CPA	✓	$N^t \cdot \text{poly}(\lambda)$	✓	LWE	✓
Our TD₁	$O(N^t)$	CCA	✓	$N^t \cdot \sqrt{\ell} \cdot \text{poly}(\lambda)$	✓	LWE	✓

(b) Non-interactive TD schemes from lattices in the *synchronous* setting.

TD Schemes	#Shares	Security	Adaptive Corruptions	Modulus	Robustness	Assumption	Standard Model
OT25 [46], OT26 [47]	$O(1)$	CPA*	×	$\text{superpoly}(\lambda)$	×	LWE/MLWE	✓
CPS26 [55]	$O(1)$	CPA*	×	$\text{superpoly}(\lambda)$	×	LWE	×
Our TD₂	$O(1)$	CCA	✓	$\sqrt{\ell} \cdot \text{poly}(\lambda)$	×	LWE	×

- To decrypt $\text{ct} = (\mathbf{c}, d, \pi)$, the decryptor first verifies π with Vrfy of NIZK, computes $h := d - \mathbf{c}^\top \text{sk}$ and obtains $\mu := \text{Decode}(h)$.

We briefly recall its CCA-security proof, which goes with four main steps. The goal is to show that an adversary $\mathcal{A}$ can hardly tell whether a challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ encrypts μ_0 or μ_1 , even if $\mathcal{A}$ gets $\text{pk} = \mathbf{A}\mathbf{k}$ and can obtain decryptions of any ciphertexts $\text{ct} = (\mathbf{c}, d, \pi) \neq \text{ct}^*$.

- Firstly, we use the zero-knowledge simulator of NIZK to generate π^* in ct^* , without using the witness for $\mathbf{c}^*$.
- Secondly, we change $\mathbf{c}^*$ in ct^* to a random one based on the LWE assumption.
- Thirdly, we resort to the OTSS of NIZK to ensure that the decryption succeeds only when $\mathbf{c}$ in ct is an LWE instance (since $\mathcal{A}$ can hardly forge a valid

proof π for a non-LWE instance $\mathbf{c}$, even if it sees a simulated π^*). This can be used to limit the information about $\mathbf{sk}$ leaked from the decryptions.

- Finally, we use the remaining entropy in $\mathbf{sk}$ to show that d^* in $\mathbf{ct}^*$ can be changed to uniformly random, so $\mathbf{ct}^*$ is totally independent of μ_0 and μ_1 .

Note that there is only one security reduction with the LWE assumption (i.e., the second step) and this security reduction does not involve the secret key $\mathbf{sk} = \mathbf{k}$. Therefore, the reduction algorithm can always sample $\mathbf{sk}$ by itself. This fact lays a very good foundation for threshold decryption to achieve CCA-security under adaptive corruptions, since the reduction algorithm can always generate $\mathbf{sk}$ and the key shares to freely deal with adaptive corruptions from adversaries.

THRESHOLDIZING THE SCHEME VIA LINEAR SECRET SHARING & OBSTACLES IN LIMITING THE INFORMATION OF $\mathbf{sk}$ LEAKED FROM PARTIAL DECRYPTIONS.

We will construct TD schemes based on the above CCA-secure PKE. Observe that its decryption is just a linear operation with $\mathbf{sk}$, so it is friendly to linear secret sharing. Let us share $\mathbf{sk} = \mathbf{k}$ with a linear secret sharing scheme (e.g., Shamir secret sharing [52]), and produce N secret key shares $\mathbf{sk}_1, \dots, \mathbf{sk}_N$. For partial decryption, each decryptor i may compute a decryption share

$$h_i := \mathbf{c}^\top \mathbf{sk}_i \quad (1)$$

and then a combiner can combine the decryption shares from set S as

$$\sum_{i \in S} \lambda_{i,S} \cdot h_i = \sum_{i \in S} \lambda_{i,S} \cdot (\mathbf{c}^\top \mathbf{sk}_i) = \mathbf{c}^\top (\sum_{i \in S} \lambda_{i,S} \cdot \mathbf{sk}_i) = \mathbf{c}^\top \mathbf{sk}, \quad (2)$$

and then recover μ as before. Here $\{\lambda_{i,S}\}$ are the combining coefficients determined by S depending on the secret sharing scheme (e.g., $\{\lambda_{i,S}\}$ are just the Lagrange interpolation coefficients for Shamir secret sharing).

Though the above idea works correctly, it is completely insecure, since the partial decryption $h_i := \mathbf{c}^\top \mathbf{sk}_i$ in (1) leaks too much information about $\mathbf{sk}_i$: collecting enough equations of h_i w.r.t different $\mathbf{c}$ can completely solve $\mathbf{sk}_i$.

To circumvent this problem, one needs to find a way to limit the leakage of $\mathbf{sk}_i$ while preserving the correctness of partial decryptions. There are generically two solutions.

- **Noise flooding: relying on super-polynomial modulus.** One common way to address this problem is adding some noise flooding term y_i to the partial decryption (1), as in [3, 20]. The idea is that as long as $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ (which can be ensured by the OTSS-NIZK), then in the partial decryption,

$$h_i := \mathbf{c}^\top \mathbf{sk}_i + y_i = \mathbf{s}^\top \mathbf{A} \mathbf{sk}_i + \mathbf{e}^\top \mathbf{sk}_i + y_i, \quad (3)$$

we expect the noise y_i to smudge the term $\mathbf{e}^\top \mathbf{sk}_i$, so that the leakage of $\mathbf{sk}_i$ is limited in $\mathbf{A} \mathbf{sk}_i$.

However, this method has two limitations. The first is that it requires a *super-polynomial modulus*, since the added noise y_i must be sufficiently large to statistically hide $\mathbf{e}^\top \mathbf{sk}_i$. The second problem is that correctness no longer holds if the combining coefficients $\lambda_{i,S}$ of the secret sharing scheme are large

(e.g., Shamir secret sharing), since the combination now introduces a large term $\sum_{i \in S} \lambda_{i,S} \cdot y_i$ to (2), which hinders the message recovery.

- **Masking technique: not work for polynomial modulus.** Another possible solution is adding some random mask Δ_i to the partial decryption (1)

$$h_i := \mathbf{c}^\top \mathbf{sk}_i + \Delta_i = \mathbf{s}^\top \mathbf{A} \mathbf{sk}_i + \mathbf{e}^\top \mathbf{sk}_i + \Delta_i.$$

This masking technique was introduced in [18] and has been used in the design of many threshold signature schemes [18, 30] and threshold fully homomorphic encryption [55]. By the masking technique, each mask Δ_i is randomly distributed over $\mathbb{Z}_q$, so it can perfectly hide the term $\mathbf{e}^\top \mathbf{sk}_i$; moreover, the masks $\{\Delta_i\}_{i \in S}$ in set S satisfy $\sum_{i \in S} \Delta_i = 0$, so that these masks can be canceled during combination. (Technically, these masks are generated by random oracle (RO) on secret seeds $\text{seed}_{i,j}$ pre-shared between each pair of users in their secret key shares.)

A subtle problem is that, to enable the cancellation of the masks during combination, the partial decryption should compute

$$h_i := \lambda_{i,S} \cdot \mathbf{c}^\top \mathbf{sk}_i + \Delta_i, \quad (4)$$

and the combination computes

$$\sum_{i \in S} h_i = \sum_{i \in S} \lambda_{i,S} \cdot \mathbf{c}^\top \mathbf{sk}_i + \sum_{i \in S} \Delta_i = \mathbf{c}^\top \mathbf{sk}. \quad (5)$$

Namely, the combining coefficient $\lambda_{i,S}$ is “moved” from the combination to partial decryption. However, this technique in fact *does not solve the leakage problem* — although the mask Δ_i hides the leakage of $\mathbf{sk}_i$ in partial decryption (4), the cancellation of the masks during combination (5) totally leaks $\mathbf{c}^\top \mathbf{sk}$, which will leak $\mathbf{sk}$ if one gets $\mathbf{c}^\top \mathbf{sk}$ w.r.t different $\mathbf{c}$.

In summary, these existing two solutions either rely on super-polynomial modulus, or do not work at all in limiting the leakage of secret key.

In this work, we propose different new solutions, resulting in our three TD schemes TD_1 , TD_2 , TD_0 with polynomial modulus.

TECHNIQUE 1: POLYNOMIAL NOISE FLOODING & OUR TD_1 FOR SMALL N .

Roughly speaking, our scheme TD_1 is designed according to the above noise flooding method, but we extend the noise flooding technique to support polynomial modulus. Namely, the partial decryption of TD_1 is similar to (3), i.e.,

$$h_i := \mathbf{c}^\top \mathbf{sk}_i + y_i = \mathbf{s}^\top \mathbf{A} \mathbf{sk}_i + \mathbf{e}^\top \mathbf{sk}_i + y_i, \quad (6)$$

but here we use a *polynomial* noise y_i to smudge the term $\mathbf{e}^\top \mathbf{sk}_i$.

The key observation is that, for achieving CCA-security, we do not need to fully hide the term $\mathbf{e}^\top \mathbf{sk}_i$ (in which case a super-polynomial y_i is inevitable). In fact, ensuring that there are enough entropy left in $\mathbf{sk}_i$ conditioned on $\mathbf{e}^\top \mathbf{sk}_i + y_i$ (and $\mathbf{A} \mathbf{sk}_i$) is already sufficient for our CCA-security proof (in which case a polynomial y_i might be possible).

More precisely, let ℓ denote the number of queries made by a CCA adversary $\mathcal{A}$, so for each share $\mathbf{sk}_i$, $\mathcal{A}$ can learn $\mathbf{Ask}_i$ and at most ℓ linear relations of the form $\{\mathbf{e}_j^\top \mathbf{sk}_i + y_j\}_{j \in [\ell]}$ from ℓ partial decryptions. By defining $\mathbf{E} := (\mathbf{e}_1, \dots, \mathbf{e}_\ell)^\top$ and $\mathbf{y} := (y_1, \dots, y_\ell)^\top$, the linear relations can be written as $\mathbf{Esk}_i + \mathbf{y}$, which forms a so-called matrix-hint in [23, 56]. Inspired by the recent works on Hint-MLWE [32] and Matrix Hint-MLWE [23, 56], we carefully analyze the conditional distribution of $\mathbf{sk}_i$ conditioned on $(\mathbf{Ask}_i, \mathbf{Esk}_i + \mathbf{y})$ and its min-entropy, which is summarized as our main technical lemma (i.e., Lemma 6 in Subsect. 4.3). Our Lemma 6 shows that $\mathbf{y}$ of norm $O(\sqrt{\ell})$ can already ensure that enough entropy is left in $\mathbf{sk}_i$ conditioned on $(\mathbf{Ask}_i, \mathbf{Esk}_i + \mathbf{y})$, so the modulus of our TD_1 can be set to $O(\sqrt{\ell})$. To establish Lemma 6, we proceed in three steps.

- Firstly, we note that conditioned on $\mathbf{Ask}_i \in \mathbb{Z}_q^n$, the min-entropy of $\mathbf{sk}_i$ is reduced by at most $n \log q$ bits. So it suffices to analyze the conditional distribution and min-entropy of $\mathbf{sk}_i$ conditioned on $\mathbf{Esk}_i + \mathbf{y}$.
- Secondly, we show that the conditional distribution of $\mathbf{sk}_i \in \mathbb{Z}^m$ conditioned on $\mathbf{Esk}_i + \mathbf{y}$ is exactly a shifted ellipsoidal discrete Gaussian $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$ over $\mathbb{Z}^m$ with some center $\mathbf{c}$ and covariance matrix Σ .
Then by a detailed analysis of this distribution, we show that if all entries of $\mathbf{E}$ are short, given $\mathbf{y}$ of norm $O(\sqrt{\ell})$, the min-entropy of $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$ over $\mathbb{Z}^m$ is about $m \log \sigma$ for some $\sigma > 0$.²
- Finally, for the min-entropy of $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$ over $\mathbb{Z}_p^m$ for some prime p , we show that the min-entropy is at least $m \log p$ for sufficiently small p , while remains $m \log \sigma$ for sufficiently large p .

Based on Lemma 6, we can choose appropriate parameters to ensure that the conditional distribution of $\mathbf{sk}_i$ always has sufficient min-entropy (i.e., $m \log p - n \log q$ or $m \log \sigma - n \log q$) when modulo each prime factor p of the modulus q .

As mentioned above, the noise flooding method does not apply to secret sharing scheme with large combining coefficients $\lambda_{i,S}$ (e.g., Shamir secret sharing). To solve this problem, in our TD_1 , we use the replicated secret sharing (RSS) scheme [28] for sharing $\mathbf{sk}$, whose combining coefficients $\lambda_{i,S} = 1$ for all i and S . More precisely, for RSS, each share $\mathbf{sk}_i$ contains a set of vectors $\{\mathbf{k}_I\}$, and the recovery of $\mathbf{sk} = \mathbf{k}$ needs the summation of $\binom{N}{N-t+1}$ vectors $\{\mathbf{k}_I\}_I$. (We refer to Subsect. 2.2 for more details about RSS.) This means that during combination (2), the noise y_i is accumulated to $\sum_i y_i$ where the summation is over $\binom{N}{N-t+1}$ terms. So to ensure correctness, we need the modulus of our TD_1 to be $O(\binom{N}{N-t+1} \sqrt{\ell}) \approx O(N^t \sqrt{\ell})$. When N is small (like $t \leq N \leq 16$)³ and ℓ is bounded (like $\ell \leq 2^{64}$)⁴, the modulus of our TD_1 is a polynomial.

² We are aware of alternative entropy-analysis techniques in prior works [7, 4]. While these techniques could be adapted to our setting, our direct analysis yields a tighter parameter requirement. See Remark 1 in Appendix B.2 for a detailed comparison.

³ Note that the RSS scheme has been recently used in [50] for the design of compact threshold signatures for $t \leq N \leq 16$.

⁴ As noted in Footnote 1, the existing works [6, 14] take $q = O(\sqrt{\ell})$ as a polynomial modulus, and [18, 30, 49] consider $q = O(\sqrt{\ell})$ as an acceptable modulus for $\ell \leq 2^{64}$.

Our TD_1 also achieves robustness, by simply requiring each decryptor, when computing the partial decryption h_i in (6), also generate an adaptively sound NIZK proof π_i to prove the well-formedness of h_i . By verifying the validity of π_i , one can check whether h_i is indeed computed according to (6).

TECHNIQUE 2: MASKING + POLYNOMIAL NOISE FLOODING & OUR TD_2 FOR LARGE N . To enable larger N , we go back to the Shamir secret sharing scheme for sharing sk , and design TD_2 by carefully integrating our polynomial noise flooding technique with the masking technique. More precisely, the partial decryption of our TD_2 is similar to (4), but we add both a polynomial noise y_i and a random mask Δ_i to it, i.e.,

$$h_i := \lambda_{i,S} \cdot \mathbf{c}^\top \text{sk}_i + y_i + \Delta_i = \lambda_{i,S} \cdot \mathbf{s}^\top \mathbf{A} \text{sk}_i + \lambda_{i,S} \cdot \mathbf{e}^\top \text{sk}_i + y_i + \Delta_i. \quad (7)$$

Then the combination (5) becomes

$$\begin{aligned} \sum_{i \in S} h_i &= \sum_{i \in S} \lambda_{i,S} \cdot \mathbf{s}^\top \mathbf{A} \text{sk}_i + \sum_{i \in S} \lambda_{i,S} \cdot \mathbf{e}^\top \text{sk}_i + \sum_{i \in S} y_i + \sum_{i \in S} \Delta_i \\ &= \mathbf{s}^\top \mathbf{A} \text{sk} + \mathbf{e}^\top \text{sk} + \sum_{i \in S} y_i. \end{aligned} \quad (8)$$

The usage of both polynomial noise y_i and random mask Δ_i plays a crucial role in limiting the leakage of secret key for our TD_2 .

- **Using random mask Δ_i to hide leakage information in partial decryption (7).** As we use Shamir secret sharing to support larger N , the combining coefficient $\lambda_{i,S}$ is large. Consequently, a polynomial noise y_i is not enough to hide the term $\lambda_{i,S} \cdot \mathbf{e}^\top \text{sk}_i$ in (7), and thus we resort to the random mask Δ_i to hide the term $\lambda_{i,S} \cdot \mathbf{e}^\top \text{sk}_i$ in our partial decryption (7).
- **Using polynomial noise $\sum_{i \in S} y_i$ to hide leakage information in combination (8).** In our combination (8), although the random masks Δ_i cancels, the terms $\lambda_{i,S} \cdot \mathbf{e}^\top \text{sk}_i$ sums to a short $\mathbf{e}^\top \text{sk}$, thus $\sum_{i \in S} y_i$ with polynomial y_i suffices for hiding the term $\mathbf{e}^\top \text{sk}$ in (8) by our polynomial noise flooding technique (i.e., our Lemma 6), so that there are enough entropy left in sk conditioned on $\mathbf{e}^\top \text{sk} + \sum_{i \in S} y_i$ (and $\mathbf{A} \text{sk}$).

Since our TD_2 uses Shamir secret sharing instead of RSS, it supports arbitrary polynomial user number N , and a modulus of size $O(\sqrt{\ell})$ suffices for the CCA-security of TD_2 , according to our Lemma 6.

On the other hand, as we use the masking technique in [30], the CCA-security of TD_2 relies on the RO model. Moreover, it incurs complications in achieving robustness for TD_2 , since it is quite involved to propose a NIZK proof for proving the well-formedness of h_i in (7), as the mask Δ_i is computed from RO. Recently, Rafael del Pino et al. [49] designed an identifiable-abort (IA) protocol for threshold signature schemes using masking technique, where they showed how to propose NIZK proofs for relations involving RO-based masks. By adapting

⁵ As mentioned above, similar to (4), the coefficient $\lambda_{i,S}$ is “moved” from the combination to partial decryption to ensure correctness, and accordingly, the partial decryption of our TD_2 needs to know the set S to compute the coefficient $\lambda_{i,S}$.

their technique, we may also obtain a NIZK proof for proving the relation (7) and hence achieve robustness for our TD_2 .

OUR CPA-SECURE TD_0 FOR UNBOUNDED QUERIES. We also propose a CPA-secure TD scheme TD_0 whose modulus gets rid of $O(\sqrt{\ell})$, thus supporting an arbitrary polynomial number ℓ of queries. To this end, we no longer use our polynomial noise flooding technique (i.e., Lemma 6), but utilize the ReRand algorithm [31, 20] to compute the partial decryption as follows:

$$h_i \leftarrow \text{ReRand}(\text{sk}_i, \mathbf{c}, \sigma, \zeta), \quad (9)$$

where σ, ζ are some Gaussian width parameters. According to the property of ReRand (cf. Lemma 16 in Appendix F), if $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ where $\mathbf{e}^\top$ follows a discrete Gaussian distribution of width σ , then the h_i generated by ReRand in (9) is statistically close to $h_i := \mathbf{s}^\top \mathbf{A} \cdot \text{sk}_i + y_i$, where y_i is distributed as a discrete Gaussian of width $2\sigma\zeta$. So the partial decryptions automatically hide information about sk_i beyond Ask_i .

Note that the above argument requires the $\mathbf{e}^\top$ in $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ be distributed as a discrete Gaussian, and this requirement holds for honestly generated ciphertexts (but may not hold for those ciphertexts submitted by $\mathcal{A}$ to a decryption oracle). So our TD_0 achieves CPA-security rather than CCA-security.

Moreover, similar to our TD_1 scheme, to ensure correctness of decryption, we cannot use Shamir secret sharing whose combining coefficients $\lambda_{i,S}$ are large, but use the RSS with $\lambda_{i,S} = 1$. Accordingly, the modulus of our TD_0 is $O(\binom{N}{N-t+1}) \approx O(N^t)$, which is a polynomial for small N . As TD_0 can be viewed as a simplification of TD_1 , we defer its full construction and security analysis to Appendix F.

A SUBTLE PROBLEM: HOW TO CHANGE $\mathbf{c}^*$ IN ct^* FROM AN LWE INSTANCE TO RANDOM. Finally, we explain the remaining arguments for the CCA-security of our TD_1, TD_2 schemes and for the CPA-security of our TD_0 scheme. That is, given that there are enough entropy left in $\text{sk} = \mathbf{k}$ (and sk_i), how to ensure that the $(\mathbf{c}^{*\top} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top, d^* = \mathbf{s}^\top \mathbf{pk} + e' + \text{Encode}(\mu))$ in the challenge ciphertext ct^* hides the message $\mu \in \{\mu_0, \mu_1\}$ that it encrypts.

A natural idea is to first change $\mathbf{c}^*$ to a random vector by the LWE assumption, and then show that d^* hides $\text{Encode}(\mu)$. However, observing that

$$\begin{aligned} d^* &= \mathbf{s}^\top \mathbf{A} \mathbf{k} + e' + \text{Encode}(\mu) = (\mathbf{c}^{*\top} - \mathbf{e}^\top) \mathbf{k} + e' + \text{Encode}(\mu) \\ &= \mathbf{c}^{*\top} \mathbf{k} - \mathbf{e}^\top \mathbf{k} + e' + \text{Encode}(\mu), \end{aligned} \quad (10)$$

we can not immediately change $\mathbf{c}^* = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ to random based on LWE since d^* involves $\mathbf{e}^\top \mathbf{k}$, which depends on the error term $\mathbf{e}$ in $\mathbf{c}^*$.

An easy solution is to use a large enough noise e' , so that it can smudge $\mathbf{e}^\top \mathbf{k}$ in (10). However, this introduces a super-polynomial modulus again.

To rescue it with a polynomial modulus, we make use of a new Matrix-Hint LWE assumption in a recent work [56], which shows that the standard LWE implies the Matrix-Hint LWE assumption with the same polynomial modulus.

The Matrix-Hint LWE assumption implies that $\mathbf{c}^{*\top} = \mathbf{A}^\top \mathbf{s} + \mathbf{e}^\top$ is computationally indistinguishable from a random vector, even conditioned on a noisy hint $\mathbf{e}' - \mathbf{e}^\top \mathbf{k}$ of $\mathbf{e}$. So the Matrix-Hint LWE perfectly solves this subtle problem.

After changing $\mathbf{c}^*$ to random, by the leftover hash lemma, the uniformly random $(\mathbf{c}^{*\top} - \mathbf{e}^\top)$ in (10) can extract the entropy left in $\mathbf{s}\mathbf{k} = \mathbf{k}$, so that $(\mathbf{c}^{*\top} - \mathbf{e}^\top)\mathbf{k}$ in (10) is statistically close to uniform. Consequently, d^* statistically hides the encrypted message $\mu \in \{\mu_0, \mu_1\}$, and this completes the high-level overview of the security proof of our schemes.

ON THE COMPATIBILITY OF MODULE-LWE. We construct our TD schemes from LWE rather than module-LWE mainly because our constructions (in particular, TD_1) require one-time simulation-sound (OTSS) NIZK in the standard model to achieve CCA-security. To the best of our knowledge, currently the (only) known method for constructing lattice-based OTSS-NIZK in the standard model is through the approach proposed in [36], which is based on LWE rather than module-LWE. To build our TD schemes, we directly adopt the OTSS-NIZK approach of [36] (cf. Appendix E), and construct the remaining components accordingly in the LWE setting. This keeps the presentation simple and allows us to focus on the main technical contributions of our work, i.e., constructing non-interactive adaptively CCA-secure TD schemes with polynomial modulus. On the other hand, since our use of OTSS-NIZK is entirely black-box, any future module-LWE-based OTSS-NIZK could be incorporated into our schemes to further improve efficiency.

2 Preliminaries

Notations. Let $\lambda \in \mathbb{N}$ denote the security parameter throughout the paper, and all algorithms, distributions, functions and adversaries take 1^λ as an implicit input. If x is defined by y or the value of y is assigned to x , we write $x := y$. For $i, j \in \mathbb{N}$ with $i < j$, define $[i, j] := \{i, i+1, \dots, j\}$ and $[j] := \{1, 2, \dots, j\}$. For a set $\mathcal{X}$, denote by $x \leftarrow_s \mathcal{X}$ the procedure of sampling x from $\mathcal{X}$ uniformly at random. If $\mathcal{D}$ is distribution, $x \leftarrow \mathcal{D}$ means that x is sampled according to $\mathcal{D}$. All our algorithms are probabilistic unless stated otherwise. We use $y \leftarrow_s \mathcal{A}(x)$ to define the random variable y obtained by executing algorithm $\mathcal{A}$ on input x . If $\mathcal{A}$ is deterministic we write $y \leftarrow \mathcal{A}(x)$. “PPT” abbreviates probabilistic polynomial-time. Denote by negl some negligible function.

For a vector $\mathbf{v}$, let $\|\mathbf{v}\|$ (resp., $\|\mathbf{v}\|_\infty$) denote the ℓ_2 (resp., infinity) norm of $\mathbf{v}$. For a matrix $\mathbf{A}$, let $\|\mathbf{A}\|$ (resp., $\|\mathbf{A}\|_\infty$) denote the largest ℓ_2 (resp., infinity) norm of $\mathbf{A}$ ’s columns, and let $s_1(\mathbf{A}) := \max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\| = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ denote the largest singular value (spectral norm) of $\mathbf{A}$. A distribution χ is B -bounded if its support is limited to $[-B, B]$. Let $\mathbb{Z}_q$ be the ring of integers modulo q , and its elements are represented by the integers in $(-q/2, q/2]$.

Let X and Y be two random variables. The min-entropy of X is defined as $\mathbf{H}_\infty(X) := -\log(\max_x \Pr[X = x])$, the average min-entropy of X given Y is defined as $\tilde{\mathbf{H}}_\infty(X | Y) := -\log[\mathbb{E}_{y \leftarrow Y}(\max_x \Pr[X = x | Y = y])]$, and the statistical distance of X and Y is defined as $\Delta(X, Y) := \frac{1}{2} \sum_x |\Pr[X = x] - \Pr[Y = x]|$.

If $\Delta(X, Y) \leq \text{negl}(\lambda)$, X and Y are called statistically indistinguishable, and denoted by $X \approx_s Y$. We will use the following variant of the leftover hash lemma.

Lemma 1 (Particular case of [40, Lemma 2.3]). *Let $n, m, q \in \mathbb{N}$ be integers and $\epsilon \in (0, 1)$. Suppose $\mathbf{k}$ is chosen from some distribution over $\mathbb{Z}_q^m$, and Y is an arbitrary random variable (possibly) correlated with $\mathbf{k}$. If $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid Y) \geq 2n \log q + 2 \log(\frac{1}{\epsilon})$ for any q 's prime factor p , then $\Delta((\mathbf{A}, \mathbf{A}\mathbf{k}), (\mathbf{A}, \mathbf{u}) \mid Y) \leq \epsilon$ where $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{n \times m}$ and $\mathbf{u} \leftarrow_s \mathbb{Z}_q^n$.*

In Appendix A, we present some additional preliminaries, including the syntax of collision-resistant hash functions in Appendix A.1 and the syntax of tag-based quasi-adaptive non-interactive zero-knowledge argument (QA-NIZK) for gap language and its one time simulation-soundness (OTSS) in Appendix A.2.

2.1 Lattice Backgrounds

The Gaussian function with parameter s and center $\mathbf{c} \in \mathbb{R}^n$ is defined as $\rho_{s,\mathbf{c}}(\mathbf{x}) := e^{-\pi \|\mathbf{x}-\mathbf{c}\|^2/s^2}$ for $\mathbf{x} \in \mathbb{R}^n$. For a countable set $\Lambda \subset \mathbb{R}^n$, the discrete Gaussian distribution $D_{\Lambda,s,\mathbf{c}}$ parameterized with s and $\mathbf{c}$ is defined as $D_{\Lambda,s,\mathbf{c}}(\mathbf{x}) := \rho_{s,\mathbf{c}}(\mathbf{x})/\rho_{s,\mathbf{c}}(\Lambda)$ for $\mathbf{x} \in \Lambda$ and $D_{\Lambda,s,\mathbf{c}}(\mathbf{x}) := 0$ for $\mathbf{x} \notin \Lambda$, where $\rho_{s,\mathbf{c}}(\Lambda) := \sum_{\mathbf{x} \in \Lambda} \rho_{s,\mathbf{c}}(\mathbf{x})$. We omit s when $s = 1$ and omit $\mathbf{c}$ when $\mathbf{c} = \mathbf{0}$.

Let Λ be a non-zero lattice. The *smoothing parameter* $\eta_\epsilon(\Lambda)$ of Λ is the smallest real $s > 0$ such that $\rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon$, where $\Lambda^* = \{\mathbf{b}^* \in \text{span}_{\mathbb{R}}(\Lambda) : \mathbf{b}^{*\top} \cdot \Lambda \subseteq \mathbb{Z}\}$ refers to the dual of Λ .

Definition 1 (LWE Assumption). *Let $q, m, n \in \mathbb{N}$ and χ be a distribution over $\mathbb{Z}_q^m$. The $\text{LWE}_{q,m,n,\chi}$ -assumption holds if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{q,m,n,\chi,\mathcal{A}}^{\text{LWE}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}^\top) = 1]| \leq \text{negl}(\lambda)$, where $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{n \times m}$, $\mathbf{s} \leftarrow_s \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow \chi$, $\mathbf{u} \leftarrow_s \mathbb{Z}_q^m$.*

Definition 2 (Matrix-Hint LWE, adapted from [56, Definition 5.1]).

Let $q, m, n \in \mathbb{N}$, and let χ_0, χ_1 be distributions over $\mathbb{Z}^m, \mathbb{Z}$, respectively. Let $\mathcal{S} \subseteq \mathbb{Z}^m$ be a set. The $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ -assumption holds if for any PPT $\mathcal{A}$ and any $\mathbf{k} \in \mathcal{S}$, it holds that

$$\text{Adv}_{\text{para}, \mathcal{A}}^{\text{Mat-hLWE}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top, \mathbf{k}, h) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}^\top, \mathbf{k}, h) = 1]| \leq \text{negl}(\lambda),$$

where $\text{para} = ((q, m, n, \chi_0), (\chi_1, \mathcal{S}))$, $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{m \times n}$, $\mathbf{s} \leftarrow_s \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow \chi_0$, $\mathbf{u} \leftarrow_s \mathbb{Z}_q^m$, $e' \leftarrow \chi_1$ and $h = \mathbf{k}^\top \mathbf{e} + e'$.

As shown in [56], the LWE assumption implies the matrix-hint LWE assumption under certain parameters.

Lemma 2 (LWE $\Rightarrow$ Matrix Hint-LWE, adapted from [56, Theorem 5.2]). *Let $m, n, q \in \mathbb{N}$ and $\mathcal{S} \subseteq \mathbb{Z}^m$. Let $B > 0$ be a real number such that $\|\mathbf{k}\|^2 \leq B$ for any $\mathbf{k} \in \mathcal{S}$. Let $\sigma_0, \sigma_1, \sigma, \delta > 0$ be real numbers satisfying $\frac{1}{(1+\delta)\sigma^2 + \delta_0^2} \geq \frac{1}{\sigma_0^2} + \frac{B}{\sigma_1^2}$ and $\sigma \geq \sqrt{1 + 1/\delta} \cdot \eta_\epsilon(\mathbb{Z}^m)$, where $\delta_0 := \sqrt{\frac{\ln(2m)+4}{\pi}}$. Let $\chi_0 := D_{\mathbb{Z}^m, \sigma_0}$, $\chi_1 := D_{\mathbb{Z}, \sigma_1}$, $\chi := D_{\mathbb{Z}^m, \sigma}$ be distributions over $\mathbb{Z}^m, \mathbb{Z}, \mathbb{Z}^m$, respectively.*

Then for any PPT adversary $\mathcal{A}$ against the $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ assumption, there exists a PPT $\mathcal{B}$ against the $\text{LWE}_{q,m,n,\chi}$ assumption, s.t. $\text{Adv}_{\text{para}_0, \mathcal{A}}^{\text{Mat-hLWE}}(\lambda) \leq \text{Adv}_{\text{para}_1, \mathcal{B}}^{\text{LWE}}(\lambda) + 2\epsilon$, where $\text{para}_0 = ((q, m, n, \chi_0), (\chi_1, \mathcal{S}))$ and $\text{para}_1 = (q, m, n, \chi)$.

The parameter settings in Lemma 2 slightly differ from those employed in [56]. This is because we focus on the standard LWE, where the secret vector $\mathbf{s}$ is sampled uniformly from $\mathbb{Z}_q^n$, and only a hint about the noise $\mathbf{e} \in \mathbb{Z}^m$ is required. For completeness, we refer to Appendix D.1 for a proof of Lemma 2.⁶

In [1, 41], an algorithm named **TrapGen** was proposed to sample a “nearly” uniform matrix $\mathbf{A}$ along with a low-norm trapdoor matrix $\mathbf{T}_\mathbf{A}$ s.t. $\mathbf{A} \cdot \mathbf{T}_\mathbf{A} = \mathbf{0}$ (cf. Lemma 3). Meanwhile, another algorithm called **Invert** was proposed to use $\mathbf{T}_\mathbf{A}$ to invert an LWE sample $(\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top)$ to obtain $\mathbf{s}$ and $\mathbf{e}$ (cf. Lemma 4).

Lemma 3 ([1, 41]). *There exists a PPT algorithm **TrapGen** that takes as input positive integers n, q ($q \geq 2$) and a sufficiently large $m = O(n \log q)$, outputs a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and an invertible trapdoor $\mathbf{T}_\mathbf{A} \in \mathbb{Z}_q^{m \times m}$ such that $\mathbf{A}$ is statistically close to the uniform distribution, $\mathbf{A} \cdot \mathbf{T}_\mathbf{A} = \mathbf{0}$, and $\|\tilde{\mathbf{T}}_\mathbf{A}\| \leq O(\sqrt{n \log q})$, where $\tilde{\mathbf{T}}_\mathbf{A}$ denotes the Gram-Schmidt orthogonalization of $\mathbf{T}_\mathbf{A}$.*

Lemma 4 ([41, Theorem 5.4]). *There exists a deterministic polynomial-time algorithm **Invert** that takes as inputs the trapdoor information $\mathbf{T}_\mathbf{A}$ and a vector $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ with $\mathbf{s} \in \mathbb{Z}_q^n$ and $\|\mathbf{e}\| \leq q/(10 \cdot \sqrt{m})$, and outputs $\mathbf{s}$ and $\mathbf{e}$.*

Lemma 5 (Tail Bound [39]). *For any $k > 1$, it holds that $\Pr[\|\mathbf{x}\| > k\sqrt{m}\sigma \mid \mathbf{x} \leftarrow D_\sigma^m] < k^m \exp(\frac{m}{2}(1 - k^2))$. In particular, for $k \geq \omega(\sqrt{\log \lambda})$, both $|x| \geq k \cdot \sigma$ and $\|\mathbf{x}\| \geq k \cdot \sigma\sqrt{m}$ happen with a negligible probability.*

2.2 Secret Sharing

Replicated Secret Sharing. The replicated secret sharing (RSS) [28] is an additive t -out-of- N threshold secret sharing scheme with *short* secret share, as recalled in $\text{RSS} = (\text{Share}, \text{Select}, \text{Recover})$ in Fig. 1.

More precisely, let $\mathcal{I} := \{I \mid I \subseteq [N], |I| = N - t + 1\}$ be the collection of sets $I \subseteq [N]$ of size $N - t + 1$ and χ be a distribution over $\mathbb{Z}^m$ (e.g., $\chi = D_{\mathbb{Z}^m, \sigma}$). In this construction, the secret $\mathbf{k}$ is the summation of $|\mathcal{I}| = \binom{N}{N-t+1}$ individual keys $\{\mathbf{k}_I \leftarrow \chi\}_{I \in \mathcal{I}}$, i.e., $\mathbf{k} = \sum_{I \in \mathcal{I}} \mathbf{k}_I$, and each share $\mathbf{sk}_i$ ($i \in [N]$) consists of all individual keys $\mathbf{k}_I$ satisfying $i \in I$ (cf. **Share** in Fig. 1). Given a set $S \subseteq [N]$ of shares $\{\mathbf{sk}_i\}_{i \in S}$ with $|S| \geq t$, for all $I \in \mathcal{I}$, it must hold that $I \cap S \neq \emptyset$ (since $|I| = N - t + 1$ and $|S| \geq t$), so $\mathbf{sk}_i$ for any $i \in I \cap S \subseteq S$ contains $\mathbf{k}_I$, and by combining all $\{\mathbf{k}_I\}_{I \in \mathcal{I}}$, one can recover the secret $\mathbf{k}$ (cf. **Select** and **Recover** in Fig. 1). Consequently, the RSS scheme satisfies correctness, i.e.,

- **Correctness.** For any $S \subseteq [N]$ with $|S| \geq t$, $(\mathbf{k}, (\mathbf{sk}_1, \dots, \mathbf{sk}_N)) \leftarrow \text{Share}(N, t, \chi)$, it holds that $\text{Recover}(S, \{\mathbf{sk}_i\}_{i \in S}) = \mathbf{k}$.

⁶ We note that similar lemmas have been proven in [22, 34], and we include a proof of Lemma 2 in Appendix D.1 mainly for completeness.

On the other hand, given $\{\text{sk}_i\}_{i \in S}$ for any set S with $|S| < t$, there must be (at least) one $I \in \mathcal{I}$ so that $I \cap S = \emptyset$, so all sk_i for $i \in S$ do not contain this specific individual key $\mathbf{k}_I$, and it fails to recover the whole secret $\mathbf{k}$.

$\text{Share}(N, t, \chi) \rightarrow (\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N)):$ χ is a distribution over $\mathbb{Z}^m$ $(\text{sk}_1, \dots, \text{sk}_N) := (\emptyset, \dots, \emptyset)$ $\mathcal{I} := \{I \mid I \subseteq [N], I = N - t + 1\}$ $\mathbf{k} := \mathbf{0}$ For $I \in \mathcal{I}$: $\mathbf{k}_I \leftarrow \chi$ $\mathbf{k} := \mathbf{k} + \mathbf{k}_I$ For each $i \in I$: $\text{sk}_i \leftarrow \text{sk}_i \cup \{\mathbf{k}_I\}$ Return $(\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N))$	$\text{Select}(S) \rightarrow \mathbf{w}:$ $\ S \subseteq [N]$ $\mathcal{I} := \{I \mid I \subseteq [N], I = N - t + 1\}$ $\mathbf{w} := (w_i)_{i \in S} = (\emptyset, \dots, \emptyset)$ For $I \in \mathcal{I}$: //choose a party for share $\mathbf{k}_I$ $i' := \max(I \cap S)$ $w_{i'} := w_{i'} \cup \{\mathbf{k}_I\}$ Return $\mathbf{w}$ $\text{Recover}(S, \{\text{sk}_i\}_{i \in S}) \rightarrow \mathbf{k}$ If $ S < t$: Return $\perp$ $(w_i)_{i \in S} \leftarrow \text{Select}(S)$ Return $\mathbf{k} := \sum_{i \in S} \sum_{I \in w_i} \mathbf{k}_I$
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Fig. 1. The replicated secret sharing scheme $\text{RSS} = (\text{Share}, \text{Select}, \text{Recover})$ in [28].

Shamir Secret Sharing. The Shamir secret sharing [52] is a linear t -out-of- N threshold secret sharing scheme. Let $q > N$ be a positive integer such that $i - j$ is invertible in $\mathbb{Z}_q$ for all distinct $i, j \in [N]$. Let $S \subseteq [N]$ be a subset with $|S| \geq t$. For any $i \in S$, the Lagrange coefficient $\lambda_{S,i}$ is defined by $\lambda_{S,i} := \prod_{j \in S \setminus \{i\}} \frac{-j}{i-j}$.

Let $k \in \mathbb{Z}_q$ be a secret, and let $f \in \mathbb{Z}_q[X]$ be a polynomial of degree $t - 1$ satisfying $f(0) = k$. Given any $S \subseteq [N]$ with $|S| \geq t$, the secret can be reconstructed via $k = \sum_{i \in S} \lambda_{S,i} \cdot f(i)$ from the corresponding shares $\{f(i)\}_{i \in S}$.

The Shamir secret sharing of $k \in \mathbb{Z}_q$ can be easily extended to $\mathbf{k} \in \mathbb{Z}_q^m$ by applying the technique component-wisely to $\mathbf{k}$. For simplicity, we denote the Shamir secret sharing algorithm by $\{\mathbf{k}_i\}_{i \in [N]} \leftarrow \text{SShare}(N, t, \mathbf{k})$.

3 Definition of Threshold Decryption

In this section, we present the definition of non-interactive public key threshold decryption (TD) scheme in the asynchronous setting and its security model. Due to space limitations, we postpone the definition in the synchronous setting and its security model to Appendix C.

Definition 3 (Threshold Decryption). Let $t, N \in \mathbb{N}$ with $t < N$. A (t, N) -threshold decryption scheme TD for message space $\mathcal{M}$ is defined by a tuple of PPT algorithms $\text{TD} = (\text{Setup}, \text{KGen}, \text{Enc}, \text{PDec}, \text{PVer}, \text{Comb})$.

- $\text{pp} \leftarrow \text{Setup}$: The setup algorithm outputs a public parameter pp , which serves as an implicit input of other algorithms.
- $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$: The key generation algorithm outputs a public key pk and a set of public-secret key share pairs $\{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}$, where each party $i \in [N]$ receives $(\text{pk}_i, \text{sk}_i)$.

- $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$: The encryption algorithm takes as input a public key pk and a message $\mu \in \mathcal{M}$, and outputs a ciphertext ct .
- $(h_i, \pi_i) \leftarrow \text{PDec}(\text{sk}_i, \text{ct})$: The partial decryption algorithm takes as input a secret key share sk_i and a ciphertext ct , and outputs a partial decryption h_i together with a proof π_i for h_i .
- $0/1 \leftarrow \text{PVer}(\text{pk}_i, \text{ct}, h_i, \pi_i)$: The partial verification algorithm takes as input a public key share pk_i , a ciphertext ct , a partial decryption h_i and a proof π_i , and outputs 1 or 0 indicating the validity of h_i .
- $\mu/\perp \leftarrow \text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{(h_i, \pi_i)\}_{i \in S})$: The combination algorithm takes as input a set of public key shares $\{\text{pk}_i\}_{i \in [N]}$, a ciphertext ct , a subset $S \subseteq [N]$ with $|S| \geq t$, and a set of partial decryption and proof pairs $\{(h_i, \pi_i)\}_{i \in S}$, and outputs a message μ or $\perp$ indicating combination failure.

Correctness. For $\forall \mu \in \mathcal{M}$, $\forall S \subseteq [N]$ with $|S| \geq t$, let $\text{pp} \leftarrow \text{Setup}$, $(\text{pk}, \{\text{pk}_i, \text{sk}_i\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$, $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$, $\{(h_i, \pi_i) \leftarrow \text{PDec}(\text{sk}_i, \text{ct})\}_{i \in S}$, then $\text{PVer}(\text{pk}_i, \text{ct}, h_i, \pi_i) = 1$ for all $i \in S$ and $\text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{(h_i, \pi_i)\}_{i \in S}) = \mu$ hold with probability $1 - \text{negl}(\lambda)$.

Next we define the IND-TCCA/IND-TCPA security (indistinguishability against chosen-ciphertext/ chosen-plaintext attacks under adaptive corruptions) and the RB security (robustness) for TD schemes.

Definition 4 (IND-TCCA/IND-TCPA security). A (t, N) -TD scheme TD is IND-TCCA secure, if for any PPT $\mathcal{A}$, it holds $\text{Adv}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda) := |\Pr[\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda) \Rightarrow 1] - \frac{1}{2}| \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}$ is defined in Fig. 2.

If $\mathcal{A}$ is not allowed to access $\mathcal{O}_{\text{DEC}}$ (and correspondingly $S^* = \emptyset$), then TD is IND-TCPA secure, and the advantage is denoted as $\text{Adv}_{\text{TD}, \mathcal{A}}^{\text{CPA}}(\lambda)$.

Definition 5 (Robustness (RB)). A (t, N) -TD scheme TD is RB secure, if for any PPT $\mathcal{A}$, it holds $\text{Adv}_{\text{TD}, \mathcal{A}}^{\text{RB}}(\lambda) := \Pr[\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{RB}}(\lambda) \Rightarrow 1] \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{RB}}$ is defined in Fig. 2.

For CCA security, we adopt the strongest security model under adaptive corruptions in [10, Fig. 1]. During the game, $\mathcal{A}$ is asked to distinguish the encryption of μ_0 and μ_1 , and can adaptively query $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ oracles. Through $\mathcal{O}_{\text{COR}}(i)$, $\mathcal{A}$ can adaptively obtain user i 's the secret key sk_i . Through $\mathcal{O}_{\text{ENC}}(\mu)$, $\mathcal{A}$ can obtain the partial decryption (h_i, π_i) for all users $i \in [N]$ w.r.t. an honestly generated encryption of μ . Through $\mathcal{O}_{\text{DEC}}(i, \text{ct})$, $\mathcal{A}$ can obtain the partial decryption (h_i, π_i) w.r.t. a ciphertext ct chosen by itself. This differs from the $\mathcal{O}_{\text{DEC}}$ in [16] where $\mathcal{A}$ is given only h_i . Throughout the game, we require that the total number of secret keys corrupted by $\mathcal{A}$ and the partial decryptions w.r.t. the challenge ciphertext ct^* queried by $\mathcal{A}$ are less than t to avoid trivial attacks.

For robustness, we adopt the notion of consistency from [20] instead of their definition of robustness, since their robustness is tailored for deterministic decryption schemes and does not apply to lattice-based constructions. During the game, $\mathcal{A}$ is given all public keys and secret key shares, and is challenged to generate two partial decryption sets S, S' for the same ciphertext ct that yield

$\text{Exp}_{\text{TD},\mathcal{A}}^{\text{CCA}}(\lambda):$ $\text{Cor} := \emptyset, \text{Hon} := [N], S^* := \emptyset$ $\text{pp} \leftarrow \text{Setup}, (\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$ $\mathcal{O} := \{\mathcal{O}_{\text{Cor}}, \mathcal{O}_{\text{Enc}}, \mathcal{O}_{\text{Dec}}\}$ $(\mu_0, \mu_1, st) \leftarrow \mathcal{A}^{\mathcal{O}}(\text{pp}, \text{pk}, \{\text{pk}_i\}_{i \in [N]})$ $\beta \leftarrow \{0, 1\}, \text{ct}^* \leftarrow \text{Enc}(\text{pk}, \mu_\beta)$ $\beta' \leftarrow \mathcal{A}^{\mathcal{O}}(\text{ct}^*, st)$ If $\beta' = \beta$: Return 1; Else: Return 0	$\mathcal{O}_{\text{Cor}}(i):$ If $ S^* \cup \text{Cor} \geq t - 1$: Return $\perp$ //avoid trivial attacks $\text{Cor} := \text{Cor} \cup \{i\}, \text{Hon} := \text{Hon} \setminus \{i\}$ Return sk_i $\mathcal{O}_{\text{Enc}}(\mu):$ $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$ For $i \in [N]$: $(h_i, \pi_i) \leftarrow \text{PDec}(\text{sk}_i, \text{ct})$ Return $(\text{ct}, \{h_i, \pi_i\}_{i \in [N]})$ $\mathcal{O}_{\text{Dec}}(i, \text{ct}):$ If $\text{ct} = \text{ct}^* \wedge S^* \cup \text{Cor} \geq t - 1$: Return $\perp$ //avoid trivial attacks $(h_i, \pi_i) \leftarrow \text{PDec}(\text{sk}_i, \text{ct})$ If $\text{ct} = \text{ct}^*$: $S^* := S^* \cup \{i\}$ Return (h_i, π_i)
$\text{Exp}_{\text{TD},\mathcal{A}}^{\text{RB}}(\lambda):$ $\text{pp} \leftarrow \text{Setup}, (\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$ $\text{inp} := (\text{pp}, \text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]})$ $(\text{ct}, S, \{h_i, \pi_i\}_{i \in S}, S', \{h_i, \pi_i\}_{i \in S'}) \leftarrow \mathcal{A}(\text{inp})$ $\mu \leftarrow \text{Comb}(\{(\text{pk}_i)_{i \in [N]}, \text{ct}, S, \{h_i, \pi_i\}_{i \in S}\})$ $\mu' \leftarrow \text{Comb}(\{(\text{pk}_i)_{i \in [N]}, \text{ct}, S', \{h_i, \pi_i\}_{i \in S'}\})$ If $\mu' = \perp$ or $\mu = \perp$: Return 0 If $\mu' \neq \mu$: Return 1; Else: Return 0	

Fig. 2. The IND-TCCA security experiment $\text{Exp}_{\text{TD},\mathcal{A}}^{\text{CCA}}$ (upper left) and the robustness experiment $\text{Exp}_{\text{TD},\mathcal{A}}^{\text{RB}}$ (bottom left) for (t, N) -TD scheme TD, where the oracles $\mathcal{O} = \{\mathcal{O}_{\text{Cor}}, \mathcal{O}_{\text{Enc}}, \mathcal{O}_{\text{Dec}}\}$ are shown in the right part of the figure.

inconsistent messages when combined. We note that our robustness is stronger than that in [16], since in our security model, $\mathcal{A}$ can obtain all the secret keys $\{\text{sk}_i\}_{i \in [N]}$ and can designate both decryption sets S, S' .

4 CCA-Secure Asynchronous TD Scheme for Small N

In this section, we propose a lattice-based asynchronous threshold decryption scheme TD_1 for small N with polynomial modulus, and prove its CCA-security and robustness based on the LWE assumption in the standard model.

4.1 Parameters and Building Blocks

Let t, N be integers, where $t \leq N$. The construction of TD_1 uses the following parameters, and we give a suggestion for asymptotic parameter setting in Table 2,⁷ so that all requirements in this section are met.

- Let $n, m, q, p, M \in \mathbb{N}$, where $q = M \cdot p$ for primes M, p .
- Let $\sigma_k, \sigma_0, \sigma_1, \sigma_y \in \mathbb{R}$ and $B_k, B_0, B_1, B_y \in \mathbb{R}$, s.t. $B_k \geq \alpha \sigma_k, B_0 \geq \alpha \sigma_0, B_1 \geq \alpha \sigma_1, B_y \geq \alpha \sigma_y$ for $\alpha = \omega(\sqrt{\log \lambda})$. According to Lemma 5, B_k, B_0, B_1, B_y are tail bounds of the distributions $D_{\mathbb{Z}, \sigma_k}, D_{\mathbb{Z}, \sigma_0}, D_{\mathbb{Z}, \sigma_1}, D_{\mathbb{Z}, \sigma_y}$, respectively.

⁷ We set $n = \lambda \log \lambda$ to guarantee the asymptotic λ -bit security of our TD_1 scheme, according to the state-of-the-art hardness results summarized in [48, 57]. By [48] and our Lemma 2, the Multi-Hint LWE with parameters in Table 2 is as hard as quantumly solving GapSVP_γ with approximation factor $\gamma = \tilde{O}(\sqrt{\ell} \cdot N^t \lambda^6)$. Then accordingly to [57] (page 8), solving GapSVP_γ with the above γ requires time $2^{\tilde{O}(n/\log \gamma)} = 2^{\tilde{O}(\lambda \log \lambda / \log \lambda)} = 2^{\tilde{O}(\lambda)}$, thus ensuring the asymptotic λ -bit security. In Table 2, p is decided by $\max(4N^t \lambda^{6.5} \log^8 \lambda, 4N^t \sqrt{\ell} \lambda^{4.5} \log \lambda^6)$ and we take the latter one, assuming that $\sqrt{\ell} \geq \lambda^2 \log^2 \lambda$.

Table 2. Asymptotic parameter setting for TD_1 , where the $O(\cdot)$ notation is omitted and ℓ denotes a bound for query number.

Para.	n	m	q	p	M
Setting	$\lambda \log \lambda$	$\log \lambda$	$\sqrt{\ell} N^t \cdot \lambda^{5.5} \log^6 \lambda$	$\sqrt{\ell} N^t \cdot \lambda^{4.5} \log^6 \lambda$	λ

Para.	σ_k	σ_0	σ_1	σ_y	B_k	B_0
Setting	$\lambda^{1.5}$	λ	$\lambda^{3.5} \log^{1.5} \lambda$	$\sqrt{\ell} \cdot \lambda^{4.5} \log^4 \lambda$	$\lambda^{1.5} \log \lambda$	$\lambda \log \lambda$

Para.	B_1	B_y	$\tilde{B}_k$	$\tilde{B}_0$	$\tilde{B}_y$
Setting	$N^{t/2} \cdot \lambda^{3.5} \log^{2.5} \lambda$	$\sqrt{\ell} \cdot \lambda^{4.5} \log^5 \lambda$	$\lambda^{2.5} \log^{3.5} \lambda$	$\lambda^2 \log^{3.5} \lambda$	$\sqrt{\ell} \cdot \lambda^{4.5} \log^6 \lambda$

Our scheme TD_1 also employs the following building blocks.

- The RSS scheme $\text{RSS} = (\text{Share}, \text{Select}, \text{Recover})$ (cf. Fig. 1 in Subsect. 2.2).
- An OTSS tag-based QA-NIZK scheme $\Pi = (\text{CRSGen}, \text{Prove}, \text{Vrfy}, \text{SimGen}, \text{Sim})$ (cf. Def. 8 in Appendix A.2) for tag space $\mathcal{T}$ and gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ defined as follows, with language parameter $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$:

$$\begin{aligned} \mathcal{L}_{\mathbf{A}}^{(\text{LWE})} &:= \{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-B_0, B_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \}, \\ \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})} &:= \{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \}. \end{aligned}$$

- An adaptively sound NIZK scheme $\overline{\Pi} = (\overline{\text{CRSGen}}, \overline{\text{Prove}}, \overline{\text{Vrfy}}, \overline{\text{SimGen}}, \overline{\text{Sim}})$ (cf. Def. 7 in Appendix A.2) for gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{lin})} = (\mathcal{L}_{\mathbf{A}}^{(\text{lin})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})})$ ⁸ defined as follows, with language parameter $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and $B^* = m\tilde{B}_0\tilde{B}_k + \tilde{B}_y$:

$$\begin{aligned} \mathcal{L}_{\mathbf{A}}^{(\text{lin})} &:= \left\{ (\mathbf{pk}, h, \mathbf{c}) \mid \begin{array}{l} \exists (\mathbf{k} \in [-B_k, B_k]^m, \\ y \in [-B_y, B_y]) \end{array} \text{ s.t. } \begin{array}{l} \mathbf{pk} = \mathbf{A}\mathbf{k} \wedge \\ h = \mathbf{c}^\top \mathbf{k} + y \end{array} \right\}, \\ \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})} &:= \left\{ (\mathbf{pk}, h, \mathbf{c}) \mid \begin{array}{l} \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m, \\ y \in [-B^*, B^*]) \end{array} \text{ s.t. } \begin{array}{l} \mathbf{c} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \wedge \\ h = \mathbf{s}^\top \mathbf{pk} + y \end{array} \right\}. \end{aligned}$$

- A family of collision-resistant hash functions $\mathcal{H} = \{H: \{0, 1\}^* \rightarrow \mathcal{T}\}$.

We provide instantiations of OTSS tag-based QA-NIZK Π for $\mathcal{GL}_{\rho}^{(\text{LWE})}$ in Appendix E.2 and adaptively sound NIZK $\overline{\Pi}$ for $\mathcal{GL}_{\rho}^{(\text{lin})}$ in Appendix E.3, respectively. Roughly speaking, we obtain the above two NIZK by first constructing trapdoor Σ -protocol for $\mathcal{GL}_{\rho}^{(\text{LWE})}$ and $\mathcal{GL}_{\rho}^{(\text{lin})}$, then compile the corresponding trapdoor Σ -protocol to QA-NIZK by the generic construction in [36], with the help of lossy PKE, pseudorandom function and correlation intractable hash functions. We emphasize that the generic transformation will not introduce exponential modulus, as long as the parameters in this section are polynomial.

4.2 Construction of TD_1

Our (t, N) -threshold decryption scheme $\text{TD}_1 = (\text{Setup}, \text{KGen}, \text{Enc}, \text{PDec}, \text{PVer}, \text{Comb})$ for small N is shown in Fig. 3, which works in the asynchronous setting.

⁸ Note that $\mathcal{L}_{\mathbf{A}}^{(\text{lin})}$ might not always be contained in $\tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$. We refer to Appendix E.3 for more discussions of the language setting.

<u>pp</u> $\leftarrow$ Setup: $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$ $\text{crs} \leftarrow \text{CRSGen}(\mathbf{A})$, $\overline{\text{crs}} \leftarrow \overline{\text{CRSGen}}(\mathbf{A})$ $\mathbf{H} \leftarrow \mathcal{H}$ Return $\text{pp} := (\mathbf{A}, \text{crs}, \overline{\text{crs}}, \mathbf{H})$ <u>(pk, {(pk_i, sk_i)}_{i ∈ [N]})</u> $\leftarrow$ KGen(pp): $(\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N)) \leftarrow \text{Share}(N, t, D_{\mathbb{Z}^m, \sigma_k})$ $\text{pk} := \mathbf{A}\mathbf{k} \in \mathbb{Z}_q^n$, $\text{sk} := \mathbf{k} \in \mathbb{Z}_q^m$ $\mathcal{I} := \{I \mid I \subseteq [N], I = N - t + 1\}$ For $i \in [N]$: Parse $\text{sk}_i = \{\mathbf{k}_I \mid I \in \mathcal{I}, i \in I\}$ $\text{pk}_i := \{\text{pk}_{i,I} := \mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_i}$ Return $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]})$ <u>ct</u> $\leftarrow$ Enc(pk, $\mu \in \mathbb{Z}_M$): $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$, $\mathbf{e}' \leftarrow D_{\mathbb{Z}, \sigma_1}$ $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ $d := \mathbf{s}^\top \text{pk} + \mathbf{e}' + p \cdot \mu \in \mathbb{Z}_q$ $\tau := \mathbf{H}(\mathbf{c}, d)$ $\pi \leftarrow \text{Prove}(\text{crs}, \tau, \mathbf{c} \in \mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, (\mathbf{s}, \mathbf{e}))$ Return $\text{ct} := (\mathbf{c}, d, \pi)$	<u>(h_i, π_i)</u> $\leftarrow$ PDec(sk_i, ct = (c, d, π)): If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ for $\tau := \mathbf{H}(\mathbf{c}, d)$: Return $\perp$ For each $\mathbf{k}_I \in \text{sk}_i$: $y_{i,I} \leftarrow D_{\mathbb{Z}, \sigma_y}$ $h_{i,I} := \mathbf{c}^\top \mathbf{k}_I + y_{i,I}$ $\pi_{i,I} \leftarrow \overline{\text{Prove}}(\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}) \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}, (\mathbf{k}_I, y_{i,I}))$ Return $(h_i := \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}, \pi_i := \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i})$ <u>0/1</u> $\leftarrow$ PVer(pk_i, ct = (c, d, π), h_i, π_i): Parse $h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ and $\pi_i = \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ For each $\text{pk}_{i,I} \in \text{pk}_i$: If $\overline{\text{Vrfy}}(\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}), \pi_{i,I}) = 0$: Return 0 Return 1 <u>μ</u> $\leftarrow$ Comb({pk_i}_{i ∈ [N]}, ct = (c, d, π), S, {(h_i, π_i)}_{i ∈ S}): If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ for $\tau := \mathbf{H}(\mathbf{c}, d)$: Return $\perp$ $T := \emptyset$ For each $i \in S$: If $\text{PVer}(\text{pk}_i, \text{ct}, h_i, \pi_i) = 1$: $T := T \cup \{i\}$ If $T < t$: Return $\perp$ $\mathbf{w} = (w_i)_{i \in T} \leftarrow \text{Select}(T)$ For $i \in T$: Parse $h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ $h := \sum_{i \in T} \sum_{I \in w_i} h_{i,I}$ Return $\mu := \lceil (d - h)/p \rceil$
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Fig. 3. Our (t, N) -TD scheme TD₁ for small N in the asynchronous setting.

Correctness. The correctness of our scheme TD₁ follows from the completeness of Π and $\overline{\Pi}$, the correctness of RSS (cf. Subsect. 2.2), and our parameter selection. Firstly, it holds that Vrfy and $\overline{\text{Vrfy}}$ algorithms in PDec, PVer and Comb will output 1 except with negligible probability, by the completeness of Π and $\overline{\Pi}$.

Next, we show that the Comb algorithm can recover the message μ correctly except with negligible probability. For any message $\mu \in \mathbb{Z}_M$, the encryption algorithm Enc(pk, μ) generates ciphertext $\text{ct} = (\mathbf{c}, d, \pi)$ by computing $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, $d := \mathbf{s}^\top \mathbf{A}\mathbf{k} + \mathbf{e}' + p \cdot \mu$ with $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$, $\mathbf{e}' \leftarrow D_{\mathbb{Z}, \sigma_1}$. Then for $(h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}, \pi_i)$ generated by the partial decryption algorithm PDec(sk_i, ct), each $h_{i,I}$ is computed by $h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}$ with $y_{i,I} \leftarrow D_{\mathbb{Z}, \sigma_y}$. Since $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, for any $S \subseteq [N]$ with $|S| \geq t$, in the combination algorithm Comb({pk_i}_{i ∈ [N]}, ct, S, {(h_i, π_i)}_{i ∈ S}), we have $T = S$ and

$$\begin{aligned}
h &:= \sum_{i \in S} \sum_{I \in w_i} h_{i,I} = \sum_{i \in S} \sum_{I \in w_i} (\mathbf{s}^\top \mathbf{A}\mathbf{k}_I + \mathbf{e}^\top \mathbf{k}_I + y_{i,I}) \\
&= \mathbf{s}^\top \mathbf{A} (\sum_{i \in S} \sum_{I \in w_i} \mathbf{k}_I) + \sum_{i \in S} \sum_{I \in w_i} (\mathbf{e}^\top \mathbf{k}_I + y_{i,I}) = \mathbf{s}^\top \mathbf{A}\mathbf{k} + \sum_{i \in S} \sum_{I \in w_i} (\mathbf{e}^\top \mathbf{k}_I + y_{i,I}),
\end{aligned}$$

where the last equality is due to the correctness of RSS. Therefore, we have $d - h = p \cdot \mu + \mathbf{e}' - \sum_{i \in S} \sum_{I \in w_i} (\mathbf{e}^\top \mathbf{k}_I + y_{i,I})$. By our parameter setting in Table 2 and the tail bound (Lemma 5), the noise term $\mathbf{e}' - \sum_{i \in S} \sum_{I \in w_i} (\mathbf{e}^\top \mathbf{k}_I + y_{i,I})$ is bounded by $B_1 + C(mB_0B_k + B_y) \leq p/2$ with $C := \binom{N}{N-t+1}$ except with negligible probability. Consequently, $\lceil (d - h)/p \rceil$ correctly recovers μ .

4.3 Security Proof of TD_1

Theorem 1 (IND-TCCA Security of TD_1). Assume that (i) Π has zero-knowledge and OTSS, (ii) $\mathcal{H}$ is collision-resistant, (iii) the matrix hint-LWE assumption holds for the parameters chosen above, then the proposed scheme TD_1 in Fig. 3 is IND-TCCA secure.

More precisely, for any PPT adversary $\mathcal{A}$, there exists PPT $\mathcal{B}_i$ for $i \in [4]$ s.t.

$$\begin{aligned} \text{Adv}_{\text{TD}_1, \mathcal{A}}^{\text{CCA}}(\lambda) &\leq \text{Adv}_{\Pi, \mathcal{B}_1}^{\text{zk}}(\lambda) + \text{Adv}_{q, m, n, D_{\mathbb{Z}^m}, \sigma_0/2, \mathcal{B}_2}^{\text{LWE}}(\lambda) \\ &\quad + \text{Adv}_{\mathcal{H}, \mathcal{B}_3}^{\text{cr}}(\lambda) + \text{Adv}_{\Pi, \mathcal{B}_4}^{\text{OTSS}}(\lambda) + \text{negl}(\lambda). \end{aligned}$$

Proof of Theorem 1. Let $\mathcal{A}$ be an adversary against the IND-TCCA security. We use a series of games G_0 - G_7 to show the advantage $\text{Adv}_{\text{TD}_1, \mathcal{A}}^{\text{CCA}}(\lambda)$ is negligible, where G_0 is the original IND-TCCA experiment and $\mathcal{A}$ has no advantage in G_8 .

Game G_0 : This is the original IND-TCCA experiment $\text{Exp}_{\text{TD}_1, \mathcal{A}}^{\text{CCA}}$ (cf. Fig. 2). By definition, we have $\text{Adv}_{\text{TD}_1, \mathcal{A}}^{\text{CCA}}(\lambda) = |\Pr[\text{G}_0 = 1] - \frac{1}{2}|$.

Let $\text{pp} = (\mathbf{A}, \text{crs}, \overline{\text{crs}}, H)$. The challenger invokes $(\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N)) \leftarrow_{\$} \text{RSS}(N, t, D_{\mathbb{Z}^m, \sigma_k})$, sets $\text{pk} := \mathbf{A}\mathbf{k}$, $\text{pk}_i := \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_i}$ for each $i \in [N]$, and returns $(\text{pp}, \text{pk}, \{\text{pk}_i\}_{i \in [N]})$ to $\mathcal{A}$. Then it answers $\mathcal{A}$'s $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$, $\mathcal{O}_{\text{DEC}}$ queries adaptively and generates a challenge ciphertext ct^* for $\mathcal{A}$ following Fig. 3. Due to space limitations, we refer to Appendix D.3 for a full description of G_0 .

Game G_1 : It is the same as G_0 except that when generating the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$, the challenger invokes $\pi^* \leftarrow_{\$} \text{Sim}(\text{crs}, \tau^*, \mathbf{c}^*)$ instead of $\pi^* \leftarrow_{\$} \text{Prove}(\text{crs}, \tau^*, \mathbf{c}^*, (\mathbf{s}, \mathbf{e}))$.

By the zero-knowledge of Π , $|\Pr[\text{G}_0 = 1] - \Pr[\text{G}_1 = 1]| \leq \text{Adv}_{\Pi, \mathcal{B}_1}^{\text{zk}}(\lambda)$.

Game G_2 : It is the same as G_1 except that when generating the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$, the challenger computes $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta$ instead of $d^* := \mathbf{s}^\top \text{pk} + e' + p \cdot \mu_\beta$.

Since $\text{pk} = \mathbf{A}\mathbf{k}$ and $\mathbf{c}^{*\top} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, the d^* in G_1 equals $\mathbf{s}^\top \mathbf{A}\mathbf{k} + e' + p \cdot \mu_\beta = (\mathbf{c}^{*\top} - \mathbf{e}^\top) \mathbf{k} + e' + p \cdot \mu_\beta$, which is just the d^* in G_2 . So the change is conceptual and we have $\Pr[\text{G}_1 = 1] = \Pr[\text{G}_2 = 1]$.

Game G_3 : It is the same as G_2 except that when generating the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$, the challenger samples $\mathbf{c}^{*\top} \leftarrow_{\$} \mathbb{Z}_q^{1 \times m}$ uniformly, instead of computing $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.

By the Matrix-hint LWE assumption (cf. Def. 2), the $\mathbf{c}^{*\top} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ in G_2 is computationally indistinguishable from the $\mathbf{c}^{*\top} \leftarrow_{\$} \mathbb{Z}_q^{1 \times m}$ in G_3 , even conditioned on the $e' - \mathbf{e}^\top \cdot \mathbf{k}$ involved in the computation of d^* . Formally, we have the following claim with proof postponed to Appendix D.2.

Claim 1. $|\Pr[\text{G}_2 = 1] - \Pr[\text{G}_3 = 1]| \leq \text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda)$, where $\text{para} := ((q, m, n, D_{\mathbb{Z}^m, \sigma_0}), (D_{\mathbb{Z}, \sigma_1}, [-CB_k, CB_k]^m))$ with $C := \binom{N}{N-t+1}$. By Lemma 2 our parameter setting, there exist $\mathcal{B}_2'$ s.t. $\text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda) \leq \text{Adv}_{q, m, n, D_{\mathbb{Z}^m}, \sigma_0/2, \mathcal{B}_2'}^{\text{LWE}}(\lambda)$.

Game G_4 : It is the same as G_3 , except that when replying to $\mathcal{A}$'s partial decryption queries $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi))$, the challenger introduces an additional rejection rule: If $\text{ct} \neq \text{ct}^* \wedge (\tau, \mathbf{c}, \pi) = (\tau^*, \mathbf{c}^*, \pi^*)$, return $\perp$ directly.

Since $\text{ct} = (\mathbf{c}, d, \pi) \neq (\mathbf{c}^*, d^*, \pi^*) = \text{ct}^* \wedge (\tau, \mathbf{c}, \pi) = (\tau^*, \mathbf{c}^*, \pi^*)$ implies that $d \neq d^*$ but $\tau = H(\mathbf{c}, d) = H(\mathbf{c}^*, d^*) = \tau^*$, it yields a collision of H . So $|\Pr[G_3 = 1] - \Pr[G_4 = 1]| \leq \text{Adv}_{\mathcal{H}, \mathcal{B}_3}^{\text{cr}}(\lambda)$ for PPT $\mathcal{B}_3$ against the collision-resistance of $\mathcal{H}$.

Game G_5 : It is the same as G_4 , except that, at the beginning of the game, when generating the $\mathbf{A}$ in pp , the challenger invokes $(\mathbf{A}, \mathbf{T}_{\mathbf{A}}) \leftarrow_s \text{TrapGen}(n, q, m)$, instead of $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{n \times m}$.

By Lemma 3, the $\mathbf{A}$ generated by $\text{TrapGen}(n, q, m)$ is statistically indistinguishable from a uniform $\mathbf{A}$. Therefore, G_5 and G_4 are indistinguishable to adversary $\mathcal{A}$ and it holds that $|\Pr[G_4 = 1] - \Pr[G_5 = 1]| \leq \text{negl}(\lambda)$.

Game G_6 : It is the same as G_5 , except that when replying to $\mathcal{A}$'s partial decryption queries $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi))$, the challenger introduces an additional rejection rule: If $\text{ct} \neq \text{ct}^* \wedge \mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, return $\perp$ directly.

For any $\mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, it holds that $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ with $\mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m$. Since $\|\mathbf{e}\| \leq \sqrt{m}\tilde{B}_0 \leq q/(10 \cdot \sqrt{m})$ by our parameter setting in Table 2, the challenger can efficiently check whether $\mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$ with the help of the trapdoor $\mathbf{T}_{\mathbf{A}}$, by using the Invert algorithm in Lemma 4.

Claim 2. $|\Pr[G_5 = 1] - \Pr[G_6 = 1]| \leq \text{Adv}_{\Pi, \mathcal{B}_4}^{\text{OTSS}}(\lambda)$.

Proof. It is easy to check that G_5 and G_6 are the same, until the following event Bad happens:

$$\text{ct} \neq \text{ct}^* \wedge \mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})} \wedge (\tau, \mathbf{c}, \pi) \neq (\tau^*, \mathbf{c}^*, \pi^*) \wedge \text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 1.$$

So $|\Pr[G_5 = 1] - \Pr[G_6 = 1]| \leq \Pr[\text{Bad}]$, and it remains to bound $\Pr[\text{Bad}]$. Note that Bad implies that $(\tau, \mathbf{c}, \pi)$ is new tuple (i.e., different from $(\tau^*, \mathbf{c}^*, \pi^*)$) for $\mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, but it passes $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 1$. By the OTSS of Π (cf. Def. 8), we can build a PPT $\mathcal{B}_4$ such that $\Pr[\text{Bad}] \leq \text{Adv}_{\Pi, \mathcal{B}_4}^{\text{OTSS}}(\lambda)$. Consequently, we have $|\Pr[G_5 = 1] - \Pr[G_6 = 1]| \leq \Pr[\text{Bad}] \leq \text{Adv}_{\Pi, \mathcal{B}_4}^{\text{OTSS}}(\lambda)$. $\blacksquare$

Game G_7 : It is the same as G_6 except that when generating the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$, the challenger samples $d^* \leftarrow_s \mathbb{Z}_q$ uniformly, instead of computing $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta$.

Claim 3. $|\Pr[G_6 = 1] - \Pr[G_7 = 1]| \leq 1/2^\lambda$.

Proof. Let $\mathcal{I} := \{I \mid I \subseteq [N], |I| = N - t + 1\}$. Below we first analyze the information about $\text{sk} = \mathbf{k}$ and its shares $\text{sk}_i = \{\mathbf{k}_I \mid I \in \mathcal{I}, i \in I\}$ ($i \in [N]$) that might be leaked to $\mathcal{A}$ in G_6 and G_7 (except for the challenge ciphertext ct^*).

- Firstly, $\mathcal{A}$ obtains $\text{pk} = \mathbf{A}\mathbf{k}$ and $\text{pk}_1 = \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_1}, \dots, \text{pk}_N = \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_N}$. Since $\cup_{i \in [N]} \text{sk}_i = \{\mathbf{k}_I\}_{I \in \mathcal{I}}$, $\mathcal{A}$ in fact gets the values of $\mathbf{A}\mathbf{k}$ and $\{\mathbf{A}\mathbf{k}_I\}_{I \in \mathcal{I}}$.

- Through $\mathcal{O}_{\text{COR}}$ queries, $\mathcal{A}$ obtains $\text{sk}_i = \{\mathbf{k}_I \mid i \in I, I \in \mathcal{I}\}$ for $i \in \text{Cor}$.
- Through $\mathcal{O}_{\text{ENC}}(\mu)$ queries, $\mathcal{A}$ can learn partial decryptions $h_i = \{h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ for all $i \in [N]$ of honestly generated $\text{ct} = (\mathbf{c}, \mathbf{d}, \pi)$. Note that for each $I \in \mathcal{I}$, $\mathbf{k}_I \in \text{sk}_i$ holds if and only if $i \in I$ holds, so $\mathcal{A}$ learns $\{h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}\}_{i \in I}$ for each $I \in \mathcal{I}$. Since the ciphertext ct is honestly generated, we have $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ with $\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and it follows

$$h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I} = \mathbf{s}^\top \mathbf{A} \mathbf{k}_I + \mathbf{e}^\top \mathbf{k}_I + y_{i,I} = \mathbf{s}^\top (\mathbf{A} \mathbf{k}_I) + \mathbf{e}^\top \mathbf{k}_I + y_{i,I}. \quad (11)$$

Consequently, in addition to $\{\mathbf{A} \mathbf{k}_I\}_{I \in \mathcal{I}}$, $\mathcal{A}$ obtains at most $|I| = N - t + 1$ linear relations about $\mathbf{k}_I$ of the form $\{\mathbf{e}^\top \mathbf{k}_I + y_{i,I}\}_{i \in I}$ for each $\mathbf{k}_I$.

- Through $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, \mathbf{d}, \pi))$ queries, $\mathcal{A}$ can learn partial decryption $h_i = \{h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ of arbitrary ciphertext $\text{ct} = (\mathbf{c}, \mathbf{d}, \pi)$. There are two cases: (i) If $\text{ct} = \text{ct}^*$, then i is added to S^* , and the partial decryption h_i might leak the whole information about $\text{sk}_i = \{\mathbf{k}_I \mid i \in I, I \in \mathcal{I}\}$. (ii) If $\text{ct} \neq \text{ct}^*$, by the change in $\mathbf{G}_6$, $\mathcal{A}$ learns $h_i = \{h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ only when $\mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, in which case $\mathbf{c}$ can be written as $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ for some $\mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m$. Then by a similar analysis as in (11), $\mathcal{A}$ obtains one linear relation of the form $\mathbf{e}^\top \mathbf{k}_I + y_{i,I}$ for each $\mathbf{k}_I$ with $i \in I$.⁹

Let Q_e and Q_d denote the number of $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ queries made by $\mathcal{A}$, respectively. Overall, $\mathcal{A}$ gets $\mathbf{A} \mathbf{k}$ and $\{\mathbf{A} \mathbf{k}_I\}_{I \in \mathcal{I}}$, might learn $\{\text{sk}_i\}_{i \in S^* \cup \text{Cor}}$ (through $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ in case (i)), and learn at most $\ell := (N - t + 1) \cdot Q_e + Q_d$ linear relations of the form $\{\mathbf{e}^\top \mathbf{k}_I + y_{i,I}\}_{i \in I}$ for each $\mathbf{k}_I$ (through $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ in case (ii)). Given that $|S^* \cup \text{Cor}| \leq t - 1$ (to avoid trivial attacks), there must exist $\tilde{I} \in \mathcal{I}$ such that $\mathbf{k}_{\tilde{I}} \notin \{\text{sk}_i\}_{i \in S^* \cup \text{Cor}}$. So the information about sk and its shares $\{\text{sk}_i\}_{i \in [N]}$ that $\mathcal{A}$ might learn is upper-bounded by $\text{View}_{\mathcal{A}} = (\mathbf{A} \mathbf{k}, \{\mathbf{A} \mathbf{k}_I\}_{I \in \mathcal{I}}, \{\mathbf{k}_I\}_{I \in \mathcal{I}, I \neq \tilde{I}}, \{\mathbf{e}_j^\top \mathbf{k}_{\tilde{I}} + y_j\}_{j \in [\ell]})$ with $\|\mathbf{e}_j\|_\infty < \tilde{B}_0$ and $y_j \leftarrow D_{\mathbb{Z}, \sigma_y}$.

Next we will show that conditioned on $\mathcal{A}$'s view $\text{View}_{\mathcal{A}}$ in $\mathbf{G}_7/\mathbf{G}_6$, the min-entropy of $\mathbf{k}_{\tilde{I}}$ is at least $2n \log q + 2\lambda$, i.e.,

$$\tilde{\mathbf{H}}_\infty(\mathbf{k}_{\tilde{I}} \mid \text{View}_{\mathcal{A}}) \geq 2n \log q + 2\lambda. \quad (12)$$

To see this, note that $\mathbf{A} \mathbf{k} = \sum_{I \in \mathcal{I}} \mathbf{A} \mathbf{k}_I$ and $\mathbf{k}_{\tilde{I}}$ is independent of $\{\mathbf{k}_I\}_{I \in \mathcal{I}, I \neq \tilde{I}}$, so $\tilde{\mathbf{H}}_\infty(\mathbf{k}_{\tilde{I}} \mid \text{View}_{\mathcal{A}}) = \tilde{\mathbf{H}}_\infty(\mathbf{k}_{\tilde{I}} \mid \mathbf{A} \mathbf{k}_{\tilde{I}}, \{\mathbf{e}_j^\top \mathbf{k}_{\tilde{I}} + y_j\}_{j \in [\ell]}) = \tilde{\mathbf{H}}_\infty(\mathbf{k}_{\tilde{I}} \mid \mathbf{A} \mathbf{k}_{\tilde{I}}, \mathbf{E} \mathbf{k}_{\tilde{I}} + \mathbf{y})$, where we compactly write $\mathbf{E} := (\mathbf{e}_1, \dots, \mathbf{e}_\ell)^\top \in \mathbb{Z}^{\ell \times m}$ and $\mathbf{y} := (y_1, \dots, y_\ell)^\top \in \mathbb{Z}^\ell$. We leverage the following lemma to analyze its lower bound.

Lemma 6 (Conditional Min-Entropy Lower Bound with Polynomial Noise Flooding). *Let $c, m, n, q, p, \ell \in \mathbb{Z}$, let $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, and let $\sigma, \sigma_k, \sigma_y, \epsilon \in \mathbb{R}$ satisfying $\sigma_y \geq \sqrt{c} \cdot \sqrt{m\ell} B \cdot \sigma_k$. Let $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}, \mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sigma_y}$ and $\mathbf{R} \in \mathbb{Z}^{\ell \times m}$ satisfying $\|\mathbf{R}\|_\infty \leq B$. Then conditioned on $Y = (\mathbf{A}, \mathbf{A} \mathbf{k}, \mathbf{h} = \mathbf{R} \mathbf{k} + \mathbf{y})$, we consider the min-entropy of $\mathbf{k} \bmod p$ in two cases:*

⁹ We note that the challenger will generate π_i with sk_i , however, this will not decrease the min-entropy of sk_i since $\overline{\Pi}$ has witness indistinguishability. We refer to Remark 2 in Appendix E.3 for more discussions.

- If $\sigma_k \geq \sqrt{1+1/c} \cdot p \sqrt{\ln(2m(1+1/\epsilon))/\pi}$, it holds that $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid Y) \geq m \log p - n \log q - \log(1+4\epsilon)$.
- If $\sigma_k \geq \sqrt{1+1/c} \cdot \sqrt{\ln(2m(1+1/\epsilon))/\pi}$ and $p/2 \geq \sqrt{m \log \sigma_k} \cdot (2\sqrt{m} + \sqrt{\ell}) \cdot \sigma_k$, it holds that $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid Y) \geq m \log \sigma_k - m/2 \cdot \log(1+1/c) - n \log q - \log(1+2\epsilon)$.

We refer to the technical overview in Sect. 1 for an intuition of the proof of Lemma 6 and postpone the formal proof to Appendix B.2.

By Lemma 6 and our parameter setting in Table 2, for q 's prime factors M and p , we have $\tilde{\mathbf{H}}_\infty(\mathbf{k}_I \bmod p \mid \mathbf{A}\mathbf{k}_I, \mathbf{E}\mathbf{k}_I + \mathbf{y}) \geq \tilde{\mathbf{H}}_\infty(\mathbf{k}_I \bmod M \mid \mathbf{A}\mathbf{k}_I, \mathbf{E}\mathbf{k}_I + \mathbf{y}) \geq m \log M - n \log q - \log(1+4\epsilon) \geq 2n \log q + 2\lambda$. This yields (12).

Finally, based on (12), we will show that the d^* in $\mathbf{G}_6$ is statistically close to the $d^* \leftarrow_s \mathbb{Z}_q$ in $\mathbf{G}_7$. Note that in $\mathbf{G}_6$, we have

$$\begin{aligned} d^* &= \mathbf{c}^{*\top} \cdot \mathbf{k} + e' - \mathbf{e}^\top \cdot \mathbf{k} + p \cdot \mu_\beta = (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k} + e' + p \cdot \mu_\beta \\ &= (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}_I + \sum_{I \in \mathcal{I}, I \neq \bar{I}} (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}_I + e' + p \cdot \mu_\beta. \end{aligned} \quad (13)$$

Recall that $\mathbf{c}^*$ is randomly chosen from $\mathbb{Z}_q^m$ and thus $\mathbf{c}^* - \mathbf{e}$ is also uniformly random over $\mathbb{Z}_q^m$. According to Leftover Hash Lemma (cf. Lemma 1), since (12) holds, it follows that $\Delta((\mathbf{c}^* - \mathbf{e}, (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}_I), (\mathbf{c}^* - \mathbf{e}, u) \mid \text{View}_{\mathcal{A}}) \leq 1/2^\lambda$, where $u \leftarrow_s \mathbb{Z}_q$. So the $(\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}_I$ in (13) is statistically close to uniform from the view of $\mathcal{A}$, and consequently, the d^* in $\mathbf{G}_6$ is statistically close to the $d^* \leftarrow_s \mathbb{Z}_q$ in $\mathbf{G}_7$. This shows that $|\Pr[\mathbf{G}_6 = 1] - \Pr[\mathbf{G}_7 = 1]| \leq 1/2^\lambda$ and Claim 3 follows. $\blacksquare$

In $\mathbf{G}_7$, we have $d^* \leftarrow_s \mathbb{Z}_q$, so the challenge bit β is totally hidden from $\mathcal{A}$ and it holds that $\Pr[\mathbf{G}_7 = 1] = \frac{1}{2}$. Putting all things together, Theorem 1 follows. $\square$

We present the following theorem to show the robustness of our TD_1 .

Theorem 2 (Robustness of TD_1). *Assume that (i) Π has adaptive soundness, (ii) $\overline{\Pi}$ has adaptive soundness, (iii) $2CB^* \leq p/2$, then the proposed scheme TD_1 in Fig. 3 is RB secure.*

Intuitively, NIZKs Π and $\overline{\Pi}$ ensure that only valid ciphertext and partial decryptions are accepted by the challenger, in which case we can argue that the noises added in the ciphertext and partial decryptions are small by the adaptive soundness of Π and $\overline{\Pi}$. So when combining the two sets of partial decryptions, the aggregated values differ at most $2CB^* \leq p/2$, and consequently, they always decrypt to a unique message μ . We refer to Appendix D.4 for a formal proof.

5 CCA-Secure Synchronous TD Scheme for Polynomial N

In this section, we propose a lattice-based synchronous threshold decryption scheme TD_2 with polynomial modulus, which supports any polynomial number of parties N , and prove its CCA-security based on LWE in the RO model.

5.1 Parameters and Building Blocks

Let t, N be integers, where $t \leq N$. The construction of TD_2 uses the following parameters, and we give a suggestion for asymptotic parameter setting in Table 3, so that all requirements in this section are met.

- Let $n, m, q, p, M \in \mathbb{N}$, where $q = M \cdot p$ for primes M, p s.t. $p > M > N$.¹⁰
- Let $\sigma_k, \sigma_0, \sigma_1, \sigma_y, B_k, B_0, B_1, B_y \in \mathbb{R}$, s.t. $B_k \geq \alpha \sigma_k, B_0 \geq \alpha \sigma_0, B_1 \geq \alpha \sigma_1, B_y \geq \alpha \sigma_y$ for $\alpha = \omega(\sqrt{\log \lambda})$. According to Lemma 5, B_k, B_0, B_1, B_y are tail bounds of the distributions $D_{\mathbb{Z}, \sigma_k}, D_{\mathbb{Z}, \sigma_0}, D_{\mathbb{Z}, \sigma_1}, D_{\mathbb{Z}, \sigma_y}$, respectively.
- Let $\tilde{B}_0 \in \mathbb{R}$, s.t. $\tilde{B}_0 \geq \gamma B_0$ with $\gamma = \omega(\log \lambda \sqrt{\log m}) \cdot m$.

Table 3. Asymptotic parameter setting for TD_2 , where the $O(\cdot)$ notation is omitted and ℓ denotes a bound for query number.

Para.	n	m	q	p	M	σ_k	σ_0
Setting	$\lambda \log \lambda$	$\lambda^2 \log \lambda$	$\sqrt{\ell} N \cdot \lambda^{5.5} \log^6 \lambda$	$\sqrt{\ell} N \cdot \lambda^{4.5} \log^6 \lambda$	λ	$\lambda^{1.5}$	λ

Para.	σ_1	σ_y	B_k	B_0	B_1	B_y	$\tilde{B}_0$
Setting	$\lambda^{3.5} \log^{1.5} \lambda$	$\sqrt{\ell} \cdot \lambda^{4.5} \log^4 \lambda$	$\lambda^{1.5} \log \lambda$	$\lambda \log \lambda$	$\lambda^{3.5} \log^{2.5} \lambda$	$\sqrt{\ell} \cdot \lambda^{4.5} \log^5 \lambda$	$\lambda^2 \log^{3.5} \lambda$

Our scheme TD_2 also employs the following building blocks.

- The share algorithm SShare of Shamir secret sharing (cf. Subsect. 2.2).
- An OTSS tag-based QA-NIZK scheme $\Pi = (\text{CRSGen}, \text{Prove}, \text{Vrfy}, \text{SimGen}, \text{Sim})$ (cf. Def. 8 in Appendix A.2) for tag space $\mathcal{T}$ and gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ defined as follows, with language parameter $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$:

$$\begin{aligned} \mathcal{L}_{\mathbf{A}}^{(\text{LWE})} &:= \left\{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-B_0, B_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \right\}, \\ \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})} &:= \left\{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \right\}. \end{aligned}$$

- A family of collision-resistant hash functions $\mathcal{H} = \{\mathbf{H}: \{0, 1\}^* \rightarrow \mathcal{T}\}$.
- A random oracle $\mathbf{H}_{\text{mask}}: \{0, 1\}^* \rightarrow \mathbb{Z}_q$.

We provide instantiations of OTSS tag-based QA-NIZK Π for $\mathcal{GL}_{\rho}^{(\text{LWE})}$ in Appendix E.4. Roughly speaking, we obtain such a NIZK by first constructing Σ -protocol for $\mathcal{GL}_{\rho}^{(\text{LWE})}$, then use the Fiat-Shamir transformation in the RO model to compile it to NIZK. We prove that the resulting NIZK satisfies all the properties we need in this section. We emphasize that this generic transformation will not introduce exponential modulus, as long as the parameters in this section are polynomial.

¹⁰ The condition $M > N$ is to ensure that for any distinct $i, j \in [N]$, $M, p \nmid i - j$ and hence $i - j$ is invertible over $\mathbb{Z}_q$, so that we can use Shamir secret sharing over $\mathbb{Z}_q$.

5.2 Construction of TD_2

Our construction of (t, N) -threshold decryption $\text{TD}_2 = (\text{Setup}, \text{KGen}, \text{Enc}, \text{PDec}, \text{Comb})$ for polynomial N is shown in Fig. 4. Note that our TD_2 scheme works in the *synchronous* setting, where the algorithm PDec takes a specific decryption set $S \subseteq [N]$ as an additional input and the algorithm PVer is removed since we do not consider the robustness for TD_2 . We refer to Appendix C for the formal definitions of TD schemes in the synchronous setting and its IND-TCCA security model (where the encryption oracle $\mathcal{O}_{\text{ENC}}$ and the partial decryption oracle $\mathcal{O}_{\text{DEC}}$ also allow the adversary to designate the set S for which PDec is invoked).

$\text{pp} \leftarrow \text{Setup}$: $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$ $\text{crs} \leftarrow \text{CRSGen}(\mathbf{A})$ $\mathcal{H} \leftarrow \mathcal{H}$ // collision-resistant hash function $\text{H}_{\text{mask}} : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ // modeled as a random oracle Return $\text{pp} := (\mathbf{A}, \text{crs}, \mathcal{H}, \text{H}_{\text{mask}})$ $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$: $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$ $(\mathbf{k}_1, \dots, \mathbf{k}_N) \leftarrow \text{SShare}(N, t, \mathbf{k})$ $\text{pk} := \mathbf{A}\mathbf{k} \in \mathbb{Z}_q^n, \text{sk} := \mathbf{k} \in \mathbb{Z}_q^m$ For $i \in [N]$ and $j \in [N]$: $\text{rand}_{i,j} \leftarrow \{0, 1\}^\lambda, \text{seed}_{i,j} := i \ j \ \text{rand}_{i,j}$ For $i \in [N]$: $\text{pk}_i := \mathbf{A}\mathbf{k}_i, \text{sk}_i := (\mathbf{k}_i, \{\text{seed}_{i,j}, \text{seed}_{j,i}\}_{j \in [N]})$ Return $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]})$ $\Delta_{i,S} \leftarrow \text{ZeroShare}(\text{sk}_i, \text{ct}, S)$ // internal algorithm for PDec Parse $\text{sk}_i = (\mathbf{k}_i, \{\text{seed}_{i,j}, \text{seed}_{j,i}\}_{j \in [N]})$ $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} \text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ $\quad - \sum_{j \in S \setminus \{i\}} \text{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S)) \in \mathbb{Z}_q$ Return $\Delta_{i,S}$	$\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu \in \mathbb{Z}_M)$: $\mathbf{s} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}, \mathbf{e}' \leftarrow D_{\mathbb{Z}, \sigma_1}$ $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ $d := \mathbf{s}^\top \text{pk} + \mathbf{e}' + p \cdot \mu \in \mathbb{Z}_q$ $\tau := \mathcal{H}(\mathbf{c}, d)$ $\pi \leftarrow \text{Prove}(\text{crs}, \tau, \mathbf{c} \in \mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, (\mathbf{s}, \mathbf{e}))$ Return $\text{ct} = (\mathbf{c}, d, \pi)$ $h_{i,S} \leftarrow \text{PDec}(\text{sk}_i, \text{ct} = (\mathbf{c}, d, \pi), S)$: Parse $\text{sk}_i = (\mathbf{k}_i, \{\text{seed}_{i,j}, \text{seed}_{j,i}\}_{j \in [N]})$ If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ for $\tau := \mathcal{H}(\mathbf{c}, d)$: Return $\perp$ $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, \text{ct}, S) \in \mathbb{Z}_q$ $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$ $h_{i,S} := \mathbf{c}^\top (\lambda_{i,S} \cdot \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$ Return $h_{i,S}$ $\mu / \perp \leftarrow \text{Comb}(\text{ct} = (\mathbf{c}, d, \pi), S, \{h_{i,S}\}_{i \in S})$: If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ for $\tau := \mathcal{H}(\mathbf{c}, d)$: Return $\perp$ $h := \sum_{i \in S} h_{i,S}$ $\mu := \lceil (d - h) / p \rceil$ Return μ
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Fig. 4. Our (t, N) -TD scheme TD_2 for polynomial N in the synchronous setting.

Due to space limitations, we postpone the formal **Correctness** analysis of TD_2 to Appendix D.5 and refer to technical overview in Sect. 1 for an intuition.

5.3 Security Proof of TD_2

Theorem 3 (IND-TCCA Security of TD_2). *Assume that (i) Π has zero-knowledge and OTSS, (ii) $\mathcal{H}$ is collision-resistant, (iii) the matrix hint-LWE assumption holds for the parameters chosen above, (iv) H_{mask} is a random oracle, then the proposed scheme TD_2 in Fig. 4 is IND-TCCA secure.*

More precisely, for any PPT adversary $\mathcal{A}$, there exists PPT $\mathcal{B}_i$ for $i \in [3]$ s.t.

$$\text{Adv}_{\text{TD}_2, \mathcal{A}}^{\text{CCA}}(\lambda) \leq \text{Adv}_{\Pi, \mathcal{B}_1}^{\text{zk}}(\lambda) + \text{Adv}_{q, m, n, D_{\mathbb{Z}^m, \sigma_0/2}, \mathcal{B}_2}^{\text{LWE}}(\lambda) + \text{Adv}_{\Pi, \mathcal{B}_3}^{\text{OTSS}}(\lambda) + \text{negl}(\lambda).$$

Proof of Theorem 3. Let $\mathcal{A}$ be an adversary against the IND-TCCA security. We use a series of games G_0 - G_{14} to show the advantage $\text{Adv}_{\text{TD}_2, \mathcal{A}}^{\text{CCA}}(\lambda)$ is negligible, where G_0 is the original IND-TCCA experiment and $\mathcal{A}$ has no advantage in G_{14} .

Game G_0 : This is the IND-TCCA experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}$ for threshold decryption scheme in the synchronous setting (cf. Fig. 6 in Appendix C). By definition, we have $\text{Adv}_{\text{TD}_2, \mathcal{A}}^{\text{CCA}}(\lambda) = |\Pr[G_0 = 1] - \frac{1}{2}|$.

Let $\text{pp} = (\mathbf{A}, \text{crs}, \mathbf{H}, \mathbf{H}_{\text{mask}})$. The challenger samples $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$, invokes $(\mathbf{k}_1, \dots, \mathbf{k}_N) \leftarrow \text{SShare}(N, t, \mathbf{k})$, and sets $\text{pk} := \mathbf{A}\mathbf{k}$, $\text{pk}_i := \mathbf{A}\mathbf{k}_i$, $\text{sk}_i := (\mathbf{k}_i, \{\text{seed}_{i,j}, \text{seed}_{j,i}\}_{j \in [N]})$ for each $i \in [N]$. The challenger sends $(\text{pp}, \text{pk}, \{\text{pk}_i\}_{i \in [N]})$ to $\mathcal{A}$ and initializes $\text{Cor} := \emptyset$, $\text{Hon} := [N]$, $\mathcal{Q}_{\text{DEC}}[\cdot] := \emptyset$, $\mathcal{Q}_{\text{MASK}}[\cdot] := \emptyset$. Then it answers $\mathcal{A}$'s $\mathbf{H}_{\text{mask}}$, $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$, $\mathcal{O}_{\text{DEC}}$ queries adaptively and generates a challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ for $\mathcal{A}$ following Fig. 4. Due to space limitations, we refer to Appendix D.6 for a full description of G_0 .

Games G_1 - G_6 : These games are essentially the same as those in the proof of Theorem 1 for the scheme TD_1 in Sect. 4. Here we briefly describe them and refer to the proof of Theorem 1 for the reasoning of their indistinguishability.

- G_1 : π^* in $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ is generated by $\pi^* \leftarrow \text{Sim}(\text{crs}, \tau^*, \mathbf{c}^*)$.
- G_2 : d^* in $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ is generated by $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta$.
- G_3 : $\mathbf{c}^*$ in $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ is generated by $\mathbf{c}^{*\top} \leftarrow \mathbb{Z}_q^{1 \times m}$.
- G_4 : for $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi), S)$, if $\text{ct} \neq \text{ct}^* \wedge (\tau, \mathbf{c}, \pi) = (\tau^*, \mathbf{c}^*, \pi^*)$, return $\perp$.
- G_5 : $\mathbf{A}$ in pp is generated by $(\mathbf{A}, \mathbf{T}_{\mathbf{A}}) \leftarrow \text{TrapGen}(n, q, m)$.
- G_6 : for $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi), S)$, if $\text{ct} \neq \text{ct}^* \wedge \mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, return $\perp$.

Games G_7 - G_{12} : We gradually modify the replies to $\mathcal{O}_{\text{DEC}}$ and $\mathcal{O}_{\text{COR}}$. In particular, in G_{12} , the replies to both oracles can be simulated using only a partial information about the secret key $\mathbf{k}$ as well as the corrupted secret key shares $\{\mathbf{k}_j\}_{j \in \text{Cor}}$, so that there are enough min-entropy left in the secret key $\mathbf{k}$. Leveraging the masking technique, these changes remain indistinguishable to $\mathcal{A}$. Due to space limitations, we only provide an overview of G_7 - G_{12} below and postpone the full descriptions and analysis to Appendix D.7.

Game G_7 : In this game, the challenger introduces a set $\mathcal{Q}_{\text{CT}}$ to record the challenge ciphertext ct^* , the ciphertexts ct generated in $\mathcal{O}_{\text{ENC}}$ and the ciphertexts ct that $\mathcal{A}$ has queried $\mathcal{O}_{\text{DEC}}(\cdot, \text{ct}, \cdot)$ or $\mathbf{H}_{\text{mask}}(\cdot, (\text{ct}, \cdot))$ thus far. When generating a ciphertext ct in $\mathcal{O}_{\text{ENC}}$ and when generating the challenge ciphertext ct^* , if $\text{ct} \in \mathcal{Q}_{\text{CT}}$ or $\text{ct}^* \in \mathcal{Q}_{\text{CT}}$, the challenger directly returns $\perp$. This change is indistinguishable to $\mathcal{A}$, since honestly generated ciphertexts have sufficient min-entropy and collisions occur only with negligible probability.

Game G_8 : From now on, for $S \subseteq [N]$, we denote the honest (resp. corrupted) users in S by $\text{HS} := S \cap \text{Hon}$ (resp. $\text{CS} := S \cap \text{Cor}$). Moreover, in this game, the challenger introduces a set $\mathcal{Q}_{\text{Dec}}$ and two tables UnDecHS , Mask as follows:

- QDec records the tuples (ct, S) for which $\mathcal{A}$ has queried $\mathcal{O}_{\text{DEC}}(\cdot, \text{ct}, S)$.
- $\text{UnDecHS}[(\text{ct}, S)]$ for $(\text{ct}, S) \in \text{QDec}$ stores the users $i \in \text{HS}$ that $\mathcal{A}$ does not issue $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$. We also denote $\widetilde{\text{HS}} := \text{UnDecHS}[(\text{ct}, S)]$ when (ct, S) is clear from the context.
- $\text{Mask}[(\text{ct}, S), i]$ stores the mask $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, \text{ct}, S)$. In addition, when answering $\mathcal{O}_{\text{COR}}(i)$, the challenger precomputes $\Delta_{i,S}$ for each $(\text{ct}, S) \in \text{QDec}$ such that $i \in S$, and stores it in $\text{Mask}[(\text{ct}, S), i]$.

It is a conceptual change, so we have $\Pr[\text{G}_7 = 1] = \Pr[\text{G}_8 = 1]$.

Game G_9 : It is the same as G_8 , except that when computing $\Delta_{i,S}$ in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$, the challenge computes $\Delta_{i,S}$ directly (instead of invoking ZeroShare), i.e., computing $\mathbf{m}_{i,j} := \text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ and $\mathbf{m}_{j,i} := \text{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))$ for $j \in S \setminus \{i\}$, then setting $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i})$. The change is conceptual and we have $\Pr[\text{G}_8 = 1] = \Pr[\text{G}_9 = 1]$.

Game G_{10} : It is the same as G_9 , except for the reply of H_{mask} and the computation of $\Delta_{i,S}$ in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$. In this game, the challenger returns $\perp$ if $\mathcal{A}$ queries H_{mask} with a $\text{seed}_{i,j}$ in the case of both $i, j \in \text{Hon}$. Moreover, when computing $\Delta_{i,S}$ in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$, the challenger samples $\mathbf{m}_{i,j}$ and $\mathbf{m}_{j,i}$ at random for $j \in \widetilde{\text{HS}} \setminus \{i\}$ (and programs H_{mask} accordingly). This change is indistinguishable to $\mathcal{A}$ since $\text{seed}_{i,j}$ is hidden when both $i, j \in \text{Hon}$.

Game G_{11} : It is the same as G_{10} , except that the challenger randomly samples all masks except for the last honest user, and the last mask is computed deterministically by other masks in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$. More precisely, the challenger samples $\Delta_{i,S} \leftarrow \mathbb{Z}_q$ except when $\text{HS} = \{i\}$, i.e. i is the last honest user w.r.t. (ct, S) . If i is the last honest user, it sets

$$\Delta_{i,S} = - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S} = - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \text{Mask}[(\text{ct}, S), j].$$

Additionally, H_{mask} is not programmed until the corresponding user is corrupted, after which it is programmed via an additional helper algorithm ProgramZeroShare . We refer to the formal proof in Appendix D.7 for the details of ProgramZeroShare .

This change is indistinguishable to $\mathcal{A}$ because: (1) If i is not the last user in $\widetilde{\text{HS}}$, then $\widetilde{\text{HS}} \setminus \{i\}$ contains at least another honest user j , so $\mathbf{m}_{i,j}$ and $\mathbf{m}_{j,i}$ are sampled at random according to the game change in G_{10} . Thus, $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i})$ is distributed uniformly randomly over $\mathbb{Z}_q$. (2) If i is the last user in $\widetilde{\text{HS}}$, then all masks are fully determined by the constrain $\sum_{i \in S} \Delta_{i,S} = 0$.

Game G_{12} : It is the same as G_{11} , except that the challenger randomly samples all partial decryption except for the last honest user, and the last partial decryption is computed from previous partial decryptions and a partial information about the secret key $\mathbf{k}$ (i.e., $\mathbf{c}^\top \mathbf{k} + y_{i,S}$ in (14) below). Additionally, the computation of masks in $\mathcal{O}_{\text{COR}}$ is changed accordingly.

More precisely, when replying to $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, \mathbf{d}), S)$, if i is not the last user in $\widetilde{\text{HS}}$, the challenger samples $y_{j,S} \leftarrow_{\$} D_{\mathbb{Z}, \sigma_y}$, $h_{i,S} \leftarrow_{\$} \mathbb{Z}_q$, stores $y_{j,S}$ and returns $h_{i,S}$. Otherwise, it sets

$$h_{i,S} := \mathbf{c}^\top \mathbf{k} + y_{i,S} + \sum_{j \in \text{HS} \setminus \{i\}} (y_{j,S} - h_{j,S}) - \sum_{j \in \text{CS}} (\Delta_{j,S} + \lambda_{j,S} \cdot \mathbf{c}^\top \mathbf{k}_j), \quad (14)$$

to guarantee the consistency.

This change is indistinguishable to $\mathcal{A}$ because: (1) If i is not the last user in $\widetilde{\text{HS}}$, the uniformity of $\Delta_{i,S}$ ensures that $h_{i,S}$ appears uniformly random. (2) If i is the last user, then our simulation strategy ensures that $\sum_{i \in \text{HS}} h_{i,S}$ is distributed identically in both $\mathbf{G}_{11}$ and $\mathbf{G}_{12}$. We refer to Appendix D.7 for more details.

Game $\mathbf{G}_{13}$: It is the same as $\mathbf{G}_{12}$, except for the reply to $\mathcal{A}$'s encryption queries $\mathcal{O}_{\text{ENC}}(\mu, S)$. The challenger now employs the same approach within $\mathcal{O}_{\text{DEC}}$ to generate the honest users' partial decryption $\{h_{i,S}\}_{i \in \text{HS}}$.

Note that this change is a relaxed version of $\mathbf{G}_8$ - $\mathbf{G}_{12}$ since the ciphertext in $\mathcal{O}_{\text{ENC}}$ is always honestly generated by the challenger. By the game change in $\mathbf{G}_7$, the ciphertexts ct generated in $\mathcal{O}_{\text{ENC}}$ will never appear in $\mathcal{Q}_{\text{CT}}$, so we can program $\mathbf{H}_{\text{mask}}$ freely. By a similar analysis, we have $|\Pr[\mathbf{G}_{12} = 1] - \Pr[\mathbf{G}_{13} = 1]| \leq \text{negl}(\lambda)$.

Game $\mathbf{G}_{14}$: It is the same as $\mathbf{G}_{13}$ except that when generating the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$, the challenger samples $d^* \leftarrow_{\$} \mathbb{Z}_q$ uniformly, instead of computing $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta$.

Claim 4. $|\Pr[\mathbf{G}_{13} = 1] - \Pr[\mathbf{G}_{14} = 1]| \leq 1/2^\lambda$.

Proof. Below we first analyze the information about $\text{sk} = \mathbf{k}$ and its shares $\text{sk}_i = \mathbf{k}_i$ ($i \in [N]$) that might be leaked to $\mathcal{A}$ in $\mathbf{G}_{13}$ and $\mathbf{G}_{14}$ (except for the challenge ciphertext ct^*).

- Firstly, $\mathcal{A}$ obtains $\text{pk} = \mathbf{A}\mathbf{k}$, and $\text{pk}_i = \mathbf{A}\mathbf{k}_i$ for $i \in [N]$.
- Through $\mathcal{O}_{\text{COR}}$ queries, $\mathcal{A}$ obtains sk_i for $i \in \text{Cor}$.
- Through $\mathcal{O}_{\text{ENC}}(\mu)$ queries, $\mathcal{A}$ can learn partial decryptions $\{h_{i,S}\}_{i \in S}$. $\mathcal{A}$ learns nothing from $\{h_{i,S}\}_{i \in \text{CS}}$ since $\mathcal{A}$ has obtained $\text{sk}_i := \mathbf{k}_i$. For $\{h_{i,S}\}_{i \in \text{HS}}$, we have that:
 - From the first $|\text{HS}| - 1$ partial decryptions $h_{i,S} \leftarrow_{\$} \mathbb{Z}_q$, the adversary learns nothing.
 - From the last $h_{i,S}$, by (14) we have

$$h_{i,S} = \mathbf{c}^\top \mathbf{k} - \sum_{j \in \text{CS}} \mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) + y_{i,S} + \sum_{j \in \text{HS} \setminus \{i\}} y_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} h_{j,S} - \sum_{j \in \text{CS}} \Delta_{j,S},$$

where $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ with $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$ and $\mathbf{e} \leftarrow_{\$} D_{\mathbb{Z}^m, \sigma_0}$. Note that (i) $\mathbf{k}_j$ and $\Delta_{j,S}$ for $j \in \text{CS}$ are all known to $\mathcal{A}$, (ii) $h_{j,S}$ for $j \in \text{HS} \setminus \{i\}$ are known to $\mathcal{A}$, (iii) there are at least one $y_{j,S}$ for $j \in \text{HS}$ that $\mathcal{A}$ can never obtain since $|\text{Cor}| \leq t - 1$. Thus, the information learned by $\mathcal{A}$ is at most $\mathbf{c}^\top \mathbf{k} + y_{j,S} = \mathbf{s}^\top (\mathbf{A}\mathbf{k}) + \mathbf{e}^\top \mathbf{k} + y_{j,S}$ for some $j \in \text{HS}$. Consequently, $\mathcal{A}$ obtains one linear relation about $\mathbf{k}$ of the form $\mathbf{e}^\top \mathbf{k} + y_{j,S}$ besides $\mathbf{A}\mathbf{k}$.

- Through $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi))$ queries, $\mathcal{A}$ can learn partial decryption $h_{i,S} = \mathbf{c}^\top \mathbf{k}_i + y_{i,S}$ of arbitrary ciphertext $\text{ct} = (\mathbf{c}, \mathbf{d}, \pi)$. There are two cases: (i) If $\text{ct} = \text{ct}^*$, then i is added to S^* , and the partial decryption h_i might leak the whole information about $\text{sk}_i = \mathbf{k}_i$. (ii) If $\text{ct} \neq \text{ct}^*$, by a similar analysis as above and by the change in $\mathbf{G}_6$, $\mathcal{A}$ obtains $\mathbf{c}^\top \mathbf{k} + y_{j,S}$ for some $j \in \text{HS}$ only when $\mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, in which case $\mathbf{c}$ can be written as $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ for some $\mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m$. Then by a similar analysis as above, $\mathcal{A}$ obtains one linear relation of the form $\mathbf{e}^\top \mathbf{k} + y_{j,S}$.

Let Q_e and Q_d denote the number of $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ queries made by $\mathcal{A}$, respectively. Overall, $\mathcal{A}$ gets $\mathbf{A}\mathbf{k}$ and $\{\mathbf{A}\mathbf{k}_i\}_{i \in [N]}$, might learn $\{\text{sk}_i\}_{i \in S^* \cup \text{Cor}}$ (through $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ in case (i)), and learn at most $\ell := Q_e + Q_d$ linear relations of the form $\mathbf{e}^\top \mathbf{k} + y_{j,S}$ (through $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ in case (ii)). Given that $|S^* \cup \text{Cor}| \leq t - 1$ (to avoid trivial attacks), $\mathcal{A}$ learns nothing about $\mathbf{k}$ from $\{\mathbf{k}_i\}_{i \in S^* \cup \text{Cor}}$. So the information about $\text{sk} = \mathbf{k}$ that $\mathcal{A}$ might learn is upper-bounded by $\text{View}_{\mathcal{A}} = (\mathbf{A}\mathbf{k}, \{\mathbf{e}_j^\top \mathbf{k} + y_j\}_{j \in [\ell]})$ with $\|\mathbf{e}_j\|_\infty < \tilde{B}_0$ and $y_j \leftarrow D_{\mathbb{Z}, \sigma_y}$.

Writing $\mathbf{E} := (\mathbf{e}_1, \dots, \mathbf{e}_\ell)^\top \in \mathbb{Z}^{\ell \times m}$ and $\mathbf{y} := (y_1, \dots, y_\ell)^\top \in \mathbb{Z}^\ell$, our goal is to analyze the min-entropy of $\mathbf{k}$ conditioned on $(\mathbf{A}\mathbf{k}, \mathbf{E}\mathbf{k} + \mathbf{y})$. By Lemma 6 and our parameter setting in Table 3, p is sufficiently large and we have

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod M \mid \mathbf{A}\mathbf{k}, \mathbf{E}\mathbf{k} + \mathbf{y}) \geq (m - n) \log M - \log(1 + 4\varepsilon) \geq 2n \log q + 2\lambda. \quad (15)$$

Finally, based on (15), we will show that the d^* in $\mathbf{G}_{13}$ is statistically close to the $d^* \leftarrow_{\$} \mathbb{Z}_q$ in $\mathbf{G}_{14}$. Note that in $\mathbf{G}_{13}$, we have

$$d^* = \mathbf{c}^{*\top} \cdot \mathbf{k} + e' - \mathbf{e}^\top \cdot \mathbf{k} + p \cdot \mu_\beta = (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k} + e' + p \cdot \mu_\beta \quad (16)$$

Recall that $\mathbf{c}^*$ is randomly chosen from $\mathbb{Z}_q^m$ and thus $\mathbf{c}^* - \mathbf{e}$ is also uniformly random over $\mathbb{Z}_q^m$. According to Leftover Hash Lemma (cf. Lemma 1), since (15) holds, it follows that $\Delta((\mathbf{c}^* - \mathbf{e}, (\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}), (\mathbf{c}^* - \mathbf{e}, u) \mid \text{View}_{\mathcal{A}}) \leq 1/2^\lambda$, where $u \leftarrow_{\$} \mathbb{Z}_q$. So the $(\mathbf{c}^* - \mathbf{e})^\top \mathbf{k}$ in (16) is statistically close to uniform from the view of $\mathcal{A}$, and consequently, the d^* in $\mathbf{G}_{13}$ is statistically close to the $d^* \leftarrow_{\$} \mathbb{Z}_q$ in $\mathbf{G}_{14}$. This shows that $|\Pr[\mathbf{G}_{13} = 1] - \Pr[\mathbf{G}_{14} = 1]| \leq 1/2^\lambda$ and Claim 4 follows. $\blacksquare$

In $\mathbf{G}_{14}$, we have $d^* \leftarrow_{\$} \mathbb{Z}_q$, so the challenge bit β is totally hidden from $\mathcal{A}$ and it holds $\Pr[\mathbf{G}_{14} = 1] = \frac{1}{2}$. Putting all things together, Theorem 3 follows. $\square$

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Supplementary Material

A Additional Preliminaries

A.1 Collision-Resistant Hash Functions

Definition 6 (Collision-Resistant Hash Functions). A family of hash functions $\mathcal{H}$ is collision-resistant, if for any PPT adversary $\mathcal{A}$, it holds that

$$\text{Adv}_{\mathcal{H}, \mathcal{A}}^{\text{cr}}(\lambda) := \Pr[H \leftarrow \mathcal{H}, (x_1, x_2) \leftarrow \mathcal{A}(H) : x_1 \neq x_2 \wedge H(x_1) = H(x_2)] \leq \text{negl}(\lambda).$$

A.2 Quasi-Adaptive Non-Interactive Zero-Knowledge

Quasi-Adaptive Non-Interactive Zero-Knowledge argument (QA-NIZK) was proposed by Jutla and Roy [29], where the common reference string (CRS) may depend on the specific language $\mathcal{L}_\rho$ for which proofs are generated. Below we formalize tag-based QA-NIZK according to [36] for gap language $\mathcal{GL}_\rho = (\mathcal{L}_\rho, \tilde{\mathcal{L}}_\rho)$, where $\mathcal{L}_\rho \subseteq \tilde{\mathcal{L}}_\rho$. Intuitively, completeness and zero-knowledge are guaranteed for instances in $\mathcal{L}_\rho$, while soundness is guaranteed for instances outside $\tilde{\mathcal{L}}_\rho$.

Definition 7 (QA-NIZK for Gap Language). A tag-based QA-NIZK scheme $\Pi = (\text{CRSGen}, \text{Prove}, \text{Vrfy}, \text{SimGen}, \text{Sim})$ for gap language $\mathcal{GL}_\rho = (\mathcal{L}_\rho, \tilde{\mathcal{L}}_\rho)$ (where $\mathcal{L}_\rho \subseteq \tilde{\mathcal{L}}_\rho \subseteq \mathcal{X}$) with tag space $\mathcal{T}$ consists of five PPT algorithms:

- $\text{crs} \leftarrow \text{CRSGen}(\rho)$: Taking as input the language parameter ρ , the CRS generation algorithm outputs a common reference string (CRS) crs .
- $\pi \leftarrow \text{Prove}(\text{crs}, \tau, x, w)$: Taking as input crs , a tag $\tau \in \mathcal{T}$ and an instance $x \in \mathcal{L}_\rho$ with witness w , the proof generation algorithm outputs a proof π .
- $0/1 \leftarrow \text{Vrfy}(\text{crs}, \tau, x, \pi)$: Taking as input crs , a tag $\tau \in \mathcal{T}$, $x \in \mathcal{X}$ and a proof π , the deterministic verification algorithm outputs a bit indicating whether π is a valid proof.
- $(\text{crs}, \text{td}_{\text{crs}}) \leftarrow \text{SimGen}(\rho)$: Taking as input the language parameter ρ , the simulated CRS generation algorithm outputs a simulated crs and a trapdoor td_{crs} .
- $\pi \leftarrow \text{Sim}(\text{crs}, \text{td}_{\text{crs}}, \tau, x)$: Taking as input crs , a trapdoor td_{crs} , a tag $\tau \in \mathcal{T}$ and $x \in \mathcal{X}$, the simulation algorithm outputs a simulated proof π .

Moreover, Π should satisfy the following properties:

- **Completeness.** For any $\text{crs} \leftarrow \text{CRSGen}(\rho)$, $\tau \in \mathcal{T}$ and $x \in \mathcal{L}_\rho$ with witness w , it holds $\Pr[\pi \leftarrow \text{Prove}(\text{crs}, \tau, x, w) : \text{Vrfy}(\text{crs}, \tau, x, \pi) = 1] \geq 1 - \text{negl}(\lambda)$.
- **Soundness.** For any PPT adversary $\mathcal{A}$, it holds that

$$\Pr[\text{crs} \leftarrow \text{CRSGen}(\rho), (x, \pi, \tau) \leftarrow \mathcal{A}(\text{crs}) : \text{Vrfy}(\text{crs}, \tau, x, \pi) = 1 \wedge x \notin \tilde{\mathcal{L}}_\rho] \leq \text{negl}(\lambda).$$

- **Zero-Knowledge.** For any PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{zk}} := \left| \Pr[\text{crs} \leftarrow \text{CRSGen}(\rho) : \mathcal{A}^{\text{Prove}(\text{crs}, \cdot, \cdot, \cdot)}(\text{crs}) = 1] - \Pr[(\text{crs}, \text{td}_{\text{crs}}) \leftarrow \text{SimGen}(\rho) : \mathcal{A}^{\mathcal{O}_{\text{SIM}}(\text{crs}, \text{td}_{\text{crs}}, \cdot, \cdot)}(\text{crs}) = 1] \right| \leq \text{negl}(\lambda),$$

where $\text{Prove}(\text{crs}, \tau, x, w)$ outputs $\perp$ if w is not a witness for $x \in \mathcal{L}_\rho$ and outputs a valid proof $\pi \leftarrow \text{Prove}(\text{crs}, \tau, x, w)$ otherwise; $\mathcal{O}_{\text{SIM}}(\text{crs}, \text{td}_{\text{crs}}, \tau, x, w)$ is an oracle that outputs $\perp$ if w is not a witness for $x \in \mathcal{L}_\rho$ and outputs a simulated proof $\pi \leftarrow \text{Sim}(\text{crs}, \text{td}_{\text{crs}}, \tau, x)$ otherwise.

Simulation-Soundness. Simulation soundness [51, 17] formalizes the inability of adversary to create a proof for a false statement x^* even after having seen simulated proofs for possibly false statements $\{x_i\}_i$ of its choice. *One-time simulation soundness* (OTSS) is a weaker notion that only allows the adversary to see one simulated proof, which is sufficient for our purpose.

Definition 8 (OTSS of QA-NIZK). A tag-based NIZK scheme Π for gap language $\mathcal{GL}_\rho = (\mathcal{L}_\rho, \tilde{\mathcal{L}}_\rho)$ has one-time simulation soundness (OTSS), if for any PPT $\mathcal{A}$, it holds that $\text{Adv}_{\Pi, \mathcal{A}}^{\text{OTSS}}(\lambda) := \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{OTSS}} \Rightarrow 1] \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{OTSS}}$ is defined in Fig. 5

$\text{Exp}_{\Pi, \mathcal{A}}^{\text{OTSS}}:$	
$(\text{crs}, \text{td}_{\text{crs}}) \leftarrow \text{SimGen}(\rho), \quad \mathcal{Q}_{\text{SIM}} := \emptyset$	
$(\tau^*, x^*, \pi^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{SIM}}(\cdot, \cdot)}(\rho, \text{td}_{\text{crs}}, \text{crs}) \quad // \text{td}_{\text{crs}} \text{ is a trapdoor for membership testing of } \tilde{\mathcal{L}}_\rho$	
If $(x^* \notin \mathcal{L}_\rho) \wedge ((\tau^*, x^*, \pi^*) \notin \mathcal{Q}_{\text{SIM}}) \wedge (\text{Vrfy}(\text{crs}, \tau^*, x^*, \pi^*) = 1)$: Return 1; Else: Return 0	
$\mathcal{O}_{\text{SIM}}(\tau, x):$	
$\pi \leftarrow \text{Sim}(\text{crs}, \text{td}_{\text{crs}}, \tau, x), \quad \mathcal{Q}_{\text{SIM}} := \mathcal{Q}_{\text{SIM}} \cup \{(\tau, x, \pi)\}$	
Return π	
$// \text{can be queried only once}$	

Fig. 5. The one-time simulation-soundness experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{OTSS}}$ for NIZK

In the above definition, in order to allow a challenger to efficiently check the winning condition $x^* \notin \tilde{\mathcal{L}}_\rho$, we restrict ourselves to trapdoor languages, where a language-specific trapdoor td_ρ enables the efficient testing of whether a given instance x^* belongs to $\tilde{\mathcal{L}}_\rho$. This restriction has no impact on our applications where we always have a membership testing trapdoor td_ρ at disposal.

B Proof of Lemma 6 (Conditional Min-Entropy Lower Bound with Polynomial Noise Flooding)

In this section, we provide the full proof of Lemma 6. To this end, we first recall some additional lattice and algebra backgrounds needed in the proof in Appendix B.1, and then prove Lemma 6 in Appendix B.2.

B.1 Lattice and Algebra Backgrounds

Let $\mathbf{B} \in \mathbb{R}^{n \times m}$ and let $\Sigma \in \mathbb{R}^{m \times m}$ be a symmetric matrix. We say a symmetric matrix Σ is positive definite (resp. semi-definite) if and only if $\mathbf{x}^\top \Sigma \mathbf{x} > 0$ (resp. ≥ 0) for all non-zero $\mathbf{x} \in \mathbb{R}^m$. The symmetric matrix $\mathbf{B}^\top \mathbf{B}$ is positive semi-definite, because $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \|\mathbf{B}\mathbf{x}\|^2 \geq 0$ for any non-zero $\mathbf{x} \in \mathbb{R}^m$, where the inequality is always strict if and only if $\mathbf{B}^\top \mathbf{B}$ is nonsingular. We say that $\mathbf{B}$ is a square root of Σ , written $\mathbf{B} := \sqrt{\Sigma}$, if $\mathbf{B}^\top \mathbf{B} = \Sigma$. Each positive semi-definite Σ has a square root, which can be computed efficiently, e.g., via the Cholesky decomposition. The (Moore-Penrose) pseudoinverse for Σ , denoted Σ^+ , is the unique matrix satisfying $(\Sigma \Sigma^+) \Sigma = \Sigma$, $\Sigma^+ (\Sigma \Sigma^+) = \Sigma^+$, and such that both $\Sigma \Sigma^+$ and $\Sigma^+ \Sigma$ are symmetric.

Let $\Sigma \in \mathbb{R}^{m \times m}$ be a positive semi-definite matrix and $\Lambda \subset \text{span}(\Lambda)$ be a lattice. We say that $\sqrt{\Sigma} \geq \eta_\epsilon(\Lambda)$ if $\rho_{\sqrt{\Sigma}^+}(\Lambda^*) \leq 1 + \epsilon$. When the covariance matrix Σ is positive definite and the lattice Λ are full-rank, $\sqrt{\Sigma} \geq \eta_\epsilon(\Lambda)$ is equivalent to $\lambda_{\min}(\Sigma) \geq \eta_\epsilon^2(\Lambda)$. Let $\lambda_{\min}(\Sigma) := \min_{\mathbf{x} \in \mathbb{R}^m, \|\mathbf{x}\|=1} \mathbf{x}^\top \Sigma \mathbf{x}$ (resp., $\lambda_{\max}(\Sigma)$) denotes the minimal (resp., maximum) eigenvalue of Σ .

Below are some useful properties of ℓ_2 -norm, spectral norm and eigenvalue.

Lemma 7 (Properties of ℓ_2 -Norm, Spectral Norm and Eigenvalue).

Let $\mathbf{B} \in \mathbb{R}^{n \times m}$, $\mathbf{C} \in \mathbb{R}^{m \times k}$ and let $\Sigma_k, \Sigma_y \in \mathbb{R}^{n \times n}$ be symmetric matrices. Let $\sigma \in \mathbb{R}$ be a real number. We have the following properties:

1. $s_1(\mathbf{B}) = \sqrt{\lambda_{\max}(\mathbf{B}^\top \mathbf{B})}$;
2. $s_1(\mathbf{B}) \leq \sqrt{m} \cdot \|\mathbf{B}\|$;
3. $s_1(\mathbf{BC}) \leq s_1(\mathbf{B}) \cdot s_1(\mathbf{C})$;
4. If Σ_k is positive definite, then $\lambda_{\min}(\Sigma_k^{-1}) = \frac{1}{\lambda_{\max}(\Sigma_k)}$;
5. $\lambda_{\max}(\Sigma_k + \Sigma_y) \leq \lambda_{\max}(\Sigma_k) + \lambda_{\max}(\Sigma_y)$.

Proof.

1. $s_1^2(\mathbf{B}) = \max_{\mathbf{x} \in \mathbb{R}^m, \|\mathbf{x}\|=1} \|\mathbf{B}\mathbf{x}\|^2 = \max_{\mathbf{x} \in \mathbb{R}^m, \|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \lambda_{\max}(\mathbf{B}^\top \mathbf{B})$.
2. Let $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{R}^m$ s.t. $\|\mathbf{x}\| = 1$ and $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_m)$, where $\mathbf{b}_i$ denotes the i -th column of $\mathbf{B}$. We have $\mathbf{B}\mathbf{x} = \sum_{i=1}^m x_i \mathbf{b}_i$. Note that $\|\mathbf{B}\mathbf{x}\| \leq \sum_{i=1}^m |x_i| \|\mathbf{b}_i\| \leq (\sum_{i=1}^m |x_i|) \cdot \max_{i \in [m]} \|\mathbf{b}_i\| = (\sum_{i=1}^m |x_i|) \cdot \|\mathbf{B}\|$. By Cauchy-Schwarz inequality, we have $\sum_{i=1}^m |x_i| \leq \sqrt{m} \cdot \|\mathbf{x}\| = \sqrt{m}$ and the statement follows.
3. Note that $s_1(\mathbf{BC}) := \max_{\mathbf{x} \in \mathbb{R}^k, \|\mathbf{x}\|=1} \|\mathbf{BC}\mathbf{x}\| \leq s_1(\mathbf{B}) \cdot \max_{\mathbf{x} \in \mathbb{R}^k, \|\mathbf{x}\|=1} \|\mathbf{C}\mathbf{x}\| = s_1(\mathbf{B}) \cdot s_1(\mathbf{C})$. That the inequality holds is because that $\|\mathbf{B}\mathbf{x}\| \leq s_1(\mathbf{B}) \|\mathbf{x}\|$ holds for any non-zero $\mathbf{x} \in \mathbb{R}^m$.
4. Let $\lambda > 0$ be an eigenvalue of Σ_k . Then $\frac{1}{\lambda}$ is an eigenvalue of Σ_k^{-1} , because there exist $\mathbf{x} \in \mathbb{R}^m$ such that $\Sigma_k \mathbf{x} = \lambda \mathbf{x} \implies \frac{1}{\lambda} \mathbf{x} = \Sigma_k^{-1} \mathbf{x}$. Thus, if $\lambda_{\max}$ is the maximal eigenvalue of Σ , then $\frac{1}{\lambda_{\max}(\Sigma)}$ is the minimal eigenvalue of $\Sigma^{-1} = \lambda_{\min}(\Sigma^{-1})$.

5. By definition, we have

$$\begin{aligned}\lambda_{\max}(\mathbf{\Sigma}_k + \mathbf{\Sigma}_y) &= \max_{\mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\|=1} (\mathbf{x}^\top (\mathbf{\Sigma}_k + \mathbf{\Sigma}_y) \mathbf{x}) \\ &\leq \max_{\mathbf{x}_0 \in \mathbb{R}^n, \|\mathbf{x}_0\|=1} (\mathbf{x}_0^\top \mathbf{\Sigma}_k \mathbf{x}_0) + \max_{\mathbf{x}_1 \in \mathbb{R}^n, \|\mathbf{x}_1\|=1} (\mathbf{x}_1^\top \mathbf{\Sigma}_y \mathbf{x}_1) = \lambda_{\max}(\mathbf{\Sigma}_k) + \lambda_{\max}(\mathbf{\Sigma}_y).\end{aligned}$$

Lemma 8 ([41, Lemma 2.5]). *Let $\Lambda \subset \mathbb{R}^n$ be any lattice. For any positive definite $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$ and $\mathbf{c} \in \mathbb{R}^n$, we have $\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda + \mathbf{c}) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda)$. Moreover, if $\sqrt{\mathbf{\Sigma}} \geq \eta_\epsilon(\Lambda)$ for some $\epsilon > 0$ and $\mathbf{c} \in \text{span}(\Lambda)$, then $\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda + \mathbf{c}) \geq \frac{1-\epsilon}{1+\epsilon} \cdot \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda)$.*

Lemma 9 ([25, Corollary 2.8]). *Let $\Lambda, \Lambda' \in \mathbb{R}^n$ be lattices, with $\Lambda' \subseteq \Lambda$. Let $\mathbf{\Sigma}$ be a positive definite symmetric matrix over $\mathbb{R}^{n \times n}$. Then for any $\epsilon \in (0, \frac{1}{2})$ and any $\mathbf{c} \in \mathbb{R}^n$, if $\sqrt{\mathbf{\Sigma}} \geq \eta_\epsilon(\Lambda')$, the distribution of $\mathcal{D}_{\Lambda, \sqrt{\mathbf{\Sigma}}, \mathbf{c}} \bmod \Lambda'$ is within statistical distance at most 2ϵ of uniform distribution over $\Lambda \bmod \Lambda'$. Specifically, for $\forall \mathbf{v} \in \Lambda$, $\Pr[\mathbf{z} = \mathbf{v} \bmod \Lambda' \mid \mathbf{z} \leftarrow D_{\Lambda, \sqrt{\mathbf{\Sigma}}, \mathbf{c}}]$ is in the range $(1 \pm 4\epsilon)/|\Lambda/\Lambda'|$.*

The following corollary gives the maximal probability of the special case of $\Lambda = \mathbb{Z}^m$ and $\Lambda' = p\mathbb{Z}^m$ for a prime p , and $\mathbf{\Sigma} = \sigma^2 \mathbf{I}_m$.

Corollary 1. *If $\sigma \geq \eta_\epsilon(p\mathbb{Z}^m)$, then for any $\mathbf{v} \in \mathbb{Z}^m$, the probability $\Pr[\mathbf{z} = \mathbf{v} \bmod p \mid \mathbf{z} \leftarrow D_{\mathbb{Z}^m, \sigma}] \leq (1 + 4\epsilon)/p^m$.*

Lemma 10 ([25, Theorem 3.1]). *Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice with basis $\mathbf{B}$, and let $\epsilon > 0$. We have $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot \sqrt{\ln(2n(1+1/\epsilon))/\pi}$, where $\tilde{\mathbf{B}}$ denotes the Gram-Schmidt orthogonalization of $\mathbf{B}$. In particular, for any $\omega(\sqrt{\log n})$ function, there is a negligible $\epsilon(n)$ for which $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot \omega(\sqrt{\log n})$.*

Taking $\Lambda = p\mathbb{Z}$ for a prime p , we have the following corollary.

Corollary 2. *Let $\epsilon > 0$. We have $\eta_\epsilon(p\mathbb{Z}^n) \leq p \cdot \sqrt{\ln(2n(1+1/\epsilon))/\pi}$.*

Lemma 11 ([21]). *Let X, Y, Z be three (possibly correlated) random variables. If Z has at most 2^λ possible values, then $\tilde{\mathbf{H}}_\infty(X \mid (Y, Z)) \geq \tilde{\mathbf{H}}_\infty(X \mid Y) - \lambda$.*

Lemma 12. *Let χ be any distribution over $\mathbb{Z}^m$, if $\Pr[\|\chi\|_\infty \geq p/2] \leq 2^{-\lambda}$, then it holds that*

$$\tilde{\mathbf{H}}_\infty(\chi \bmod p) \geq \tilde{\mathbf{H}}_\infty(\chi) - \log(1 + 2^{\tilde{\mathbf{H}}_\infty(\chi) - \lambda}).$$

Proof. Let $p_{\max}$ and $p_{\max}^*$ denote the maximum probability of χ and $\chi \bmod p$, respectively. By definition, $p_{\max} = \max_{\mathbf{x} \in \mathbb{Z}^m} \Pr[\chi = \mathbf{x}] = 2^{-\tilde{\mathbf{H}}_\infty(\chi)}$ and $p_{\max}^* = \max_{\mathbf{x} \in \mathbb{Z}_p^m} \Pr[\chi = \mathbf{x} \bmod p] = 2^{-\tilde{\mathbf{H}}_\infty(\chi \bmod p)}$. Since $\Pr[\|\chi\|_\infty \geq p/2] \leq 2^{-\lambda}$, we have $p_{\max}^* \leq p_{\max} + 2^{-\lambda}$ and Lemma 12 follows. $\blacksquare$

B.2 Proof of Lemma 6

Lemma 6 *Let $c, m, n, q, p, \ell \in \mathbb{Z}$, let $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, and let $\sigma, \sigma_k, \sigma_y, \epsilon \in \mathbb{R}$ satisfying $\sigma_y \geq \sqrt{c} \cdot \sqrt{m\ell} B \cdot \sigma_k$. Let $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$, $\mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sigma_y}$ and $\mathbf{R} \in \mathbb{Z}^{\ell \times m}$ satisfying $\|\mathbf{R}\|_\infty \leq B$. Then conditioned on $Y = (\mathbf{A}, \mathbf{A}\mathbf{k}, \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y})$, we consider the min-entropy of $\mathbf{k} \bmod p$ in two cases:*

– If $\sigma_k \geq \sqrt{1 + 1/c} \cdot p \sqrt{\ln(2m(1 + 1/\epsilon))/\pi}$, it holds that

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid Y) \geq m \log p - n \log q - \log(1 + 4\epsilon).$$

– If $\sigma_k \geq \sqrt{1 + 1/c} \cdot \sqrt{\ln(2m(1 + 1/\epsilon))/\pi}$ and $p/2 \geq \sqrt{m \log \sigma_k} \cdot (2\sqrt{m} + \sqrt{\ell}) \cdot \sigma_k$, it holds that

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid Y) \geq m \log \sigma_k - m/2 \cdot \log(1 + 1/c) - n \log q - \log(1 + 2\epsilon).$$

Proof of Lemma 6. Before presenting the formal proof, we provide an overview of the proof strategy for Lemma 6, which proceeds in three steps:

1. Since $\mathbf{A}\mathbf{k}$ takes at most q^n distinct values, conditioned on $\mathbf{A}\mathbf{k}$, the min-entropy of $\mathbf{k}$ is reduced by at most $n \log q$ bits. Therefore, it suffices to analyze $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{h})$.
2. We lower bound $\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h})$ via analyzing the conditional distribution of $\mathbf{k}$ given the linear constraint $\mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y}$;
3. Finally, we analyze $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{h})$ by considering two cases, depending on whether p is small or large.

First of all, we observe that $\mathbf{A}\mathbf{k} \in \mathbb{Z}_q^n$ has at most q^n possible values. Hence, by Lemma 11, it holds that

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{A}, \mathbf{A}\mathbf{k}, \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y}) \geq \tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{h}) - n \log q. \quad (17)$$

To further analyze the distribution of $\mathbf{k}$ conditioned on $\mathbf{h}$, we leverage Lemma 13. Let $\Sigma_y := \sigma_y^2 \mathbf{I}_\ell$ and $\Sigma_k := \sigma_k^2 \mathbf{I}_\ell$ denote the covariance matrix of $\mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sqrt{\Sigma_y}}$ and $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_k}}$, respectively. By applying Lemma 13, the conditional distribution of $\mathbf{k}$ is $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$, where

$$\Sigma = (\Sigma_k^{-1} + \mathbf{R}^\top \Sigma_y^{-1} \mathbf{R})^{-1} \text{ and } \mathbf{c} = \Sigma \mathbf{R}^\top \Sigma_y^{-1} \mathbf{h}.$$

Our next step is to analyze the min-entropy given by the distribution $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$. For any $\mathbf{k}^* \in \mathbb{Z}^m$, the probability of $\mathbf{k}^*$ is

$$\Pr[\mathbf{k} = \mathbf{k}^*] = \frac{\rho_{\mathbf{c}, \sqrt{\Sigma}}(\mathbf{k}^*)}{\rho_{\mathbf{c}, \sqrt{\Sigma}}(\mathbb{Z}^m)} = \frac{\exp[-\pi(\mathbf{k}^* - \mathbf{c})^\top \Sigma^{-1}(\mathbf{k}^* - \mathbf{c})]}{\sum_{\mathbf{k} \in \mathbb{Z}^m} \exp[-\pi(\mathbf{k} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{k} - \mathbf{c})]}. \quad (18)$$

Recall that $\Sigma_k = \sigma_k^2 \mathbf{I}_m$ and $\Sigma_y = \sigma_y^2 \mathbf{I}_\ell$, then

$$\Sigma^{-1} = \left(\frac{\mathbf{I}_m}{\sigma_k^2} + \frac{\mathbf{R}^\top \mathbf{R}}{\sigma_y^2} \right).$$

It is clear to see that Σ^{-1} is positive definite and hence the upper bound of the numerator in (18) is 1. To bound the maximal probability of $\mathbf{k}$, we need to determine the lower bound of the denominator in (18). Observe that for any

non-zero $\mathbf{k} \in \mathbb{Z}^m$, it holds that $(\mathbf{k} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{k} - \mathbf{c}) \leq \lambda_{\max}(\Sigma^{-1})(\mathbf{k} - \mathbf{c})^\top (\mathbf{k} - \mathbf{c})$. Consequently, we have

$$\sum_{\mathbf{k} \in \mathbb{Z}^m} \exp[-\pi(\mathbf{k} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{k} - \mathbf{c})] \geq \sum_{\mathbf{k} \in \mathbb{Z}^m} \exp[-\pi \lambda_{\max}(\Sigma^{-1})(\mathbf{k} - \mathbf{c})^\top (\mathbf{k} - \mathbf{c})].$$

Therefore, to lower bound $\exp[-\pi \lambda_{\max}(\Sigma^{-1})(\mathbf{k} - \mathbf{c})^\top (\mathbf{k} - \mathbf{c})]$, our target is to analyze the upper bound of $\lambda_{\max}(\Sigma^{-1})$. Since $\mathbf{I}_m$ and $\mathbf{R}^\top \mathbf{R}$ are symmetric, by Lemma 7, it holds that

$$\lambda_{\max}(\Sigma^{-1}) \leq \frac{1}{\sigma_k^2} \cdot \lambda_{\max}(\mathbf{I}_m) + \frac{1}{\sigma_y^2} \cdot \lambda_{\max}(\mathbf{R}^\top \mathbf{R}) = \frac{1}{\sigma_k^2} + \frac{1}{\sigma_y^2} \cdot s_1^2(\mathbf{R})$$

Here we have $s_1^2(\mathbf{R}) \leq m \cdot \|\mathbf{R}\|^2 \leq m \cdot \ell \cdot \|\mathbf{R}\|_\infty^2 = m\ell B^2$. Consequently, we obtain that

$$\lambda_{\max}(\Sigma^{-1}) \leq \frac{1}{\sigma_k^2} + \frac{m\ell B^2}{\sigma_y^2}.$$

By setting $\sigma_y \geq \sqrt{c} \cdot \sqrt{m\ell} B \sigma_k$, we have $\lambda_{\max}(\Sigma^{-1}) \leq (c+1)/c \cdot 1/\sigma_k^2$ and $\lambda_{\min}(\Sigma) \geq c/(c+1) \cdot \sigma_k^2$. We below analyze two cases that p is small and p is sufficient large, respectively.

We first analyze the case p is small, in which case we leverage the smoothing lemma. According to Corollary 2, $\eta_\epsilon(p\mathbb{Z}^m) \leq \frac{p}{\sqrt{\pi}}(\sqrt{\ln(2m(1+1/\epsilon))})$, and thus we can set $\lambda_{\min}(\Sigma) \geq c/(c+1) \cdot \sigma_k^2 \geq \eta_\epsilon^2(p\mathbb{Z}^m)$, i.e., $\sigma_k \geq \sqrt{1+1/c} \cdot \eta_\epsilon(p\mathbb{Z}^m)$. By Lemma 9, the distribution $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}} \bmod p\mathbb{Z}^m$ is statistically close to a uniform distribution over $\mathbb{Z}^m/p\mathbb{Z}^m = \mathbb{Z}_p^m$, and for any $\mathbf{v}^* \in \mathbb{Z}^m$, we have the probability $\Pr[\mathbf{k} = \mathbf{k}^* \bmod p] \leq \frac{1+4\epsilon}{p^m}$. Thus, the min-entropy given by $\mathbf{k} \bmod p$ is greater than $m \log p - \log(1+4\epsilon)$. Combining with (17), this completes the proof for small p .

For a sufficient large p , we hope that $\bmod p$ does not decrease the min-entropy of $\mathbf{k}$, i.e., $\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{h}) \approx \tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h})$. It suffices to require that $\|\mathbf{k}\|_\infty \leq p/2$ with overwhelming probability, then the operation of $\bmod p$ will not significantly decrease the min-entropy of $\mathbf{k}$.

We first analyze the min-entropy of $\mathbf{k}$ on $\mathbb{Z}^m$. According to Lemma 10, $\eta_\epsilon(\mathbb{Z}^m) < \sqrt{\ln(2m(1+1/\epsilon))/\pi}$, and thus $\lambda_{\min}(\Sigma) \geq c/(c+1) \cdot \sigma_k^2 \geq \eta_\epsilon^2(\mathbb{Z}^m)$. For any $\mathbf{k}^* \in \mathbb{Z}^m$ the probability

$$\begin{aligned} \Pr[\mathbf{k} = \mathbf{k}^*] &\leq \frac{1}{\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbb{Z}^m)} \leq \left(\frac{1+\epsilon}{1-\epsilon}\right) \frac{1}{\rho_{\sqrt{\Sigma}}(\mathbb{Z}^m)} \quad (\text{Lemma 8}) \\ &\leq (1+2\epsilon) \frac{1}{\sqrt{\det(\Sigma)}} \leq (1+2\epsilon) \frac{1}{\sigma_k^m} \cdot \left(1 + \frac{1}{c}\right)^{m/2} \end{aligned} \quad (19)$$

The inequity (19) holds is due to the fact that

$$\rho_{\sqrt{\Sigma}}(\mathbb{Z}^m) = \sqrt{\det(\Sigma)} \sum_{\mathbf{w} \in \mathbb{Z}^m} e^{-\pi \mathbf{w}^\top \Sigma \mathbf{w}} \geq \sqrt{\det(\Sigma)}.$$

Thus, we have

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h}) \geq m \log \sigma_k - m/2 \cdot \log(1 + 1/c) - \log(1 + 2\epsilon). \quad (20)$$

Below we analyze the tail bound of $\mathbf{k}$ for $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$ via the inequality $\|\mathbf{k}\|_\infty \leq \|\mathbf{k}\| \leq \|\mathbf{c}\| + \|\mathbf{k} - \mathbf{c}\|$.

Let $T := \sqrt{m \log \sigma_k}$. For $\mathbf{c} = \Sigma \mathbf{R} \Sigma_y^{-1} \mathbf{h}$, it holds that

$$\begin{aligned} \|\mathbf{c}\| &\leq s_1(\Sigma \mathbf{R} \Sigma_y^{-1}) \cdot \|\mathbf{h}\| \leq s_1(\Sigma) \cdot s_1(\mathbf{R}) \cdot s_1(\Sigma_y^{-1}) \cdot \|\mathbf{h}\| \\ &- s_1(\Sigma) \leq \sigma_k^2. \text{ Recall that } \Sigma^{-1} = \left(\frac{\mathbf{I}_m}{\sigma_k^2} + \frac{\mathbf{R}^\top \mathbf{R}}{\sigma_y^2} \right), \text{ where } \mathbf{I}_m \text{ is positive definite} \\ &\text{ and } \mathbf{R}^\top \mathbf{R} \text{ is semi-positive definite. Thus, } \lambda_{\min}(\Sigma^{-1}) \geq \lambda_{\min}(\frac{\mathbf{I}_m}{\sigma_k^2}) = \frac{1}{\sigma_k^2} \text{ and} \\ &\lambda_{\max}(\Sigma) \leq \sigma_k^2. \\ &- s_1(\mathbf{R}) \leq \sqrt{m\ell}B, \text{ as analyzed above.} \\ &- s_1(\Sigma_y^{-1}) = \lambda_{\max}(\Sigma_y^{-1}) = \frac{1}{\sigma_y^2}. \\ &- \|\mathbf{h}\| \leq T\sqrt{\ell} \cdot (mB\sigma_k + \sigma_y) \text{ with probability at least } 1 - 2^{-T^2}. \text{ Since } \|\mathbf{h}\| = \|\mathbf{R}\mathbf{k} + \mathbf{y}\| \leq \|\mathbf{R}\mathbf{k}\| + \|\mathbf{y}\| \leq s_1(\mathbf{R}) \cdot \|\mathbf{k}\| + \|\mathbf{y}\|, \text{ by Lemma 5 (tail bound), we} \\ &\text{ have } \|\mathbf{h}\| \leq T \cdot (\sqrt{m\ell}B \cdot \sqrt{m}\sigma_k + \sqrt{\ell}\sigma_y) \leq T \cdot \sqrt{m\ell}B \cdot (\sqrt{m} + \sqrt{\ell})\sigma_k \text{ with} \\ &\text{ probability at least } 1 - 2^{-T^2+1}. \end{aligned}$$

Overall, we have the following bound with probability at least $1 - 2^{-T^2+1}$:

$$\|\mathbf{c}\| \leq \frac{\sigma_k^2}{\sigma_y^2} \cdot \sqrt{m\ell}B \cdot T \cdot \sqrt{m\ell}B \cdot (\sqrt{m} + \sqrt{\ell})\sigma_k = T(\sqrt{m} + \sqrt{\ell})\sigma_k \quad (21)$$

For $\|\mathbf{k} - \mathbf{c}\|$, we bound it by Lemma 5 and have $\|\mathbf{k} - \mathbf{c}\| \leq T \cdot \sqrt{m} \lambda_{\max}(\Sigma) \leq T \cdot \sqrt{m}\sigma_k$ with probability at least $1 - 2^{-T^2}$. Combining with (21), we have

$$\|\mathbf{k}\|_\infty \leq \|\mathbf{k}\| \leq \|\mathbf{k} - \mathbf{c}\| + \|\mathbf{c}\| \leq T(2\sqrt{m} + \sqrt{\ell})\sigma_k.$$

with probability at least $1 - 2^{-m \log \sigma_k + 2}$.

In the case of $\|\mathbf{k}\|_\infty \leq T(2\sqrt{m} + \sqrt{\ell})\sigma_k \leq p/2$, Lemma 12 shows that

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \bmod p \mid \mathbf{h}) \geq \tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h}) - \log(1 + 2^{\mathbf{H}_\infty(\mathbf{k}) - m \log \sigma_k + 2}) \approx \tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h}). \quad (22)$$

Combining (22) with (17) and (20), we completes the proof for large p , and Lemma 6 follows. $\square$

Remark 1 (Comparison to Entropy Analyses in [7, 4]). Brakerski and Döttling [7] introduce the *gentle flooding* technique to establish the hardness of LWE with low-entropy secrets. We observe that their technique can be adapted to analyze $\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y})$ in our setting. Besides, a very recent work [4] also presents a similar analysis based on the techniques in [7]. For completeness, we briefly describe how their approach can be applied in our setting and refer the readers to [7, 4] for more details.

To analyze $\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y})$, one can proceed in two steps via their approach:

- (1) By (a discrete variant of) [7, Proposition 3.2], for any matrix $\mathbf{R} \in \mathbb{Z}^{\ell \times m}$, if $\sigma_y \geq s_1(\mathbf{R}) \cdot \sigma_{y_0}$, then the smudge noise term $\mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sigma_y}$ can be decomposed as $\mathbf{y} \stackrel{s}{\approx} \mathbf{y}_1 + \mathbf{R}\mathbf{y}_0$, where $\mathbf{y}_0 \leftarrow D_{\mathbb{Z}^m, \sigma_{y_0}}$ and $\mathbf{y}_1 \leftarrow D_{\mathbb{Z}^\ell, \sqrt{\Sigma}}$ with $\Sigma := \sigma_y^2 \mathbf{I} - \sigma_{y_0}^2 \mathbf{R}^\top \mathbf{R}$.
- (2) Consequently, it suffices to analyze

$$\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{R}\mathbf{y}_0 + \mathbf{y}_1) = \tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{k} + \mathbf{y}_0),$$

for which [7, Lemma 5.4] implies that if $\sigma_{y_0} \geq \sqrt{m} \cdot \|\mathbf{k}\|$, then it follows that $\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{k} + \mathbf{y}_0) \geq \tilde{\mathbf{H}}_\infty(\mathbf{k}) - \pi \log e$.

To compare our Lemma 6 with their approach, we can set $c = m$ in Lemma 6. Our analysis shows that $\tilde{\mathbf{H}}_\infty(\mathbf{k} \mid \mathbf{h}) \geq m \log \sigma_k - m/2 \cdot \log(1 + 1/m) - \log(1 + 2\epsilon) \approx \tilde{\mathbf{H}}_\infty(\mathbf{k}) - 1/2 \cdot \log e$, which is comparable to the entropy loss obtained above.

We note that the main difference lies in the requirement on the smudge noise width σ_y . The analysis from [7] requires that $\sigma_y \geq s_1(\mathbf{R}) \cdot \sigma_{y_0} \geq s_1(\mathbf{R}) \cdot \sqrt{m} \cdot \|\mathbf{k}\|$, and the recent analysis of [4, Lemma 3.3] inherits the same requirement, since its analysis is based on [7, Lemma 5.4]. Note that $\|\mathbf{k}\| = \Theta(\sqrt{m}\sigma_k)$ holds with overwhelming probability for $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$, thus their approach essentially requires that

$$\sigma_y \geq s_1(\mathbf{R}) \cdot \sigma_{y_0} \geq s_1(\mathbf{R}) \cdot \sqrt{m} \cdot \|\mathbf{k}\| = s_1(\mathbf{R}) \cdot \Theta(m \cdot \sigma_k).$$

In contrast, according to Lemma 6, our analysis only requires

$$\sigma_y \geq s_1(\mathbf{R}) \cdot \sqrt{m} \cdot \sigma_k,$$

thus improving the required noise width σ_y by a factor of $\sqrt{m}$.

Conceptually, our analysis is also more direct, as it relates the required Gaussian width σ_y directly to the underlying distribution parameter σ_k , rather than the norm bound $\|\mathbf{k}\|$. We attribute the improvement of the required noise width σ_y to our direct analysis of the conditional distribution of $\mathbf{k}$ given $\mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y}$.

C Def. of Threshold Decryption in Synchronous Setting

Here we present the syntax of threshold decryption scheme in the synchronous setting, which the PDec algorithm also takes a decryption set $S \subseteq [N]$ as an input. We also present the formal definition of its IND-TCCA security model, where the encryption oracle $\mathcal{O}_{\text{ENC}}$ and the partial decryption oracle $\mathcal{O}_{\text{DEC}}$ also allow the adversary to designate the set S for which PDec is invoked.

Definition 9 (Threshold Decryption in Synchronous Setting). Let $t, N \in \mathbb{N}$ with $t < N$. A (t, N) -threshold decryption scheme TD in the synchronous setting for message space $\mathcal{M}$ is defined by a tuple of PPT algorithms $\text{TD} = (\text{Setup}, \text{KGen}, \text{Enc}, \text{PDec}, \text{Comb})$.

- $\text{pp} \leftarrow \text{Setup}$: The setup algorithm outputs a public parameter pp , which serves as an implicit input of other algorithms.

- $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$: The key generation algorithm outputs a public key pk and a set of public-secret key share pairs $\{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}$, where each party $i \in [N]$ receives $(\text{pk}_i, \text{sk}_i)$.
- $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$: The encryption algorithm takes as input a public key pk and a message $\mu \in \mathcal{M}$, and outputs a ciphertext ct .
- $h_{i,S} \leftarrow \text{PDec}(\text{sk}_i, \text{ct}, S)$: The partial decryption algorithm takes as input a secret key share sk_i , a ciphertext ct and a subset $S \subseteq [N]$ with $|S| \geq t$, and outputs a partial decryption $h_{i,S}$.
- $\mu \leftarrow \text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{h_{i,S}\}_{i \in S})$: The combination algorithm takes as input a set of public key shares $\{\text{pk}_i\}_{i \in [N]}$, a ciphertext ct , a subset $S \subseteq [N]$ with $|S| \geq t$, and a set of partial decryption $\{h_{i,S}\}_{i \in S}$, and outputs a message μ .

Correctness. For $\forall \mu \in \mathcal{M}, \forall S \subseteq [N]$ with $|S| \geq t$, let $\text{pp} \leftarrow \text{Setup}$, $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$, $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$, $\{h_{i,S} \leftarrow \text{PDec}(\text{sk}_i, \text{ct}, S)\}_{i \in S}$, then $\text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{h_{i,S}\}_{i \in S}) = \mu$ holds with probability $1 - \text{negl}(\lambda)$.

Definition 10 (IND-TCCA security for Threshold Decryption in Synchronous Setting). A (t, N) -TD scheme TD in the synchronous setting is IND-TCCA secure, if for any PPT $\mathcal{A}$, it holds that $\text{Adv}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda) := |\Pr[\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda) \Rightarrow 1] - \frac{1}{2}| \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda)$ is defined in Fig. 6.

$\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda)$: $\text{Cor} := \emptyset, \text{Hon} := [N], \mathcal{Q}_{\text{Dec}}[\cdot] := \emptyset$ $\text{pp} \leftarrow \text{Setup}, (\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]}) \leftarrow \text{KGen}(\text{pp})$ $\mathcal{O} := \{\mathcal{O}_{\text{Cor}}, \mathcal{O}_{\text{Enc}}, \mathcal{O}_{\text{Dec}}\}$ $(\mu_0, \mu_1, st) \leftarrow \mathcal{A}^{\mathcal{O}}(\text{pk}, \{\text{pk}_i\}_{i \in [N]})$ $\beta \leftarrow \{0, 1\}, \text{ct}^* \leftarrow \text{Enc}(\text{pk}, \mu_\beta)$ $\beta' \leftarrow \mathcal{A}^{\mathcal{O}}(\text{ct}^*, st)$ If $\beta' = \beta$: Return 1; Else: Return 0 $\mathcal{O}_{\text{Cor}}(i)$: If $\exists S$ s.t. $ \mathcal{Q}_{\text{Dec}}[S] \cup \text{Cor} \geq t - 1$: Return $\perp$ //avoid TA $\text{Cor} := \text{Cor} \cup \{i\}, \text{Hon} := \text{Hon} \setminus \{i\}$ Return sk_i	$\mathcal{O}_{\text{Enc}}(\mu, S)$: $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu)$ For $i \in S$: $h_{i,S} \leftarrow \text{PDec}(\text{sk}_i, \text{ct}, S)$ Return $(\text{ct}, \{h_{i,S}\}_{i \in S})$ $\mathcal{O}_{\text{Dec}}(i, \text{ct}, S)$: //at most one query per distinct (i, ct, S) If $\text{ct} = \text{ct}^* \wedge \mathcal{Q}_{\text{Dec}}[S] \cup \text{Cor} \geq t - 1$: Return $\perp$ //avoid TA $h_{i,S} \leftarrow \text{PDec}(\text{sk}_i, \text{ct}, S)$ If $\text{ct} = \text{ct}^*$: $\mathcal{Q}_{\text{Dec}}[S] := \mathcal{Q}_{\text{Dec}}[S] \cup \{i\}$ Return $h_{i,S}$
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Fig. 6. The IND-TCCA security experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}$ for (t, N) -TD in the synchronous setting.

D Postponed Proofs, Analysis and Descriptions

D.1 Proof of Lemma 2

Below we first recall the core lemma established in [23, 56]. At a high level, the following lemma states that the conditional distribution of $\mathbf{k}$ given some hints $\mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y}$ turns out to be a Gaussian distribution with center $\mathbf{c}$ and a covariance parameter Σ . $\mathbf{c}$ and Σ are dependent on the public matrix $\mathbf{R}$ and the covariance parameters of $\mathbf{k}$ and $\mathbf{y}$, but independent of the secret $\mathbf{k}$.

Lemma 13 (Adapted from [56, Lemma G.2]). Let $m, \ell \in \mathbb{N}$ be integers. Let Σ_0, Σ_y be positive definite symmetric matrix over $\mathbb{R}^{m \times m}$ and $\mathbb{R}^{\ell \times \ell}$. Let $\mathbf{R} \in \mathbb{Z}^{\ell \times m}$ be an integer matrix. Denote $\Sigma := (\Sigma_0^{-1} + \mathbf{R}^\top \Sigma_y^{-1} \mathbf{R})^{-1}$. Then, the following two distributions over $\mathbb{Z}^{m+\ell}$ are statistically identical:

$$\begin{aligned} & \left\{ (\mathbf{k}, \mathbf{h}) \mid \mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_0}}, \mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sqrt{\Sigma_y}}, \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y} \right\} \\ \equiv & \left\{ (\hat{\mathbf{k}}, \mathbf{h}) \mid \begin{array}{l} \mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_0}}, \mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sqrt{\Sigma_y}}, \mathbf{h} = \mathbf{R}\mathbf{k} + \mathbf{y}, \\ \mathbf{c} = \Sigma \mathbf{R}^\top \Sigma_y^{-1} \mathbf{h}, \hat{\mathbf{k}} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}} \end{array} \right\} \end{aligned}$$

Proof of Lemma 13. For any $(\mathbf{s}, \mathbf{t}) \in \mathbb{Z}^m \times \mathbb{Z}^\ell$, the probability of $\mathbf{k} = \mathbf{s}, \mathbf{h} = \mathbf{t}$ can be written as follows:

$$\begin{aligned} & \Pr[\mathbf{k} = \mathbf{s}, \mathbf{R}\mathbf{k} + \mathbf{y} = \mathbf{t} \mid \mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_0}}, \mathbf{y} \leftarrow D_{\mathbb{Z}^\ell, \sqrt{\Sigma_y}}] \\ &= D_{\mathbb{Z}^m, \sqrt{\Sigma_0}}(\mathbf{s}) \cdot D_{\mathbb{Z}^\ell, \sqrt{\Sigma_y}}(\mathbf{t} - \mathbf{R}\mathbf{s}) \\ &\propto \exp \left[-\pi (\mathbf{s}^\top \Sigma_0^{-1} \mathbf{s} + (\mathbf{t} - \mathbf{R}\mathbf{s})^\top \Sigma_y^{-1} (\mathbf{t} - \mathbf{R}\mathbf{s})) \right] \\ &= \exp \left[-\pi (\mathbf{s}^\top \Sigma_0^{-1} \mathbf{s} + \mathbf{t}^\top \Sigma_y^{-1} \mathbf{t} - 2(\mathbf{R}\mathbf{s})^\top \Sigma_y^{-1} \mathbf{t} + (\mathbf{R}\mathbf{s})^\top \Sigma_y^{-1} \mathbf{R}\mathbf{s}) \right] \\ &= \exp \left[-\pi (\mathbf{s}^\top (\Sigma_0^{-1} + \mathbf{R}^\top \Sigma_y^{-1} \mathbf{R}) \mathbf{s} - 2\mathbf{s}^\top \mathbf{R}^\top \Sigma_y^{-1} \mathbf{t} + \mathbf{t}^\top \Sigma_y^{-1} \mathbf{t}) \right] \\ &= \exp \left[-\pi (\mathbf{s}^\top \Sigma^{-1} \mathbf{s} - 2\mathbf{s}^\top \Sigma^{-1} \mathbf{c} + \mathbf{c}^\top \Sigma^{-1} \mathbf{c} - \mathbf{c}^\top \Sigma^{-1} \mathbf{c} + \mathbf{t}^\top \Sigma_y^{-1} \mathbf{t}) \right] \\ &= \exp \left[-\pi ((\mathbf{s} - \mathbf{c})^\top \Sigma^{-1} (\mathbf{s} - \mathbf{c}) - \mathbf{c}^\top \Sigma^{-1} \mathbf{c} + \mathbf{t}^\top \Sigma_y^{-1} \mathbf{t}) \right]. \end{aligned}$$

Note that the term $-\mathbf{c}^\top \Sigma^{-1} \mathbf{c} + \mathbf{t}^\top \Sigma_y^{-1} \mathbf{t}$ is a constant that does not depend on $\mathbf{s}$ and the conditional probability $\Pr[\mathbf{k} = \mathbf{s} \mid \mathbf{R}\mathbf{k} + \mathbf{y} = \mathbf{t}]$ is proportional to $\exp \left[-\pi ((\mathbf{s} - \mathbf{c})^\top \Sigma^{-1} (\mathbf{s} - \mathbf{c})) \right]$, it implies

$$\Pr[\mathbf{k} = \mathbf{s} \mid \mathbf{R}\mathbf{k} + \mathbf{y} = \mathbf{t}] \equiv \rho_{\sqrt{\Sigma}}(\mathbf{s} - \mathbf{c}) \equiv \Pr[\hat{\mathbf{k}} = \mathbf{s} \mid \mathbf{R}\mathbf{k} + \mathbf{y} = \mathbf{t}].$$

Thus, the given two distributions are statistically identical. $\square$

Lemma 2 Let $m, n, q \in \mathbb{N}$ and $\mathcal{S} \subseteq \mathbb{Z}^m$. Let $B > 0$ be a real number such that $\|\mathbf{k}\|^2 \leq B$ for any possible $\mathbf{k} \in \mathcal{S}$. Let $\sigma_0, \sigma_1, \sigma, \delta > 0$ be real numbers satisfying $\frac{1}{(1+\delta)\sigma^2 + \delta_0^2} \geq \frac{1}{\sigma_0^2} + \frac{B}{\sigma_1^2}$ and $\sigma \geq \sqrt{1 + 1/\delta} \cdot \eta_\epsilon(\mathbb{Z}^m)$, where $\delta_0 := \sqrt{\frac{\ln(2m)+4}{\pi}}$. Let $\chi_0 := D_{\mathbb{Z}^m, \sigma_0}, \chi_1 := D_{\mathbb{Z}, \sigma_1}, \chi := D_{\mathbb{Z}^m, \sigma}$ be distributions over $\mathbb{Z}^m, \mathbb{Z}, \mathbb{Z}^m$, respectively.

Then for any PPT adversary $\mathcal{A}$ against the $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ assumption, there exists a PPT adversary $\mathcal{B}$ against the $\text{LWE}_{q,m,n,\chi}$ assumption, such that

$$\text{Adv}_{\text{para}_0, \mathcal{A}}^{\text{Mat-hLWE}}(\lambda) \leq \text{Adv}_{\text{para}_1, \mathcal{B}}^{\text{LWE}}(\lambda) + 2\epsilon,$$

where $\text{para}_0 = ((q, m, n, \chi_0), (\chi_1, \mathcal{S}))$ and $\text{para}_1 = (q, m, n, \chi)$.

Proof of Lemma 2. The proof is based on the proof of [56, Theorem G.3]. We provide an overview here and refer to [56] for more detailed analysis.

Below we show that, with an adversary $\mathcal{A}$ against $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ assumption, how to construct an algorithm $\mathcal{B}$ to break $\text{LWE}_{q,m,n,\chi}$ assumption. Given an $\text{LWE}_{q,m,n,\chi}$ instance $(\mathbf{A}, \mathbf{b}^\top) \in \mathbb{Z}^{n \times m} \times \mathbb{Z}^{1 \times m}$, $\mathcal{B}$ construct an instance of $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ as follows.

Let $\Sigma_0 := \sigma_0^2 \cdot \mathbf{I}_m$. For any $\mathbf{k} \in \mathcal{S}$, $\mathcal{B}$ sets $\Sigma := (\Sigma_0^{-1} + \sigma_1^{-2} \cdot \mathbf{k}\mathbf{k}^\top)^{-1}$. Then it samples $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$, $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$, computes $h = \mathbf{k}^\top \mathbf{e} + e'$, and samples $\mathbf{t} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma - \sigma^2 \mathbf{I}_m}, \mathbf{c}}$, where $\mathbf{c} := \sigma_1^{-2} h \cdot \Sigma \mathbf{k}$. By setting the parameter satisfying $\frac{1}{(1+\delta)\sigma^2 + \delta_0^2} \geq \frac{1}{\sigma_0^2} + \frac{B}{\sigma_1^2}$, we can efficiently sample $\mathbf{t} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma - \sigma^2 \mathbf{I}_m}, \mathbf{c}}$.

Now we claim that $\mathcal{B}$ can use the sampled elements to transform the given $\text{LWE}_{q,m,n,\chi}$ instance $(\mathbf{A}, \mathbf{b}^\top)$ to a $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$, and utilizes the reply from $\mathcal{A}$ to break LWE assumption.

The $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ instance constructed by $\mathcal{B}$ is $(\mathbf{A}, \mathbf{b}^\top + \mathbf{t}^\top, \mathbf{k}, h)$. In the case of $\mathbf{b}^\top \leftarrow \mathbb{Z}_q^m$, then the distribution $(\mathbf{A}, \mathbf{b}^\top + \mathbf{t}^\top, \mathbf{k}, h)$ is identical to the uniform case of the $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$ assumption.

Suppose $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_c^\top$ where $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e}_c \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sigma}$. Then by setting $\sigma \geq \sqrt{1 + 1/\delta} \cdot \eta_\epsilon(\mathbb{Z}^m)$, the distribution $\mathcal{D}_{\mathbb{Z}^m, \sigma} + D_{\mathbb{Z}^m, \sqrt{\Sigma - \sigma^2 \mathbf{I}_m}, \mathbf{c}}$ is within statistical distance 2ϵ of $D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$. Thus, $\mathbf{t} + \mathbf{e}_c$ is statistically indistinguishable from $\hat{\mathbf{e}} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}, \mathbf{c}}$.

Next, we consider a special case of Lemma 13 to analyze the distribution of $\mathbf{e}$ conditioned on h . Precisely, we take $m = n, \ell = 1, \Sigma_0 := \sigma_0^2 \cdot \mathbf{I}_m, \Sigma_y = \sigma_1^2$ and obtain the following Corollary 3.

Corollary 3 (Special case of Lemma 13). *Let $n \in \mathbb{N}$ be integer. Let Σ_0 be positive definite symmetric matrix over $\mathbb{R}^{n \times n}$ and σ_1 be a positive real number. Let $\mathbf{k} \in \mathbb{Z}^n$ be an integer vector. Denote $\Sigma := (\Sigma_0^{-1} + \sigma_1^{-2} \cdot \mathbf{k}\mathbf{k}^\top)^{-1}$. Then, the following two distributions over $\mathbb{Z}^{n+1}$ are statistically identical:*

$$\begin{aligned} & \{(\mathbf{e}, h) \mid \mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sqrt{\Sigma_0}}, e' \leftarrow D_{\mathbb{Z}, \sigma_1}, h = \mathbf{k}^\top \mathbf{e} + e'\} \\ & \approx \left\{(\hat{\mathbf{e}}, h) \left| \begin{array}{l} \mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sqrt{\Sigma_0}}, e' \leftarrow D_{\mathbb{Z}, \sigma_1}, h = \mathbf{k}^\top \mathbf{e} + e', \\ \mathbf{c} = \sigma_1^{-2} h \cdot \Sigma \mathbf{k}, \hat{\mathbf{e}} \leftarrow D_{\mathbb{Z}^n, \sqrt{\Sigma}, \mathbf{c}} \end{array} \right. \right\} \end{aligned}$$

By Corollary 3, the distribution of $\mathbf{e}$ conditioned on $h = \mathbf{k}^\top \mathbf{e} + e'$ is statistically identical to $\hat{\mathbf{e}} \leftarrow D_{\mathbb{Z}^n, \sqrt{\Sigma}, \mathbf{c}}$. Consequently, we have

$$\begin{aligned} (\mathbf{A}, \mathbf{b}^\top + \mathbf{t}^\top, \mathbf{k}, h) &= (\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}_c^\top + \mathbf{t}^\top, \mathbf{k}, h) \\ &\approx_{2\epsilon} (\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \hat{\mathbf{e}}, \mathbf{k}, h) \approx (\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}, \mathbf{k}, h). \end{aligned}$$

This shows in the case of $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_c^\top$, $\mathcal{B}$'s construction is statically close to an instance of $\text{Mat-hLWE}_{q,m,n,\chi_0}^{\chi_1, \mathcal{S}}$. Combining the above two cases, we complete the proof of Lemma 2. $\square$

D.2 Proof of Claim 1

Claim 1. $|\Pr[\mathbf{G}_2 = 1] - \Pr[\mathbf{G}_3 = 1]| \leq \text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda)$, where $\text{para} := ((q, m, n, D_{\mathbb{Z}^m, \sigma_0}), (D_{\mathbb{Z}, \sigma_1}, [-CB_k, CB_k]^m))$ with $C := \binom{N}{N-t+1}$.

Proof. To bound the difference between G_3 and G_2 , we construct an adversary $\mathcal{B}_2$ against the Matrix-Hint LWE assumption. $\mathcal{B}_2$ invokes $(\mathbf{k}, (\mathbf{sk}_1, \dots, \mathbf{sk}_N)) \leftarrow_{\$} \text{RSS}(N, t, D_{\mathbb{Z}^m, \sigma_k})$ and sets $\mathbf{pk} := \mathbf{A}\mathbf{k}$, $\mathbf{pk}_i := \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \mathbf{sk}_i}$ for each $i \in [N]$. Then $\mathcal{B}_2$ is given $(\mathbf{A}, \mathbf{u}^\top, -\mathbf{k}, e' - \mathbf{e}^\top \mathbf{k})$, and aims to decide whether $\mathbf{u}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ or $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^m$, where $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$. $\mathcal{B}_2$ sends $(\mathbf{pp}, \mathbf{pk}, \{\mathbf{pk}_i\}_{i \in [N]})$ and simulates G_2/G_3 for $\mathcal{A}$ as follows:

- $\mathcal{B}_2$ can answer $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{DEC}}$ queries on its own with the help of $\mathbf{pk}$, $\{(\mathbf{pk}_i, \mathbf{sk}_i)\}_{i \in [N]}$.
- On receiving (μ_0, μ_1) , $\mathcal{B}_2$ randomly chooses $\beta \leftarrow_{\$} \{0, 1\}$ and generates the challenge ciphertext $\mathbf{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ for $\mathcal{A}$ as follows:
 - Set $\mathbf{c}^* := \mathbf{u}$ using the $\mathbf{u}$ contained in $\mathcal{B}_2$'s input.
 - Compute $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta \in \mathbb{Z}_q$ using the $e' - \mathbf{e}^\top \mathbf{k}$ contained in $\mathcal{B}_2$'s input.
 - Compute $\tau^* := H(\mathbf{c}^*, d^*)$, and invoke $\pi^* \leftarrow_{\$} \text{Sim}(\text{crs}, \tau^*, \mathbf{c}^*)$.
- Finally, $\mathcal{B}_2$ receives a bit β' from $\mathcal{A}$ and returns $\beta = \beta'$.

Clearly, if $\mathbf{u}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, $\mathcal{B}_2$ simulates G_2 for $\mathcal{A}$; if $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^m$, $\mathcal{B}_2$ simulates G_3 for $\mathcal{A}$. Thus, we have $|\Pr[G_2 = 1] - \Pr[G_3 = 1]| \leq \text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda)$. ■

D.3 Full Description of G_0 in Theorem 1

Game G_0 : This is the original IND-TCCA experiment $\text{Exp}_{\text{TD1}, \mathcal{A}}^{\text{CCA}}$ (cf. Fig. 2). By definition, we have $\text{Adv}_{\text{TD1}, \mathcal{A}}^{\text{CCA}}(\lambda) = |\Pr[G_0 = 1] - \frac{1}{2}|$.

Let $\mathbf{pp} = (\mathbf{A}, \text{crs}, \overline{\text{crs}}, \mathbf{H})$. The challenger invokes $(\mathbf{k}, (\mathbf{sk}_1, \dots, \mathbf{sk}_N)) \leftarrow_{\$} \text{RSS}(N, t, D_{\mathbb{Z}^m, \sigma_k})$, sets $\mathbf{pk} := \mathbf{A}\mathbf{k}$, $\mathbf{pk}_i := \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \mathbf{sk}_i}$ for each $i \in [N]$, and returns $(\mathbf{pp}, \mathbf{pk}, \{\mathbf{pk}_i\}_{i \in [N]})$ to $\mathcal{A}$. The challenger initializes $\text{Cor} := \emptyset$, $\text{Hon} := [N]$, $S^* := \emptyset$. Then it answers $\mathcal{A}$'s $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$, $\mathcal{O}_{\text{DEC}}$ queries adaptively and generates the challenge ciphertext $\mathbf{ct}^*$ as follows.

- For corruption query $\mathcal{O}_{\text{COR}}(i)$, the challenger adds i to Cor and returns $\mathbf{sk}_i$.
- For encryption query $\mathcal{O}_{\text{ENC}}(\mu)$, the challenger generates $\mathbf{ct} = (\mathbf{c}, d, \pi)$ as follows:
 - Sample $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d := \mathbf{s}^\top \mathbf{pk} + e' + p \cdot \mu \in \mathbb{Z}_M$.
 - Compute $\tau := H(\mathbf{c}, d)$, and generate a proof $\pi \leftarrow_{\$} \text{Prove}(\text{crs}, \tau, \mathbf{c}, (\mathbf{s}, \mathbf{e}))$.

Then, for each $i \in [N]$, the challenger invokes $\text{PDec}(\mathbf{sk}_i, \mathbf{ct})$ to obtain (h_i, π_i) as follows.

- For each $\mathbf{k}_I \in \mathbf{sk}_i$, compute $h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}$ with $y_{i,I} \leftarrow D_{\mathbb{Z}, \sigma_y}$, and generate a proof $\pi_{i,I} \leftarrow \overline{\text{Prove}}(\overline{\text{crs}}, (\mathbf{pk}_{i,I}, h_{i,I}, \mathbf{c}), (\mathbf{k}_I, y_{i,I}))$.
- Set $h_i := \{h_{i,I}\}_{\mathbf{k}_I \in \mathbf{sk}_i}$ and $\pi_i := \{\pi_{i,I}\}_{\mathbf{k}_I \in \mathbf{sk}_i}$.

Finally, the challenger returns $(\mathbf{ct}, \{h_i, \pi_i\}_{i \in [N]})$ to $\mathcal{A}$.

- For partial decryption query $\mathcal{O}_{\text{DEC}}(i, \mathbf{ct} = (\mathbf{c}, d, \pi))$, the challenger adds i to S^* if $\mathbf{ct} = \mathbf{ct}^*$ and computes partial decryption as follows.
 - If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ with $\tau := H(\mathbf{c}, d)$, the challenger returns $\perp$.

- For each $\mathbf{k}_I \in \text{sk}_i$, the challenger computes $h_{i,I} = \mathbf{c}^\top \mathbf{k}_I + y_{i,I}$ with $y_{i,I} \leftarrow D_{\mathbb{Z}, \sigma_y}$, and generates a proof $\pi_{i,I} \leftarrow \overline{\text{Prove}}(\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}), (\mathbf{k}_I, y_{i,I}))$. Then it returns $h_i := \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ and $\pi_i := \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ to $\mathcal{A}$.
- On receiving (μ_0, μ_1) , the challenger chooses $\beta \leftarrow_{\$} \{0, 1\}$ and generates the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as an encryption of μ_β as follows:
 - Sample $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^{*\top} := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d^* := \mathbf{s}^\top \mathbf{pk} + e' + p \cdot \mu_\beta \in \mathbb{Z}_q$.
 - Compute $\tau^* := H(\mathbf{c}^*, d^*)$ and generate a proof $\pi^* \leftarrow_{\$} \text{Prove}(\text{crs}, \tau^*, \mathbf{c}^*, (\mathbf{s}, \mathbf{e}))$.

Recall that to avoid trivial attacks, we require $|S^* \cup \text{Cor}| \leq t - 1$.

D.4 Proof of Theorem 2

Theorem 2 Assume that (i) Π has adaptive soundness, (ii) $\overline{\Pi}$ has adaptive soundness, (iii) $2CB^* \leq p/2$, then the proposed scheme TD_1 in Fig. 3 is RB secure.

Proof of Theorem 2. Let $\text{crs} = \mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and $\{\text{pk}_i\}_{i \in [N]}$ be the public keys. Let $(\text{ct} = (\mathbf{c}, d, \pi), S, \{h_i, \pi_i\}_{i \in S}, S', \{h_i, \pi_i\}_{i \in S'})$ be the output of $\mathcal{A}$. Let μ, μ' be the outputs of Comb w.r.t. $\mathcal{A}$'s output, i.e., $\mu \leftarrow \text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{(h_i, \pi_i)\}_{i \in S})$ and $\mu' \leftarrow \text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S', \{(h_i, \pi_i)\}_{i \in S'})$. $\mathcal{A}$ wins the game if $\mu, \mu' \neq \perp$ and $\mu \neq \mu'$. In our scheme, when combining, we will invoke $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi)$ for $\tau = H(\mathbf{c}, d)$ to check the validity of ct , and invoke $\text{PVer}(\text{pk}_i, \text{ct}, h_i, \pi_i)$ to check the correctness of each h_i .

We claim that $\mathcal{A}$ wins the game only in the case of $\mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, i.e., $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ for some $\mathbf{s} \in \mathbb{Z}_q^n$ and $\mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m$. If $\mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$ but $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 1$, then we can construct an algorithm $\mathcal{B}_1$ to break the adaptive soundness of Π . In the case of $\mathbf{c} \in \tilde{\mathcal{L}}_{\rho}^{(\text{LWE})}$, we claim that $\mathcal{A}$ wins the game only in the case of $(\text{pk}_{i,I}, h_{i,I}, \mathbf{c}) \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$ for each sk_i and $\mathbf{k}_I \in \text{sk}_i$. If $(\text{pk}_{i,I}, h_{i,I}, \mathbf{c}) \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$ but $\text{PVer}(\text{pk}_i, \text{ct}, h_i, \pi_i) = 1$, then we can construct an algorithm $\mathcal{B}_2$ to break the adaptive soundness of $\overline{\Pi}$.

In the case of $\mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$ and $(\text{pk}_{i,I}, h_{i,I}, \mathbf{c}) \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$, for each sk_i and $\mathbf{k}_I \in \text{sk}_i$, it holds that $h_{i,I} = \mathbf{s}^\top \text{pk}_{i,I} + y_{i,I}$ for $\mathbf{s} \in \mathbb{Z}_q^n$ and $y_{i,I} \in [-B^*, B^*]$. Let $h := \sum_{i \in T} \sum_{i \in w_i} h_{i,I}$ and $h' := \sum_{i \in T'} \sum_{i \in w'_i} h'_{i,I}$. By the correctness of RSS (cf. Subsect. 2.2), it holds that $\sum_{i \in T} \sum_{i \in w_i} \text{pk}_{i,I} = \mathbf{pk}$ and $\sum_{i \in T'} \sum_{i \in w'_i} \text{pk}_{i,I} = \mathbf{pk}$. Consequently, we have $h - h' = \sum_{i \in T} \sum_{i \in w_i} y_{i,I} - \sum_{i \in T'} \sum_{i \in w'_i} y'_{i,I} \leq 2CB^* \leq p/2$ and thus $d - h$ and $d - h'$ has a difference at most $p/2$. Finally, we have $\mu = \lceil (d - h)/p \rceil = \lceil (d - h')/p \rceil = \mu'$. Taking all things together, Theorem 2 follows. $\square$

D.5 Correctness Analysis of TD_2

Correctness. The correctness of TD_2 follows from the completeness of Π , the correctness of Shamir secret sharing (cf. Subsect. 2.2) and our parameter selection. Firstly, it is clear that Vrfy algorithms in PDec , Comb will output 1 except with negligible probability by the completeness of Π .

Next, we show that the **Comb** algorithm can recover the message μ correctly except with negligible probability. To this end, we first note that for any ct and S , it holds that $\sum_{i \in S} \Delta_{i,S} = 0$ where $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, \text{ct}, S)$. This is because, according to the definition of **ZeroShare** in Fig. 4, we have

$$\begin{aligned} \sum_{i \in S} \Delta_{i,S} &= \sum_{i \in S} \text{ZeroShare}(\text{sk}_i, \text{ct}, S) \\ &= \sum_{i \in S} \sum_{j \in S \setminus \{i\}} (\text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S)) - \text{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))) \\ &= \sum_{i,j \in S, i \neq j} \text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S)) - \sum_{i,j \in S, i \neq j} \text{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S)) = 0. \end{aligned} \quad (23)$$

For any message $\mu \in \mathbb{Z}_M$, the encryption algorithm $\text{Enc}(\text{pk}, \mu)$ generates ciphertext $\text{ct} = (\mathbf{c}, d, \pi)$ by computing $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, $d := \mathbf{s}^\top \mathbf{A} \mathbf{k} + e' + p \cdot \mu$ with $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$, $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$. Then for $h_{i,S}$ generated by the partial decryption algorithm $\text{PDec}(\text{sk}_i, \text{ct}, S)$, we have $h_{i,S} := \mathbf{c}^\top (\lambda_{i,S} \cdot \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$, where $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$. Thus, for any $S \subseteq [N]$ with $|S| \geq t$, in the combination algorithm $\text{Comb}(\text{ct}, S, \{h_{i,S}\}_{i \in S})$, we have

$$h := \sum_{i \in S} h_{i,S} = \mathbf{c}^\top (\sum_{i \in S} \lambda_{i,S} \mathbf{k}_i) + \sum_{i \in S} y_{i,S} + \sum_{i \in S} \Delta_{i,S} = \mathbf{c}^\top \mathbf{k} + \sum_{i \in S} y_{i,S},$$

where the last equality holds due to $\sum_{i \in S} \Delta_{i,S} = 0$ and $\sum_{i \in S} \lambda_{i,S} \mathbf{k}_i = \mathbf{k}$ by the correctness of Shamir secret sharing (cf. Subsect. 2.2). Therefore,

$$d - h = \mathbf{s}^\top \mathbf{A} \mathbf{k} + e' + p \cdot \mu - \mathbf{c}^\top \mathbf{k} - \sum_{i \in S} y_{i,S} = p \cdot \mu + e' - \mathbf{e}^\top \mathbf{k} - \sum_{i \in S} y_{i,S}.$$

By our parameter setting and the tail bound (Lemma 5), the noise term $e' - \mathbf{e}^\top \mathbf{k} - \sum_{i \in S} y_{i,S}$ is bounded by $B_1 + mB_0B_k + NB_y \leq p/2$ except with negligible probability. Consequently, $\lceil (d - h)/p \rceil$ correctly recovers μ .

D.6 Full Description of $\mathbf{G}_0$ in Theorem 3

Game $\mathbf{G}_0$: This is the IND-TCCA experiment $\text{Exp}_{\text{TD}, \mathcal{A}}^{\text{CCA}}$ for threshold decryption scheme in the synchronous setting (cf. Fig. 6 in Appendix C). By definition, we have $\text{Adv}_{\text{TD}, \mathcal{A}}^{\text{CCA}}(\lambda) = |\Pr[\mathbf{G}_0 = 1] - \frac{1}{2}|$.

Let $\text{pp} = (\mathbf{A}, \text{crs}, \text{H}, \text{H}_{\text{mask}})$. The challenger samples $\mathbf{k} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$, invokes $(\mathbf{k}_1, \dots, \mathbf{k}_N) \leftarrow \text{SShare}(N, t, \mathbf{k})$, and sets $\text{pk} := \mathbf{A} \mathbf{k}$, $\text{pk}_i := \mathbf{A} \mathbf{k}_i$, $\text{sk}_i := (\mathbf{k}_i, \{\text{seed}_{i,j}, \text{seed}_{j,i}\}_{j \in [N]})$ for each $i \in [N]$. The challenger sends $(\text{pp}, \text{pk}, \{\text{pk}_i\}_{i \in [N]})$ to $\mathcal{A}$ and initializes $\text{Cor} := \emptyset$, $\text{Hon} := [N]$, $\mathcal{Q}_{\text{DEC}}[\cdot] := \emptyset$, $\mathcal{Q}_{\text{MASK}}[\cdot] := \emptyset$. Then it answers $\mathcal{A}$'s H_{mask} , $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$, $\mathcal{O}_{\text{DEC}}$ queries adaptively and generates the challenge ciphertext ct^* as follows.

- For random oracle query $\text{H}_{\text{mask}}(\text{input})$, if $\mathcal{Q}_{\text{MASK}}[\text{input}] = \perp$, the challenger samples $\mathcal{Q}_{\text{MASK}}[\text{input}] \leftarrow \mathbb{Z}_q$ uniformly. Then it returns $\mathcal{Q}_{\text{MASK}}[\text{input}]$ to $\mathcal{A}$.
- For corruption query $\mathcal{O}_{\text{COR}}(i)$, the challenger adds i to Cor and returns sk_i .
- For encryption query $\mathcal{O}_{\text{ENC}}(\mu, S)$, the challenger generates $\text{ct} = (\mathbf{c}, d, \pi)$ as follows.
 - Sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d := \mathbf{s}^\top \text{pk} + e' + p \cdot \mu \in \mathbb{Z}_q$.

- Compute $\tau := H(\mathbf{c}, d)$, and generate a proof $\pi \leftarrow \text{Prove}(\text{crs}, \tau, \mathbf{c}, (\mathbf{s}, \mathbf{e}))$. Then, for each $i \in S$, the challenger invokes $\text{PDec}(\text{sk}_i, \text{ct}, S)$ to obtain $h_{i,S}$, by computing $h_{i,S} := \mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$ with $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$ and $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, \text{ct}, S)$. Finally, the challenger returns $(\text{ct}, \{h_{i,S}\}_{i \in S})$ to $\mathcal{A}$.
- For partial decryption queries $\mathcal{O}_{\text{DEC}}(i, \text{ct} = (\mathbf{c}, d, \pi), S)$, the challenger adds i to $\mathcal{Q}_{\text{DEC}}[S]$ if $\text{ct} = \text{ct}^*$ and computes partial decryption $\text{PDec}(\text{sk}_i, \text{ct}, S)$:
 - If $\text{Vrfy}(\text{crs}, \tau, \mathbf{c}, \pi) = 0$ with $\tau := H(\mathbf{c}, d)$, the challenger returns $\perp$.
 - The challenger computes $h_{i,S} := \mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$ with $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$ and $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, \text{ct}, S)$, and returns $h_{i,S}$ to $\mathcal{A}$.
Without loss of generality, we may assume $i \in \text{Hon}$, since otherwise if $i \in \text{Cor}$, $\mathcal{A}$ already obtains sk_i and can invoke $\text{PDec}(\text{sk}_i, \text{ct}, S)$ itself.
- On receiving (μ_0, μ_1) , the challenger chooses $\beta \leftarrow \{0, 1\}$ and generates the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as an encryption of μ_β as follows:
 - Sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^{*\top} := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d^* := \mathbf{s}^\top \mathbf{pk} + e' + p \cdot \mu \in \mathbb{Z}_q$.
 - Compute $\tau^* := H(\mathbf{c}^*, d^*)$ and generate $\pi^* \leftarrow \text{Prove}(\text{crs}, \tau^*, \mathbf{c}^*, (\mathbf{s}, \mathbf{e}))$.

Recall that to avoid trivial attacks, we require $|\mathcal{Q}_{\text{DEC}}[S] \cup \text{Cor}| \leq t - 1$ for any $S \subseteq [N]$.

D.7 Full Descriptions and Analysis of $\mathbf{G}_7$ - $\mathbf{G}_{12}$ in Theorem 3

Game $\mathbf{G}_7$: In this game, the challenger introduces a set $\mathcal{Q}_{\text{CT}}$ as defined below:

- $\mathcal{Q}_{\text{CT}}$ records the ciphertexts ct generated in $\mathcal{O}_{\text{ENC}}$ thus far, the challenge ciphertext ct^* if it has been generated, and the ciphertexts ct that $\mathcal{A}$ has queried $\mathcal{O}_{\text{DEC}}(\cdot, \text{ct}, \cdot)$ or $\text{H}_{\text{mask}}(\cdot, (\text{ct}, \cdot))$ thus far.

Now when generating a ciphertext ct in $\mathcal{O}_{\text{ENC}}$ and when generating the challenge ciphertext ct^* , the challenger adds a rejection rule:

- If $\text{ct} \in \mathcal{Q}_{\text{CT}}$ or $\text{ct}^* \in \mathcal{Q}_{\text{CT}}$, return $\perp$ directly.

Let Q denote the number of $\mathcal{O}_{\text{ENC}}, \mathcal{O}_{\text{DEC}}, \text{H}_{\text{mask}}$ queries made by $\mathcal{A}$. So $|\mathcal{Q}_{\text{CT}}| \leq Q+1$. Since $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ in ct has min-entropy $\geq n \log q$ due to the randomness of $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, and $\mathbf{c}^{*\top} \leftarrow \mathbb{Z}_q^{1 \times m}$ in ct^* has min-entropy $m \log q \geq n \log q$, the probability that the challenger returns $\perp$ is bounded by $(Q+1)^2/q^n$, which is negligible in λ . Thus, we have $|\Pr[\mathbf{G}_6 = 1] - \Pr[\mathbf{G}_7 = 1]| \leq \text{negl}(\lambda)$.

With the change in $\mathbf{G}_7$, we ensure that for honestly generated ciphertexts ct and ct^* , none of the value $\text{H}_{\text{mask}}(\cdot, (\text{ct}/\text{ct}^*, \cdot))$ have been previously fixed, thereby preparing for the following games.

Game $\mathbf{G}_8$: In this game (and the subsequent games), for $S \subseteq [N]$, we denote the honest (resp. corrupted) users in S by $\text{HS} := S \cap \text{Hon}$ (resp. $\text{CS} := S \cap \text{Cor}$). Moreover, in this game, the challenger introduces a set $\mathcal{Q}_{\text{Dec}}$ and two tables $\text{UnDecHS}, \text{Mask}$ as defined below:

- QDec records the tuples (ct, S) for which $\mathcal{A}$ has queried $\mathcal{O}_{\text{DEC}}(\cdot, ct, S)$.
- $\text{UnDecHS}[(ct, S)]$ for $(ct, S) \in \text{QDec}$ stores the users $i \in \text{HS}$ that $\mathcal{A}$ does not issue $\mathcal{O}_{\text{DEC}}(i, ct, S)$. We also denote $\widetilde{\text{HS}} := \text{UnDecHS}[(ct, S)]$ when (ct, S) is clear from the context.
- $\text{Mask}[(ct, S), i]$ stores the mask $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, ct, S)$.

Now the challenger answers $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ queries according to Fig. 7, where we highlight the changes in gray boxes. Note that the change of G_8 does not affect the view of $\mathcal{A}$, so we have $\Pr[G_7 = 1] = \Pr[G_8 = 1]$.

G₈: $\mathcal{O}_{\text{COR}}(i)$: If $\exists S$ s.t. $\mathcal{Q}_{\text{DEC}}[S] \cup \text{Cor} \geq t - 1$: Return $\perp$ For $(ct, S) \in \text{QDec}$ s.t. $(i \in S) \wedge (\text{Mask}[(ct, S), i] = \perp)$: $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, ct, S)$ $\text{Mask}[(ct, S), i] := \Delta_{i,S}$ $\text{UnDecHS}[(ct, S)] := \text{UnDecHS}[(ct, S)] \setminus \{i\}$ Cor := Cor $\cup \{i\}$, Hon := Hon $\setminus \{i\}$ Return sk_i $\text{CheckDec}(i, ct = (c, d, \pi), S)$: //internal alg. for $\mathcal{O}_{\text{DEC}}$ If any of the following condition holds, return 0: (1) $ct = ct^* \wedge \mathcal{Q}_{\text{DEC}}[S] \cup \text{Cor} \geq t - 1$; (2) $i \in \text{Cor}$; (3) $ct \neq ct^* \wedge (\tau, c, \pi) = (\tau^*, c^*, \pi^*)$ with $\tau := H(c, d)$;	(4) $ct \neq ct^* \wedge c \notin \widetilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$; (5) $\text{Vrfy}(\text{crs}, \tau, c, \pi) = 0$ for $\tau := H(c, d)$. Else: Return 1 $\mathcal{O}_{\text{DEC}}(i, ct = (c, d, \pi), S)$: If $\text{CheckDec}(i, ct, S) = 0$: Return $\perp$ If $(ct, S) \notin \text{QDec}$: $\text{QDec} := \text{QDec} \cup \{(ct, S)\}$ $\text{UnDecHS}[(ct, S)] := \text{HS}$ $\Delta_{i,S} := \text{ZeroShare}(\text{sk}_i, ct, S)$ $y_{i,S} \leftarrow D_{Z, \sigma_y}$ $h_{i,S} := c^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$ $\text{Mask}[(ct, S), i] := \Delta_{i,S}$ $\text{UnDecHS}[(ct, S)] := \text{UnDecHS}[(ct, S)] \setminus \{i\}$ If $ct = ct^*$: $\mathcal{Q}_{\text{DEC}}[S] := \mathcal{Q}_{\text{DEC}} \cup \{i\}$ Return $h_{i,S}$
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Fig. 7. The game G_8 with changes highlighted in gray boxes.

Game G_9 : It is the same as G_8 , except that when computing $\Delta_{i,S}$ in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$, the challenge computes $\Delta_{i,S}$ directly (instead of invoking ZeroShare):

- Compute $\mathbf{m}_{i,j} := H_{\text{mask}}(\text{seed}_{i,j}, (ct, S))$, $\mathbf{m}_{j,i} := H_{\text{mask}}(\text{seed}_{j,i}, (ct, S))$ for $j \in S \setminus \{i\}$, then set $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i})$.

The change is conceptual and we have $\Pr[G_8 = 1] = \Pr[G_9 = 1]$.

Game G_{10} : It is the same as G_9 , except for the reply of H_{mask} and the computation of $\Delta_{i,S}$ in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$. More precisely, the challenger answers H_{mask} , $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ queries according to Fig. 8.

Claim 5. $|\Pr[G_9 = 1] - \Pr[G_{10} = 1]| \leq \text{negl}(\lambda)$.

Proof. Let **Abort** denote the event that the hash oracle $H_{\text{mask}}(\cdot)$ outputs $\perp$. We first analyze the probability $\Pr[\text{Abort}]$. For each H_{mask} query made by $\mathcal{A}$, if $i, j \in \text{Hon}$, the value of $\text{rand}_{i,j} \in \{0, 1\}^\lambda$ is hidden from $\mathcal{A}$, so $\Pr[\text{rand} = \text{rand}_{i,j}] = \frac{1}{2^\lambda}$.

$\mathbf{G}_{10}$: $\mathbf{H}_{\text{mask}}(\text{input})$: //for $\mathcal{A}$'s queries Parse $\text{input} = (\text{seed}, (\text{ct}, S))$ Parse $\text{seed} = (i, j, \text{rand})$ If $i, j \in \text{Hon} \wedge \text{rand} = \text{rand}_{i,j}$: Return $\perp$ If $\mathcal{Q}_{\text{MASK}}[\text{input}] = \perp$: $\mathbf{m} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{input}] := \mathbf{m}$ Return $\mathcal{Q}_{\text{MASK}}[\text{input}]$ $\mathcal{O}_{\text{COR}}(i)$ and $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$: //replace any //computation of $\Delta_{i,S}$ with the following: For $j \in \text{CS}$: $\mathbf{m}_{i,j} := \mathbf{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$	$\mathbf{m}_{j,i} := \mathbf{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))$ //note that $(\text{ct}, S) \in \text{QDec}$ when computing $\Delta_{i,S}$ $\widetilde{\text{HS}} := \text{UnDecHS}[(\text{ct}, S)]$ For $j \in \text{HS} \setminus \widetilde{\text{HS}}$: $\mathbf{m}_{i,j} := \mathcal{Q}_{\text{MASK}}[\text{seed}_{i,j}, (\text{ct}, S)]$ $\mathbf{m}_{j,i} := \mathcal{Q}_{\text{MASK}}[\text{seed}_{j,i}, (\text{ct}, S)]$ For $j \in \widetilde{\text{HS}} \setminus \{i\}$: $\mathbf{m}_{i,j} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{seed}_{i,j}, (\text{ct}, S)] := \mathbf{m}_{i,j}$ $\mathbf{m}_{j,i} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{seed}_{j,i}, (\text{ct}, S)] := \mathbf{m}_{j,i}$ $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i}) \in \mathbb{Z}_q$ $\text{Mask}[(\text{ct}, S), i] := \Delta_{i,S}$
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Fig. 8. The game $\mathbf{G}_{10}$ with changes highlighted in gray boxes .

By the union bound, we have $\Pr[\text{Abort}] \leq \frac{Q_{\text{mask}}}{2^\lambda} = \text{negl}(\lambda)$, where Q_{mask} is the number of $\mathbf{H}_{\text{mask}}$ queries made by $\mathcal{A}$.

We further argue that for any $j \in \text{HS} \setminus \widetilde{\text{HS}}$, the hash values $\mathcal{Q}_{\text{MASK}}[\text{seed}_{i,j}, (\text{ct}, S)]$ and $\mathcal{Q}_{\text{MASK}}[\text{seed}_{j,i}, (\text{ct}, S)]$ have already been defined, whereas for any $j \in \widetilde{\text{HS}} \setminus \{i\}$, the corresponding hash values remain undefined.

- For any $j \in \text{HS} \setminus \widetilde{\text{HS}}$, by the definition of HS and $\widetilde{\text{HS}}$, $\mathcal{A}$ has previously queried $\mathcal{O}_{\text{DEC}}(j, \text{ct}, S)$, during which the values $\mathbf{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ and $\mathbf{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))$ have already been defined.
- For any $j \in \widetilde{\text{HS}} \setminus \{i\}$, by the definition of $\widetilde{\text{HS}}$, $\mathcal{A}$ has not queried $\mathcal{O}_{\text{DEC}}(j, \text{ct}, S)$ so far. Moreover, since $i, j \in \text{Hon}$ at this point, $\mathcal{A}$ can hardly query $\mathbf{H}_{\text{mask}}$ with honest $\text{seed}_{i,j}$ or $\text{seed}_{j,i}$. Hence, the values $\mathbf{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ and $\mathbf{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))$ have not been defined prior to this point.

Putting these observations together, Claim 5 follows. $\blacksquare$

Game $\mathbf{G}_{11}$: It is the same as $\mathbf{G}_{10}$, except that the challenger randomly samples all masks except for the last honest user, and the last mask is computed deterministically by other masks in $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$. Additionally, the oracle $\mathbf{H}_{\text{mask}}$ is not programmed accordingly until the corresponding user is corrupted. More precisely, the challenger answers oracles $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ according to Fig. 9.

Claim 6. $\Pr[\mathbf{G}_{10} = 1] = \Pr[\mathbf{G}_{11} = 1]$.

Proof. We need to show that the views of $\mathcal{A}$ in $\mathbf{G}_{10}$ and $\mathbf{G}_{11}$ are identical. Firstly, we prove that the values of $\Delta_{i,S}$ are identically distributed in both games.

- If i is not the last honest user, i.e., $\widetilde{\text{HS}} \neq \{i\}$ when querying $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$, the values $\mathbf{m}_{i,j}$ and $\mathbf{m}_{j,i}$ for $j \in \widetilde{\text{HS}} \setminus \{i\}$ are sampled at random. Thus, $\Delta_{i,S} := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i})$ in $\mathbf{G}_{10}$ is uniformly random, as it is in $\mathbf{G}_{11}$.
- If $\widetilde{\text{HS}} = \{i\}$ when querying $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$, the values $(\mathbf{m}_{i,j}, \mathbf{m}_{j,i})$ for $j \in \text{HS} \setminus \{i\}$ have already been stored in $\mathcal{Q}_{\text{MASK}}$ in $\mathbf{G}_{10}$, and hence $\text{Mask}[(\text{ct}, S), j]$ is determined. Note that $\Delta_{i,S}$ should satisfy the equation $\sum_{j \in S} \Delta_{j,S} = 0$ in both

G_{11} : <u>ProgramZeroShare</u> ((ct, S), i, Δ^* , CS, HS): Assert $\Delta^* \neq \perp \wedge \text{HS} > 1$ For $j \in \text{CS}$: $\mathbf{m}_{i,j} := \text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ $\mathbf{m}_{j,i} := \text{H}_{\text{mask}}(\text{seed}_{j,i}, (\text{ct}, S))$ Pick an arbitrary user $a \in \text{HS} \setminus \{i\}$ For $j \in \text{HS} \setminus \{i, a\}$: $\mathbf{m}_{i,j} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{seed}_{i,j}, (\text{ct}, S)] := \mathbf{m}_{i,j}$ $\mathbf{m}_{j,i} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{seed}_{j,i}, (\text{ct}, S)] := \mathbf{m}_{j,i}$ $\mathbf{m}_{i,a} \leftarrow \mathbb{Z}_q$, $\mathcal{Q}_{\text{MASK}}[\text{seed}_{i,a}, (\text{ct}, S)] := \mathbf{m}_{i,a}$ $\mathcal{Q}_{\text{MASK}}[\text{seed}_{a,i}, (\text{ct}, S)] :=$ $\Delta^* - \sum_{j \in S \setminus \{i,a\}} (\mathbf{m}_{j,i} - \mathbf{m}_{i,j}) + \mathbf{m}_{i,a}$	$\mathcal{O}_{\text{COR}}(i)$ and $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$: //Replace any computation of $\Delta_{i,S}$ with the following: $\widetilde{\text{HS}} := \text{UnDecHS}[(\text{ct}, S)]$ If $\widetilde{\text{HS}} \neq \{i\}$: $\Delta_{i,S} \leftarrow \mathbb{Z}_q$ Else: For $j \in \text{CS}$: $\Delta_{j,S} := \text{ZeroShare}(\text{sk}_j, \text{ct}, S)$ $\Delta_{i,S} := - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \text{Mask}[(\text{ct}, S), j]$ $\text{Mask}[(\text{ct}, S), i] := \Delta_{i,S}$ $\mathcal{O}_{\text{COR}}(i)$: //Add the following step before update Hon and Cor: For all $(\text{ct}, S) \in \text{QDec}$ s.t. $\text{Mask}[(\text{ct}, S), i] \neq \perp$: <u>ProgramZeroShare</u> ((ct, S), i, $\text{Mask}[(\text{ct}, S), i]$, CS, HS)
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Fig. 9. The game G_{11} with changes highlighted in gray boxes .

games (cf. (23)). We have the following equation in both G_{10} and G_{11} .

$$\Delta_{i,S} = - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S} = - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \text{Mask}[(\text{ct}, S), j].$$

Secondly, we argue that the output of H_{mask} in G_{11} and G_{10} are identically distributed.

- If $i, j \in \text{Hon} \wedge \text{seed} = \text{seed}_{i,j}$, the challenger aborts in H_{mask} in both games.
- In G_{10} $\Delta_{i,S}$ is generated via H_{mask} . In contrast, in G_{11} , $\Delta_{i,S}$ can also be viewed as $\Delta_i := \sum_{j \in S \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i}) \in \mathbb{Z}_q$ via the algorithm ProgramZeroShare, since we reprogram $\text{H}_{\text{mask}}(\text{seed}_{i,j}, (\text{ct}, S))$ only in the case of $i, j \in \text{Hon}$.

The views of $\mathcal{A}$ in both games are identical. Thus, $\Pr[G_{10} = 1] = \Pr[G_{11} = 1]$. $\blacksquare$

Game G_{12} : It is the same as G_{11} , except that the challenger randomly samples all partial decryption except for the last honest user, and the last partial decryption is computed from previous partial decryptions and a partial information about the secret key $\mathbf{k}$ (i.e., $\mathbf{c}^\top \mathbf{k} + y_{i,S}$ in Fig. 10 below). Additionally, the computation of masks in $\mathcal{O}_{\text{COR}}$ is changed accordingly. Besides, in this game, the challenger introduces two tables RandY, MaskedDec as defined below:

- RandY[(ct, S), i] records the noise term $y_{i,S}$ when answering $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$.
- MaskedDec[(ct, S), i] records the partial decryption results $h_{i,S}$ when answering $\mathcal{O}_{\text{DEC}}(i, \text{ct}, S)$.

More precisely, the challenger answers $\mathcal{O}_{\text{COR}}$ and $\mathcal{O}_{\text{DEC}}$ queries according to Fig. 10.

Claim 7. $\Pr[G_{11} = 1] = \Pr[G_{12} = 1]$.

Proof. We introduce an intermediate game $G_{11.5}$ that is identical to G_{11} except that now the challenger samples $\Delta_{i,S}^* \leftarrow \mathbb{Z}_q$ and sets $\Delta_{i,S} := \Delta_{i,S}^* - (\mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S})$ when computing $\Delta_{i,S}$, and store RandY[(ct, S), i] = $y_{i,S}$. Since both $\Delta_{i,S}$ and $\Delta_{i,S}^*$ are uniformly distributed over $\mathbb{Z}_q$, $G_{11.5}$ is identi-

$\mathcal{G}_{12}$: $\mathcal{O}_{\text{Cor}}(i)$: If $\exists S \subseteq [N]$ s.t. $ \text{Cor} \cup \mathcal{Q}_{\text{Dec}}[S] \geq t - 1$: Return $\perp$ For all $(\text{ct} = (\mathbf{c}, d), S) \in \mathcal{Q}_{\text{Dec}}$ s.t. $(i \in S) \wedge (\text{Mask}[(\text{ct}, S)] = \perp)$: If $\text{MaskDec}[(\text{ct}, S), i] \neq \perp$: $\text{Mask}[(\text{ct}, S), i] := \text{MaskDec}[(\text{ct}, S), i] - \mathbf{c}^\top \lambda_{i,S} \mathbf{k}_i - \text{RandY}[(\text{ct}, S), i]$ Else: $\widetilde{\text{HS}} := \text{UnDecHS}[(\text{ct}, S)]$ If $\widetilde{\text{HS}} \neq \{i\}$: $\text{Mask}[(\text{ct}, S), i] \leftarrow \mathbb{Z}_q$ Else: For $j \in \text{CS}$: $\Delta_{j,S} := \text{ZeroShare}(\text{sk}_j, (\text{ct}, S))$ $\Delta_{i,S} := \mathbf{c}^\top \mathbf{k} - \sum_{j \in \text{CS} \cup \{i\}} \mathbf{c}^\top \cdot \lambda_{j,S} \cdot \mathbf{k}_j + \sum_{j \in \text{HS} \setminus \{i\}} \text{RandY}[(\text{ct}, S), j]$ $- \sum_{j \in \text{HS} \setminus \{i\}} \text{MaskDec}[(\text{ct}, S), j] - \sum_{j \in \text{CS}} \Delta_{j,S}$. $\text{Mask}[(\text{ct}, S), i] := \Delta_{i,S}$ $\text{UnDecHS}[(\text{ct}, S)] := \text{UnDecHS}[(\text{ct}, S)] \setminus \{i\}$ For all $(\text{ct}, S) \in \mathcal{Q}_{\text{Dec}}$ s.t. $\text{Mask}[(\text{ct}, S), i] \neq \perp$ $\text{ProgramZeroShare}((\text{ct}, S), i, \text{Mask}[(\text{ct}, S), i], \text{CS}, \text{HS})$ $\text{Cor} := \text{Cor} \cup \{i\}$, $\text{Hon} := \text{Hon} \setminus \{i\}$ Return sk_i	$\mathcal{O}_{\text{Dec}}(i, \text{ct}, S)$: If $\text{CheckDec}(i, \text{ct}, S) = 0$: Return $\perp$ If $(\text{ct}, S) \notin \mathcal{Q}_{\text{Dec}}$: $\mathcal{Q}_{\text{Dec}} := \mathcal{Q}_{\text{Dec}} \cup \{(\text{ct}, S)\}$ $\text{UnDecHS}[(\text{ct}, S)] := \text{HS}$, $\widetilde{\text{HS}} := \text{HS}$ If $(\text{ct}, S) \in \mathcal{Q}_{\text{Dec}}$: $\widetilde{\text{HS}} := \text{UnDecHS}[(\text{ct}, S)]$ If $\widetilde{\text{HS}} \neq \{i\}$: $h_{i,S} \leftarrow \mathbb{Z}_q$, $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$, $\text{RandY}[(\text{ct}, S), i] := y_{i,S}$ Else: Parse $\text{ct} = (\mathbf{c}, d)$ For $j \in \text{Cor}$: $\Delta_{j,S} := \text{ZeroShare}(\text{seed}_j[S], (\text{ct}, S))$ $y_{i,S} \leftarrow D_{\mathbb{Z}, \sigma_y}$ $h_{i,S} := \mathbf{c}^\top \mathbf{k} + y_{i,S} + \sum_{j \in \text{HS} \setminus \{i\}} \text{RandY}[(\text{ct}, S), j]$ $- \sum_{j \in \text{HS} \setminus \{i\}} \text{MaskDec}[(\text{ct}, S), j] - \sum_{j \in \text{CS}} \lambda_{j,S} \mathbf{c}^\top \mathbf{k}_j - \sum_{j \in \text{CS}} \Delta_{j,S}$ $\text{MaskDec}[(\text{ct}, S), i] := h_{i,S}$ $\text{UnDecHS}[(\text{ct}, S)] := \text{UnDecHS}[(\text{ct}, S)] \setminus \{i\}$ If $\text{ct} = \text{ct}^*$: $\mathcal{Q}_{\text{Dec}}[S] := \mathcal{Q}_{\text{Dec}}[S] \cup \{i\}$ Return $h_{i,S}$
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Fig. 10. The game $\mathcal{G}_{12}$ with changes highlighted in gray boxes.

cally distributed to $\mathcal{G}_{11}$. Besides, observe that in $\mathcal{G}_{11.5}$, when $\widetilde{\text{HS}} \neq \{i\}$, due to $h_{i,S} := \mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} + \Delta_{i,S}$, we have $h_{i,S} = \Delta_{i,S}^*$, which will be stored in $\text{MaskDec}[(\text{ct}, S), i]$. When $\widetilde{\text{HS}} = \{i\}$, we have $\Delta_{i,S} := -\sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S}$ in $\mathcal{G}_{11.5}$ and thus

$$\begin{aligned}
h_{i,S} &:= \mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S} \\
&= \mathbf{c}^\top (\lambda_{i,S} \mathbf{k}_i) + y_{i,S} - \sum_{j \in \text{CS}} \Delta_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S}^* + \sum_{j \in \text{HS} \setminus \{i\}} (\mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) + y_{j,S}) \\
&= \mathbf{c}^\top \mathbf{k} - \sum_{j \in \text{CS}} \mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) + y_{i,S} + \sum_{j \in \text{HS} \setminus \{i\}} y_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} h_{j,S} - \sum_{j \in \text{CS}} \Delta_{j,S}. \quad (24)
\end{aligned}$$

By substituting $y_{j,S}$ by $\text{RandY}[(\text{ct}, S), j]$ and $h_{j,S}$ by $\text{MaskDec}[(\text{ct}, S), j]$ respectively, (24) is exactly the simulation of $h_{i,S}$ in $\mathcal{G}_{12}$ as shown in Fig. 10.

On the other hand, we need to check the consistency of the simulation of $\Delta_{i,S}$ in $\mathcal{O}_{\text{Cor}}(i)$ if $\widetilde{\text{HS}} = \{i\}$. In $\mathcal{G}_{11.5}$, we have $\text{Mask}[(\text{ct}, S), j] = \Delta_{j,S}^* - \mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) - y_{j,S}$ and hence

$$\begin{aligned}
\Delta_{i,S} &:= - \sum_{j \in \text{HS} \setminus \{i\}} \Delta_{j,S} - \sum_{j \in \text{CS}} \Delta_{j,S} = - \sum_{j \in \text{HS} \setminus \{i\}} (\Delta_{j,S}^* - \mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) - y_{j,S}) - \sum_{j \in \text{CS}} \Delta_{j,S} \\
&= \mathbf{c}^\top \mathbf{k} - \sum_{j \in \text{CS} \cup \{i\}} \mathbf{c}^\top (\lambda_{j,S} \mathbf{k}_j) + \sum_{j \in \text{HS} \setminus \{i\}} y_{j,S} - \sum_{j \in \text{HS} \setminus \{i\}} h_{j,S} - \sum_{j \in \text{CS}} \Delta_{j,S}. \quad (25)
\end{aligned}$$

By substituting $y_{j,S}$ by $\text{RandY}[(\text{ct}, S), j]$ and $h_{j,S}$ by $\text{MaskDec}[(\text{ct}, S), j]$ respectively, (25) is exactly the simulation of $\Delta_{i,S}$ in $\mathcal{G}_{12}$ as shown in Fig. 10. Therefore, the simulation of $\Delta_{i,S}$ in $\mathcal{G}_{11.5}$ is identical to that in $\mathcal{G}_{12}$.

Overall, we conclude that the views of $\mathcal{A}$ among $\mathcal{G}_{11}$, $\mathcal{G}_{11.5}$ and $\mathcal{G}_{12}$ are identical. Thus, $\Pr[\mathcal{G}_{11} = 1] = \Pr[\mathcal{G}_{12} = 1]$. ■

E Instantiations of NIZK from Lattices

In this section, we describe how to instantiate NIZK in our work. More precisely, we first present the definitions of trapdoor Σ -protocol and the rejection sampling technique in Appendix E.1. Then, we present the instantiations of tag-based OTSS NIZK for $\mathcal{GL}_\rho^{(\text{LWE})}$ in Appendix E.2 and adaptively sound NIZK for $\mathcal{GL}_\rho^{(\text{lin})}$ in Appendix E.3 in the standard model, respectively, which are used in our TD_1 and TD_0 . Finally, we construct a tag-based OTSS NIZK for $\mathcal{GL}_\rho^{(\text{LWE})}$ in the random oracle model in Appendix E.4, which is used in our construction TD_2 .

E.1 Building Blocks

Below we present the core building block trapdoor Σ -protocol for our NIZK constructions. We first present the formal definitions of Σ -protocol and trapdoor- Σ -protocol introduced by Canetti et al. in [11]. Differs from the usual formulation, Def. 11 relaxes the standard special soundness property in that extractability is not required.

Definition 11 (Σ -Protocol [11]). Let $\mathcal{GL}_\rho = (\mathcal{L}_\rho, \tilde{\mathcal{L}}_\rho)$ be a gap language parameterized by language parameter ρ . A Σ -protocol for gap language $\mathcal{GL}_\rho$ consists of PPT algorithms $\Sigma = (\Sigma.\text{CRSGen}, \Sigma.\text{Prove}_1, \Sigma.\text{Prove}_2, \Sigma.\text{Vrfy}, \Sigma.\text{Sim})$.

- $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$: Taking as input a language parameter ρ , the CRS generation algorithm outputs a common reference string (CRS) crs , which implicitly defines a challenge space $\mathcal{CH}$.
- **3-Move Protocol.** The 3-move protocol is executed between a “prover” and a “verifier”. The prover and the verifier both take a CRS crs and an instance $x \in \mathcal{L}_\rho$ as input. The prover also takes as input a witness w of x .
 - $(a, \text{st}) \leftarrow \Sigma.\text{Prove}_1(\text{crs}, x, w)$: The prover invokes $\Sigma.\text{Prove}_1$ to get a message a and a state st . Then the prover keeps st as its own state information and sends a to the verifier.
 - $\text{ch} \leftarrow \mathcal{CH}$: After receiving a , the verifier chooses $\text{ch} \leftarrow \mathcal{CH}$ uniformly at random as the challenge and sends ch to the prover.
 - $z \leftarrow \Sigma.\text{Prove}_2(\text{crs}, x, w, a, \text{st}, \text{ch})$: After obtaining ch , the prover invokes $\Sigma.\text{Prove}_2$ to get a message z and sends z to the verifier.
 - $0/1 \leftarrow \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z)$: After getting z , the verifier invokes $\Sigma.\text{Vrfy}$ to obtain a decision bit.
- $(\tilde{a}, \tilde{z}) \leftarrow \Sigma.\text{Sim}(\text{crs}, x, \text{ch})$: Taking as input crs , an instance x and a challenge $\text{ch} \in \mathcal{CH}$, the simulation algorithm outputs a simulated $(\tilde{a}, \tilde{z})$. Here $(\tilde{a}, \text{ch}, \tilde{z})$ serves as a simulated transcript.

The following properties are required:

- **Completeness:** For all $x \in \mathcal{L}_\rho$ with witness w and all $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$, it holds that

$$\Pr \left[\begin{array}{l} (a, \text{st}) \leftarrow \Sigma.\text{Prove}_1(\text{crs}, x, w), \\ \text{ch} \leftarrow \mathcal{CH}, \\ z \leftarrow \Sigma.\text{Prove}_2(\text{crs}, x, w, a, \text{st}, \text{ch}) \end{array} : \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z) = 1 \right] \geq 1 - \text{negl}(\lambda).$$

- **Special Soundness:** For any $x \notin \tilde{\mathcal{L}}_\rho$, any $\text{crs} \in \Sigma.\text{CRSGen}(\rho)$ and any first message a , there is at most one challenge $\text{ch} \in \mathcal{CH}$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z) = 1$ for some third message z . Moreover, we define a “bad challenge function” f with $f(\text{crs}, x, a) := \text{ch}$ if there exists such a unique ch and $f(\text{crs}, x, a) := \perp$ otherwise. Note that f might not be efficiently computable.
- **Special Zero-Knowledge:** For all $x \in \mathcal{L}_\rho$ with witness w , all $\text{crs} \leftarrow_\$ \Sigma.\text{CRSGen}(\rho)$ and all $\text{ch} \in \mathcal{CH}$, it holds that

$$\Delta((a, z), (\tilde{a}, \tilde{z})) \leq \text{negl}(\lambda),$$

where the probability is over $(a, \text{st}) \leftarrow_\$ \Sigma.\text{Prove}_1(\text{crs}, x, w)$, $z \leftarrow_\$ \Sigma.\text{Prove}_2(\text{crs}, x, w, a, \text{st}, \text{ch})$ and $(\tilde{a}, \tilde{z}) \leftarrow_\$ \Sigma.\text{Sim}(\text{crs}, x, \text{ch})$.

The trapdoor Σ -protocols as Σ -protocols where the bad challenge function is efficiently computable using a trapdoor. Here, we use a definition where the CRS and the trapdoor may depend on the language.

Definition 12. A Σ -protocol $\Sigma = (\Sigma.\text{CRSGen}, \Sigma.\text{Prove}_1, \Sigma.\text{Prove}_2, \Sigma.\text{Vrfy}, \Sigma.\text{Sim})$ with bad challenge function f for a trapdoor language $\mathcal{GL}_\rho$ (with trapdoor td_ρ being the trapdoor information for $\mathcal{GL}_\rho$) is a **trapdoor** Σ -protocol if it satisfies the properties of Def. 11 and there exist PPT algorithms $(\Sigma.\text{TrapGen}, \Sigma.\text{BadChallenge})$ with the following properties.

- $(\text{crs}, \text{td}_\Sigma) \leftarrow_\$ \Sigma.\text{TrapGen}(\rho, \text{td}_\rho)$: Taking as input a language parameter ρ and a trapdoor information td_ρ for the gap language $\mathcal{GL}_\rho$, the trapdoor CRS generation algorithm outputs a crs and a trapdoor td_Σ for the scheme.
- $\text{ch} \leftarrow_\$ \Sigma.\text{BadChallenge}(\text{crs}, \text{td}_\Sigma, x, a)$: Taking as input crs , a trapdoor td_Σ , an instance x and a first message a , the bad challenge algorithm outputs a challenge ch .

The following properties are required:

- **CRS Indistinguishability:** The crs generated by $\text{crs} \leftarrow_\$ \Sigma.\text{CRSGen}(\rho)$ is statistically close to the crs generated by $(\text{crs}, \text{td}_\Sigma) \leftarrow_\$ \Sigma.\text{TrapGen}(\rho, \text{td}_\rho)$.
- **Correctness of $\Sigma.\text{BadChallenge}$:** For all $x \notin \tilde{\mathcal{L}}_\rho$, all $(\text{crs}, \text{td}_\Sigma) \in \text{TrapGen}(\rho, \text{td}_\rho)$ and all first messages a , it holds that $\Sigma.\text{BadChallenge}(\text{crs}, \text{td}_\Sigma, x, a) = f(\text{crs}, x, a)$ if $f(\text{crs}, x, a) \neq \perp$. Here f is the bad challenge function of Σ .

To achieve statistical zero-knowledge, we leverage the rejection sampling technique introduced in [38, 39].

Lemma 14 (Rejection Sampling [39, Theorem 4.6]). Let $\mathcal{V}$ be a subset of $\mathbb{Z}^m$ in which all elements have ℓ_2 norms less than T , σ be a real number such that $\sigma = \omega(T\sqrt{\log m})$ and V be a distribution over $\mathcal{V}$. Then, there exists a real number M such that the distributions of the following algorithms $\mathcal{A}$ and $\mathcal{F}$ has statistical distance at most $\frac{2^{-\omega(\log m)}}{M}$:

- $\mathcal{A}$: sample $\mathbf{v} \leftarrow_\$ V$, $\mathbf{z} \leftarrow_\$ D_{\mathbb{Z}^m, \sigma, \mathbf{v}}$ and output $(\mathbf{z}, \mathbf{v})$ with probability

$$\min \left(\frac{D_{\mathbb{Z}^m, \sigma}(\mathbf{z})}{M \cdot D_{\mathbb{Z}^m, \sigma, \mathbf{v}}(\mathbf{z})}, 1 \right);$$

– $\mathcal{F}$: sample $\mathbf{v} \leftarrow_s V$, $\mathbf{z} \leftarrow_s D_{\mathbb{Z}^m, \sigma}$ and output $(\mathbf{z}, \mathbf{v})$ with probability $1/M$.

Moreover, the probability that $\mathcal{A}$ outputs something is at least $\frac{1-2^{-\omega(\log m)}}{M}$.

More concretely, if $\sigma = \alpha T$ for any positive α , then $M = e^{12/\alpha+1/(2\alpha^2)}$, the above statistical distance is at most $2^{-100}/M$, and the probability that $\mathcal{A}$ outputs something is at least $(1 - 2^{-100})/M$.

E.2 One-Time Simulation Sound Tag-based QA-NIZK for $\mathcal{GL}_\rho^{(\text{LWE})}$ in the Standard Model

Let $\text{pp}_{\text{LWE}} := (q, m, n, B, \tilde{B})$ be the LWE related public parameters, where $B < \tilde{B}$. More precisely, let $\rho = \mathbf{A} \in \mathbb{Z}_q^{n \times m}$ be the language parameter. Accordingly, the gap language $\mathcal{GL}_\rho^{(\text{LWE})}$ with $\rho = \mathbf{A}$ is instantiated as follows:

$$\begin{aligned}\mathcal{L}_\mathbf{A}^{(\text{LWE})} &:= \left\{ \mathbf{c} \mid \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-B, B]^m) \text{ s.t. } \mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \right\} \\ \tilde{\mathcal{L}}_\mathbf{A}^{(\text{LWE})} &:= \left\{ \mathbf{c} \mid \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m) \text{ s.t. } \mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \right\}\end{aligned}$$

The specific construction of (even unbounded simulation-sound) tag-based QA-NIZK for such gap-language in the standard modal can be found in [26, Appendix G] by first constructing a trapdoor Σ -protocol for $\mathcal{GL}_\rho^{(\text{LWE})}$ and then compiling it into a NIZK via the generic construction proposed in [36]. We emphasize that the construction of trapdoor Σ -protocol and the generic construction from trapdoor Σ -protocol to NIZK will not introduce an exponential modulus q as long as $B, \tilde{B}$ is polynomial.

E.3 QA-NIZK for $\mathcal{GL}_\rho^{(\text{lin})}$ in the Standard Model

In order to construct QA-NIZK for gap language $\mathcal{GL}_\mathbf{A}^{(\text{lin})} = (\mathcal{L}_\mathbf{A}^{(\text{lin})}, \tilde{\mathcal{L}}_\mathbf{A}^{(\text{lin})})$, in this subsection, we will first construct a trapdoor Σ -protocol for $\mathcal{GL}_\mathbf{A}^{(\text{lin})}$, then compile the trapdoor Σ -protocol to a QA-NIZK for $\mathcal{GL}_\mathbf{A}^{(\text{lin})}$ by the generic construction in [36]. Inspired by [20], we design our trapdoor Σ -protocol directly, without using the generic transformation to transform any Σ -protocol to a trapdoor Σ -protocol in [13].

Let $\text{pp} = (q, m, n, \sigma_k, \sigma_e, \tilde{B}, B_k, \tilde{B}_k, B_e, \tilde{B}_e)$ be the public parameter that serve as implicit input to all algorithms, where $B < \tilde{B}$, $B_k < \tilde{B}_k$, $B_e < \tilde{B}_e$. More precisely, let $\rho = \mathbf{A} \in \mathbb{Z}_q^{n \times m}$ be the language parameter. The language $\mathcal{L}_\rho^{(\text{lin})}$ with $\rho = \mathbf{A}$ is instantiated as follows:

$$\mathcal{L}_\mathbf{A}^{(\text{lin})} := \left\{ (\text{pk}, h, \mathbf{c}) \mid \exists (\mathbf{k} \in [-B_k, B_k]^m, \mathbf{e}' \in [-B_e, B_e]^m) \text{ s.t. } \begin{aligned} \text{pk} &= \mathbf{A}\mathbf{k} \wedge \\ h &= \mathbf{c}^\top \mathbf{k} + \mathbf{e}' \end{aligned} \right\}. \quad (26)$$

Note that when proving the robustness of our scheme, we only need to prove that the partial decryption h is well-form in the case of the ciphertext is well-form (i.e., $\mathbf{c} \in \tilde{\mathcal{L}}_\rho^{(\text{LWE})}$ in Sect. 4). We observe that, in the case of $\mathbf{c} \in \tilde{\mathcal{L}}_\rho^{(\text{LWE})}$

($\exists(\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m)$ s.t. $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$), if $(\mathbf{pk}, h, \mathbf{c}) \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}$, it always holds that $h - \mathbf{s}^\top \mathbf{pk} = \mathbf{e}^\top \mathbf{k} + e' < mBB_k + B_e$. Consequently, by defining $B^* := m\tilde{B}\tilde{B}_k + \tilde{B}_e$, $\mathcal{L}_{\mathbf{A}}^{(\text{lin})}$ is contained the following language

$$\tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})} := \left\{ (\mathbf{pk}, h, \mathbf{c}) \mid \exists(\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m, y \in [-B^*, B^*]) \text{ s.t. } \begin{cases} \mathbf{c} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \\ h = \mathbf{s}^\top \mathbf{pk} + y \end{cases} \right\}. \quad (27)$$

The Trapdoor Σ -Protocol for $\mathcal{GL}_{\mathbf{A}}^{(\text{lin})}$. The syntax of trapdoor Σ -protocol is shown in Def. 12. Below we construct an LWE-based trapdoor Σ -protocol $\Sigma = (\Sigma.\text{CRSGen}, \Sigma.\text{Prove}_1, \Sigma.\text{Prove}_2, \Sigma.\text{Vrfy}, \Sigma.\text{Sim}, \Sigma.\text{TrapGen}, \Sigma.\text{BadChallenge})$ for the gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{lin})} = (\mathcal{L}_{\mathbf{A}}^{(\text{lin})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})})$ specified by (26) and (27), with challenge set $\mathcal{CH} = \{0, 1\}$.

- $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$: On input of the language parameter $\rho = \mathbf{A}$, return $\text{crs} := \mathbf{A}$.
- $(a, \text{st}) \leftarrow \Sigma.\text{Prove}_1(\text{crs}, x, w)$: Parse $x = (\mathbf{pk}, h, \mathbf{c})$. Choose $\mathbf{k}_0 \leftarrow D_{\mathbb{Z}^m, \sigma_k}$ and $e'_0 \leftarrow D_{\mathbb{Z}, \sigma_e}$. Compute $\mathbf{pk}_0 := \mathbf{A}\mathbf{k}_0 \bmod q$, $h_0 := \mathbf{c}^\top \mathbf{k}_0 + e'_0$. Return $a := (\mathbf{pk}_0, h_0)$ and $\text{st} := (\mathbf{k}_0, e'_0)$.
- $z \leftarrow \Sigma.\text{Prove}_2(\text{crs}, x, w, a, \text{st}, \text{ch})$: Parse $w = (\mathbf{k}, e')$ and $\text{st} = (\mathbf{k}_0, e'_0)$. Compute

$$\mathbf{k}_{\text{mix}} := \mathbf{k}_0 + \text{ch} \cdot \mathbf{k}, \quad e'_{\text{mix}} := e'_0 + \text{ch} \cdot e'.$$

Return $z := (\mathbf{k}_{\text{mix}}, e'_{\text{mix}})$ with probability θ and abort otherwise, where the probability θ is defined by $\theta := \min\left(\frac{D_{\mathbb{Z}^m, \sigma_k}(\mathbf{k}_{\text{mix}})}{M \cdot D_{\mathbb{Z}^m, \sigma_k, \text{ch} \cdot \mathbf{k}}(\mathbf{k}_{\text{mix}})} \cdot \frac{D_{\mathbb{Z}, \sigma_e}(e'_{\text{mix}})}{M' \cdot D_{\mathbb{Z}, \sigma_e, \text{ch} \cdot e'}(e'_{\text{mix}})}, 1\right)$ with $M := e^{12\sqrt{m}B_k/\sigma_k + mB_k^2/(2\sigma_k^2)}$ and $M' := e^{12B_e/\sigma_e + B_e^2/(2\sigma_e^2)}$.

- $0/1 \leftarrow \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z)$: Parse $x = (\mathbf{pk}, h, \mathbf{c})$, $a = (\mathbf{pk}_0, h_0)$ and $z = (\mathbf{k}_{\text{mix}}, e'_{\text{mix}})$. Check if

$$\|\mathbf{k}_{\text{mix}}\|_\infty \leq \tilde{B}_k/2 \quad \text{and} \quad |e'_{\text{mix}}| < \tilde{B}_e/2,$$

and check if

$$\mathbf{pk}_0 + \text{ch} \cdot \mathbf{pk} = \mathbf{A}\mathbf{k}_{\text{mix}} \bmod q \quad \text{and} \quad h_0 + \text{ch} \cdot h = \mathbf{c}^\top \mathbf{k}_{\text{mix}} + e'_{\text{mix}} \bmod q. \quad (28)$$

If all the checks pass, return 1; otherwise, return 0.

- $(\tilde{a}, \tilde{z}) \leftarrow \Sigma.\text{Sim}(\text{crs}, x, \text{ch})$: Parse $x = (\mathbf{pk}, h, \mathbf{c})$. Choose $\tilde{\mathbf{k}}_{\text{mix}} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$ and $\tilde{e}'_{\text{mix}} \leftarrow D_{\mathbb{Z}, \sigma_e}$. Compute $\tilde{\mathbf{pk}}_0 := \mathbf{A}\tilde{\mathbf{k}}_{\text{mix}} - \text{ch} \cdot \mathbf{pk} \bmod q$ and $\tilde{h}_0 := \mathbf{c}^\top \tilde{\mathbf{k}}_{\text{mix}} + \tilde{e}'_{\text{mix}} - \text{ch} \cdot h \bmod q$. Return $\tilde{a} := (\tilde{\mathbf{pk}}_0, \tilde{h}_0)$ and $\tilde{z} := (\tilde{\mathbf{k}}_{\text{mix}}, \tilde{e}'_{\text{mix}})$ with probability $\tilde{\theta}$ and abort otherwise, where the probability $\tilde{\theta}$ is defined by $\tilde{\theta} := 1/(MM')$ with $M := e^{12\sqrt{m}B_k/\sigma_k + mB_k^2/(2\sigma_k^2)}$ and $M' := e^{12B_e/\sigma_e + B_e^2/(2\sigma_e^2)}$.
- $(\text{crs}, \text{td}_\Sigma) \leftarrow \Sigma.\text{TrapGen}(\rho, \text{td}_\rho)$: On input of language parameter $\rho = \mathbf{A}$ and trapdoor information $\text{td}_\rho = \mathbf{T}_{\mathbf{A}}$ for $\mathcal{GL}_{\rho}^{(\text{lin})}$, where $(\mathbf{A}, \mathbf{T}_{\mathbf{A}})$ is generated by $\text{TrapGen}(n, q, m)$ (cf. Lemma 3), return $\text{crs} := \mathbf{A}$ and $\text{td}_\Sigma := \mathbf{T}_{\mathbf{A}}$.

- $\text{ch} \leftarrow_{\$} \Sigma.\text{BadChallenge}(\text{crs}, \text{td}_{\Sigma} = \mathbf{T}_{\mathbf{A}}, x, a)$: Parse $x = (\mathbf{A}, \text{pk}, h, \mathbf{c})$ and $a = (\text{pk}_0, h_0)$. Invoke $(\mathbf{s}, \mathbf{e}) \leftarrow_{\$} \text{Invert}(\mathbf{c}, \mathbf{T}_{\mathbf{A}})$ (cf. Lemma 4). If fails, return $\perp$. For $b = \{0, 1\}$, if

$$|h_0 + b \cdot h - \mathbf{s}^{\top}(\text{pk}_0 + b \cdot \text{pk})| > B^*/2,$$

return $\text{ch} = 1 - b$. Otherwise, return $\perp$.

Theorem 4 (Trapdoor Σ -protocol for $\mathcal{GL}_{\rho}^{(\text{lin})}$). Let $\sigma_k = \sqrt{m}B_k \cdot \omega(\sqrt{\log m})$, $\sigma_e = B_e \cdot \omega(1)$, $\tilde{B}_k = 2 \cdot (\omega(\sqrt{\log \lambda}) \cdot \sigma_k + B_k)$, $\tilde{B}_e = 2 \cdot (\omega(\sqrt{\log \lambda})\sigma_e + B_e)$ and $q > 10m\tilde{B}$. Then the above construction is a trapdoor Σ -protocol for the gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{lin})} = (\mathcal{L}_{\mathbf{A}}^{(\text{lin})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})})$ specified by (26) and (27).

Proof of Theorem 4. We prove all of the properties as follows:

Completeness. For any instance $x = (\text{pk}, h, \mathbf{c}) \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}$ with witness $w = (\mathbf{k}, e')$, any proof $a = (\text{pk}_0, e'_0)$, $z = (\mathbf{k}_{\text{mix}}, e'_{\text{mix}})$ and state $\text{st} = (\mathbf{k}_0, h_0)$ generated honestly by prover, we have that $\mathbf{k}_0 \leftarrow D_{\mathbb{Z}^m, \sigma_k}$ and $e'_0 \leftarrow D_{\mathbb{Z}, \sigma_e}$. By Lemma 5 and our parameter setting, $\|\mathbf{k}_0\|_{\infty} \leq (\tilde{B}_k/2 - B_k)$ and $|e'_0| \leq (\tilde{B}_e/2 - B_e)$ except with negligible probability. Hence for any challenge $\text{ch} \in \{0, 1\}$, it holds that $\|\mathbf{k}_{\text{mix}}\|_{\infty} \leq \|\mathbf{k}_0\|_{\infty} + \|\mathbf{k}\|_{\infty} \leq \tilde{B}_k/2$ and $|e'_{\text{mix}}| \leq |e'_0| + |e'| \leq \tilde{B}_e/2$, except with negligible probability. By the construction, the equality test in (28) will always pass for an honest prover. Therefore, the verification passes except with negligible probability.

Special Soundness. Special soundness requires that for any $x \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$ and any first message a , there exists at most one challenge $\text{ch} \in \{0, 1\}$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z) = 1$ for some third message z . We consider the special soundness in the case of $\mathbf{c} \in \tilde{\mathcal{L}}_{\rho}^{(\text{LWE})}$ ($\exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m)$ s.t. $\mathbf{c}^{\top} = \mathbf{s}^{\top} \mathbf{A} + \mathbf{e}^{\top}$). Suppose, towards contradiction, there exist $x = (\text{pk}, h, \mathbf{c}) \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$ and $a = (\text{pk}_0, h_0)$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z^{(\text{ch})}) = 1$ for both $\text{ch} = 0, 1$ for some $z^{(0)} = (\mathbf{k}_{\text{mix}}^{(0)}, e'_{\text{mix}}^{(0)})$ and $z^{(1)} = (\mathbf{k}_{\text{mix}}^{(1)}, e'_{\text{mix}}^{(1)})$. Then it must hold that $\|\mathbf{k}_{\text{mix}}^{(0)}\|_{\infty}, \|\mathbf{k}_{\text{mix}}^{(1)}\|_{\infty} \leq \tilde{B}_k/2$, $|e'_{\text{mix}}^{(0)}|, |e'_{\text{mix}}^{(1)}| \leq \tilde{B}_e/2$ and

$$\text{pk}_0 + 0 \cdot \text{pk} = \mathbf{A}\mathbf{k}_{\text{mix}}^{(0)} \bmod q \quad \text{and} \quad h_0 + 0 \cdot h = \mathbf{c}^{\top} \mathbf{k}_{\text{mix}}^{(0)} + e'_{\text{mix}}^{(0)} \bmod q, \quad (29)$$

$$\text{pk}_0 + 1 \cdot \text{pk} = \mathbf{A}\mathbf{k}_{\text{mix}}^{(1)} \bmod q \quad \text{and} \quad h_0 + 1 \cdot h = \mathbf{c}^{\top} \mathbf{k}_{\text{mix}}^{(1)} + e'_{\text{mix}}^{(1)} \bmod q. \quad (30)$$

By subtracting the corresponding equations in (29) from (30), we have

$$\text{pk} = \mathbf{A}(\mathbf{k}_{\text{mix}}^{(1)} - \mathbf{k}_{\text{mix}}^{(0)}) \bmod q \quad \text{and} \quad h = \mathbf{c}^{\top}(\mathbf{k}_{\text{mix}}^{(1)} - \mathbf{k}_{\text{mix}}^{(0)}) + (e'_{\text{mix}}^{(1)} - e'_{\text{mix}}^{(0)}) \bmod q.$$

Since $\mathbf{c} \in \tilde{\mathcal{L}}_{\rho}^{(\text{LWE})}$, there exists $\mathbf{s} \in \mathbb{Z}_q^n$ and $\mathbf{e} \in [-\tilde{B}, \tilde{B}]^m$ s.t. $\mathbf{c}^{\top} = \mathbf{s}^{\top} \mathbf{A} + \mathbf{e}$. Thus, we have

$$y := h - \mathbf{s}^{\top} \text{pk} = \mathbf{e}^{\top}(\mathbf{k}_{\text{mix}}^{(1)} - \mathbf{k}_{\text{mix}}^{(0)}) + (e'_{\text{mix}}^{(1)} - e'_{\text{mix}}^{(0)}),$$

and $|y| \leq m \|\mathbf{e}^\top\|_\infty \|\mathbf{k}_{mix}^{(1)} - \mathbf{k}_{mix}^{(0)}\|_\infty + |e'_{mix}^{(1)} - e'_{mix}^{(0)}| \leq m\tilde{B}\tilde{B}_k + \tilde{B}_e = B^*$.

Consequently, $(\mathbf{s}, \mathbf{e}, y)$ constitutes a witness for $x = (\mathbf{pk}, h, \mathbf{c}) \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$, which yields a contradiction.

Special Zero-Knowledge. The special ZK property can be shown by applying Lemma 14. Given a statement $x = (\mathbf{pk}, h, \mathbf{c}) \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}$ and a challenge $\text{ch} \in \{0, 1\}$, the simulator chooses $\tilde{\mathbf{k}}_{mix} \leftarrow D_{\mathbb{Z}^m, \sigma_k}$ and $\tilde{e}'_{mix} \leftarrow D_{\mathbb{Z}, \sigma_e}$ and computes $\tilde{\mathbf{pk}}_0 := \mathbf{A}\tilde{\mathbf{k}}_{mix} - \text{ch} \cdot \mathbf{pk} \bmod q$ and $\tilde{h}_0 := \mathbf{c}^\top \tilde{\mathbf{k}}_{mix} + \tilde{e}'_{mix} - \text{ch} \cdot h \bmod q$. It outputs $\tilde{a} := (\tilde{\mathbf{pk}}_0, \tilde{h}_0)$ and $\tilde{z} := (\tilde{\mathbf{k}}_{mix}, \tilde{e}'_{mix})$ with probability $\tilde{\theta} = 1/(MM')$. By construction, the triple $(\tilde{a}, \text{ch}, \tilde{z})$ satisfies the verification conditions w.h.p.

We leverage Lemma 14 to show that the simulated $(\tilde{a}, \text{ch}, \tilde{z})$ is statistically close an honest proof (a, ch, z) . Note that $\|\text{ch} \cdot \mathbf{k}\| \leq \sqrt{m}B_k$ and $\sigma_k = \sqrt{m}B_k \cdot \omega(\sqrt{\log m})$, Lemma 14 shows that $\mathbf{k}_{mix} = \mathbf{k}_0 + \text{ch} \cdot \mathbf{k}$ in z – when output with probability $\min(\frac{D_{\mathbb{Z}^m, \sigma_k}(\mathbf{k}_{mix})}{M \cdot D_{\mathbb{Z}^m, \sigma_k, \text{ch} \cdot \mathbf{k}}(\mathbf{k}_{mix})}, 1)$ – and $\tilde{\mathbf{k}}_{mix}$ in $\tilde{z}$ – when output with probability $1/M$ – have statistical distance at most $2^{-100}/M$. A similar analysis shows that e'_{mix} and $\tilde{e}'_{mix}$ have statistical distance at most $2^{-100}/M'$.

Overall, the simulated proof $\tilde{z} = (\tilde{\mathbf{k}}_{mix}, \tilde{e}'_{mix})$ and the honest $z = (\mathbf{k}_{mix}, e'_{mix})$ have statistical distance at most $2^{-100} \cdot (1/M + 1/M')$. This completes the proof of special zero-knowledge.

CRS Indistinguishability. On input of language parameter $\rho = \mathbf{A}$, both $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$ and $(\text{crs}, \text{td}_\Sigma) \leftarrow \Sigma.\text{TrapGen}(\rho, \text{td}_\rho)$ simply set $\text{crs} = \mathbf{A}$. By Lemma 3, the matrix $\mathbf{A}$ generated by TrapGen is statistically close to a uniform $\mathbf{A}$, which implies the CRS indistinguishability.

Correctness of $\Sigma.\text{BadChallenge}$. For any $x = (\mathbf{pk}, h, \mathbf{c}) \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})}$ and any $a = (\mathbf{pk}_0, h_0)$, if the bad challenge function $f(\text{crs}, x, a) \neq \perp$, we aim to prove that $\Sigma.\text{BadChallenge}(\text{crs}, \text{td}_\Sigma, x, a) = f(\text{crs}, x, a)$. Assuming that $\mathbf{c} \in \tilde{\mathcal{L}}_\rho^{(\text{LWE})}$ ($\exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m)$ s.t. $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$), by setting $q \geq 10m\tilde{B}$, it must hold that $\text{Invert}(\mathbf{T}_{\mathbf{A}}, \mathbf{c}) = (\mathbf{s}, \mathbf{e})$ by Lemma 4 since $\|\mathbf{e}\| \leq \sqrt{m}\tilde{B} \leq q/(10\sqrt{m})$.

We next prove that, if the output of $\Sigma.\text{BadChallenge}$ is 1, then there does not exist $z = (\mathbf{k}_{mix}, e'_{mix})$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, 0, z) = 1$. Since $\Sigma.\text{BadChallenge} \Rightarrow 1$ if and only if $|h_0 - \mathbf{s}^\top \mathbf{pk}_0| > B^*/2$, towards contradiction, assume that there exist $z = (\mathbf{k}_{mix}, e'_{mix})$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, 0, z) = 1$, which implies the following condition holds:

$$\|\mathbf{k}_{mix}\|_\infty \leq \tilde{B}_k/2 \quad \text{and} \quad |e'_{mix}| < \tilde{B}_e/2,$$

$$\mathbf{pk}_0 = \mathbf{A}\mathbf{k}_{mix} \bmod q \quad \text{and} \quad h_0 = \mathbf{c}^\top \mathbf{k}_{mix} + e'_{mix} \bmod q.$$

In this case, we have $|h_0 - \mathbf{s}^\top \mathbf{pk}_0| = |\mathbf{e}^\top \mathbf{k}_{mix} + e'_{mix}| \leq m \|\mathbf{e}^\top\|_\infty \|\mathbf{k}_{mix}\|_\infty + |e'_{mix}| \leq (m\tilde{B}\tilde{B}_k)/2 + \tilde{B}_e/2 < B^*/2$, which yields a contradiction. The analysis of the case that the output of $\Sigma.\text{BadChallenge}$ is 0 is similar and the correctness of $\Sigma.\text{BadChallenge}$ follows. $\square$

We note that the trapdoor Σ -protocol presented in this section works with binary challenge, and hence, admits a soundness error of $1/2$. This error can be

made negligibly small via $O(\lambda)$ parallel repetitions. Then, using the transformation suggested in [36], we can obtain a non-interactive protocol in the standard model.

Witness Indistinguishability. As shown in [20, 53], zero-knowledge property implies witness indistinguishability (WI). Intuitively, let w_0, w_1 be two witnesses of an instance x , WI can be proved by showing that the transcripts generated with w_0, w_1 are both indistinguishable from the simulated transcripts outputted by $\text{Sim}(\text{crs}, x, \text{ch})$, and thus the transcripts of w_0 and w_1 are indistinguishable.

Remark 2 (Witness Independence or Secure Erasure). We claim that the additional NIZK proof $\overline{\Pi}$, which attests to the correctness of the decryption, does not affect the CCA security proof. This is because, throughout the reductions, $\overline{\Pi}$ is always either generated honestly or simulated on true statements $x \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}$.

A naive justification of this claim would require assuming reliable erasure of the decryption servers' internal randomness. However, as observed in [20, 53], such an erasure assumption is unnecessary provided that the proof system $\overline{\Pi}$ satisfies witness independence (witness indistinguishability in [53]). This property is implied by the zero-knowledge property of the underlying Σ -protocol. Intuitively, witness independence is a strictly weaker notion than zero-knowledge: it only requires that two transcripts generated using different witnesses w_0, w_1 for the same statement x be indistinguishable.

Consequently, in our security proof, we may conceptually rely on the existence of a straight-line but inefficient simulator that outputs simulated proofs $\tilde{\pi}_i$ which are statistically close to honestly generated proofs π_i . As a result, even if the challenger faithfully generates WI proofs of correct partial decryptions, these proofs do not reduce the min-entropy of the decryption shares from the adversary's perspective. We refer to [20, App. C, D] and [53, App. A] for a more detailed discussion.

E.4 One-Time Simulation Sound Tag-based QA-NIZK for $\mathcal{GL}_{\rho}^{(\text{LWE})}$ in the Random Oracle Model

In this subsection, we will construct an one-time simulation sound tag-based QA-NIZK for in the random oracle model. We will first construct a Σ -protocol for $\mathcal{GL}_{\rho}^{(\text{LWE})}$, then use FS-transformation to obtain a non-interactive version.

Let $\text{pp}_{\text{LWE}} := (q, m, n, \sigma, B, \tilde{B})$ be the LWE related public parameters, where $B < \tilde{B}$. More precisely, let $\rho = \mathbf{A} \in \mathbb{Z}_q^{n \times m}$ be the language parameter. Accordingly, the gap language $\mathcal{GL}_{\rho}^{(\text{LWE})}$ with $\rho = \mathbf{A}$ is instantiated as follows:

$$\mathcal{L}_{\mathbf{A}}^{(\text{LWE})} := \{ \mathbf{c} \mid \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-B, B]^m) \text{ s.t. } \mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \} \quad (31)$$

$$\tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})} := \{ \mathbf{c} \mid \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}, \tilde{B}]^m) \text{ s.t. } \mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \} \quad (32)$$

Σ -Protocol for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})}$. The syntax of Σ -protocol is shown in Def. 11. Below we construct a Σ -protocol $\Sigma = (\Sigma.\text{CRSGen}, \Sigma.\text{Prove}_1, \Sigma.\text{Prove}_2, \Sigma.\text{Vrfy}, \Sigma.\text{Sim})$

for the gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ specified by (31) and (32), with challenge set $\mathcal{CH} = \{0, 1\}$.

- $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$: On input of the language parameter $\rho = \mathbf{A}$, return $\text{crs} := \mathbf{A}$.
- $(a, \text{st}) \leftarrow \Sigma.\text{Prove}_1(\text{crs}, x, w)$: Choose $\mathbf{s}_0 \leftarrow \mathbb{Z}_q^n$ and $\mathbf{e}_0 \leftarrow D_{\mathbb{Z}^m, \sigma}$. Compute $\mathbf{c}_0^\top := \mathbf{s}_0^\top \mathbf{A} + \mathbf{e}_0^\top$. Return $a := \mathbf{c}_0$ and $\text{st} := (\mathbf{s}_0, \mathbf{e}_0)$.
- $z \leftarrow \Sigma.\text{Prove}_2(\text{crs}, x, w, a, \text{st}, \text{ch})$: Parse $w = (\mathbf{s}, \mathbf{e})$ and $\text{st} = (\mathbf{s}_0, \mathbf{e}_0)$. Compute

$$\mathbf{s}_{\text{mix}} := \mathbf{s}_0 + \text{ch} \cdot \mathbf{s}, \quad \mathbf{e}_{\text{mix}} := \mathbf{e}_0 + \text{ch} \cdot \mathbf{e}.$$

Return $z := (\mathbf{s}_{\text{mix}}, \mathbf{e}_{\text{mix}})$ with probability θ and abort otherwise, where the probability θ is defined by $\theta := \min(\frac{D_{\mathbb{Z}^m, \sigma}(\mathbf{e}_{\text{mix}})}{M \cdot D_{\mathbb{Z}^m, \sigma, \text{ch} \cdot \mathbf{e}}(\mathbf{e}_{\text{mix}})}, 1)$ with $M := e^{12\sqrt{m}B/\sigma + mB^2/(2\sigma^2)}$.

- $0/1 \leftarrow \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z)$: Parse $x = \mathbf{c}$, $a = \mathbf{c}_0$ and $z = (\mathbf{s}_{\text{mix}}, \mathbf{e}_{\text{mix}})$. Check if $\|\mathbf{e}_{\text{mix}}\| < \tilde{B}/2$ and

$$\mathbf{c}_0^\top + \text{ch} \cdot \mathbf{c}^\top = \mathbf{s}_{\text{mix}}^\top \mathbf{c}^\top + \mathbf{e}_{\text{mix}}^\top \pmod{q}. \quad (33)$$

If all the checks pass, return 1; otherwise, return 0.

- $(\tilde{a}, \tilde{z}) \leftarrow \Sigma.\text{Sim}(\text{crs}, x, \text{ch})$: Choose $\tilde{\mathbf{s}}_{\text{mix}} \leftarrow \mathbb{Z}_q^n$ and $\tilde{\mathbf{e}}_{\text{mix}} \leftarrow D_{\mathbb{Z}^m, \sigma}$. Compute $\tilde{\mathbf{c}}_0 := \tilde{\mathbf{s}}_{\text{mix}}^\top \mathbf{A} + \tilde{\mathbf{e}}_{\text{mix}}^\top - \text{ch} \cdot \mathbf{c}^\top \pmod{q}$. Return $\tilde{a} := (\tilde{\mathbf{c}}_0)$ and $\tilde{z} := (\tilde{\mathbf{s}}_{\text{mix}}, \tilde{\mathbf{e}}_{\text{mix}})$ with probability $\tilde{\theta}$ and abort otherwise, where the probability $\tilde{\theta} = 1/M$ with $M := e^{12\sqrt{m}B/\sigma + mB^2/(2\sigma^2)}$.

Theorem 5 (Σ -protocol for $\mathcal{GL}_{\rho}^{(\text{LWE})}$). Let $\sigma = \sqrt{m}B \cdot \omega(\sqrt{\log m})$, $\tilde{B} = 2 \cdot (\sigma\sqrt{m} \cdot \omega(\log \lambda) + B)$. Then the above construction is a Σ -protocol for the gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ specified by (31) and (32).

Proof of Theorem 5. We prove all of the properties as follows:

Completeness. For any instance $x = \mathbf{c} \in \mathcal{L}_{\mathbf{A}}^{(\text{LWE})}$ with witness $w = (\mathbf{s}, \mathbf{e})$, any proof $a = (\mathbf{c}_0)$, $z = (\mathbf{s}_{\text{mix}}, \mathbf{e}_{\text{mix}})$ and state $\text{st} = (\mathbf{s}_0, \mathbf{e}_0)$ generated honestly by prover, we have that $\mathbf{s} \leftarrow \mathbb{Z}^m$ and $\mathbf{e}' \leftarrow D_{\mathbb{Z}^m, \sigma}$. By Lemma 5 and our parameter setting, $\mathbf{e}_0$ is $(\tilde{B}/2 - B)$ -bounded except with negligible probability. Hence for any challenge $\text{ch} \in \{0, 1\}$, it holds that $\|\mathbf{e}_{\text{mix}}\|_\infty \leq \|\mathbf{e}_0\|_\infty + \|\mathbf{e}\|_\infty \leq \tilde{B}/2$, except with negligible probability. By the construction, the equality test in (33) will always pass for an honest prover. Therefore, the verification passes except with negligible probability.

Special Soundness. Special soundness requires that for any $x \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$ and any first message a , there exists at most one challenge $\text{ch} \in \{0, 1\}$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z) = 1$ for some third message z . Suppose, towards contradiction, there exist $x = \mathbf{c} \notin \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$ and $a = \mathbf{c}_0$ such that $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z^{(\text{ch})}) =$

1 for both $\text{ch} = 0, 1$ for some $z^{(0)} = (\mathbf{s}_{mix}^{(0)}, \mathbf{e}_{mix}^{(0)})$ and $z^{(1)} = (\mathbf{s}_{mix}^{(1)}, \mathbf{e}_{mix}^{(1)})$. Then it must hold that $\|\mathbf{e}_{mix}^{(0)}\|_\infty, \|\mathbf{e}_{mix}^{(1)}\|_\infty \leq \tilde{B}/2$ and

$$\mathbf{c}_0^\top + 0 \cdot \mathbf{c}^\top = \mathbf{s}_{mix}^{(0)\top} \mathbf{A} + \mathbf{e}_{mix}^{(0)\top} \bmod q, \quad (34)$$

$$\mathbf{c}_0^\top + 1 \cdot \mathbf{c}^\top = \mathbf{s}_{mix}^{(1)\top} \mathbf{A} + \mathbf{e}_{mix}^{(1)\top} \bmod q, \quad (35)$$

By subtracting the corresponding equations in (34) from (35), we have

$$\mathbf{c} = (\mathbf{s}_{mix}^{(1)\top} - \mathbf{s}_{mix}^{(0)\top}) \mathbf{A} + (\mathbf{e}_{mix}^{(1)\top} - \mathbf{e}_{mix}^{(0)\top}) \bmod q.$$

Since $\mathbf{s}_{mix}^{(1)} - \mathbf{s}_{mix}^{(0)} \in \mathbb{Z}_q^n$ and $\|\mathbf{e}_{mix}^{(1)\top} - \mathbf{e}_{mix}^{(0)\top}\|_\infty \leq \|\mathbf{e}_{mix}^{(0)}\|_\infty + \|\mathbf{e}_{mix}^{(1)}\|_\infty \leq \tilde{B}$, $(\mathbf{s}_{mix}^{(1)} - \mathbf{s}_{mix}^{(0)}, \mathbf{e}_{mix}^{(1)} - \mathbf{e}_{mix}^{(0)})$ constitutes a witness for $x = \mathbf{c} \in \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})}$, which yields a contradiction.

Special Zero-Knowledge. The special ZK property can be shown by applying Lemma 14. Given a statement $x = \mathbf{c} \in \mathcal{L}_{\mathbf{A}}^{(\text{LWE})}$ and a challenge $\text{ch} \in \{0, 1\}$, the simulator chooses $\tilde{\mathbf{s}}_{mix} \leftarrow \mathbb{Z}_q^n$ and $\tilde{\mathbf{e}}_{mix} \leftarrow D_{\mathbb{Z}^m, \sigma}$ and computes $\tilde{\mathbf{c}}_0^\top := \tilde{\mathbf{s}}_{mix}^\top \mathbf{A} + \tilde{\mathbf{e}}_{mix}^\top - \text{ch} \cdot \mathbf{c}^\top \bmod q$. It outputs $\tilde{a} := (\tilde{\mathbf{c}}_0)$ and $\tilde{z} := (\tilde{\mathbf{s}}_{mix}, \tilde{\mathbf{e}}_{mix})$ with probability $\theta = 1/M$. By construction, the triple $(\tilde{a}, \text{ch}, \tilde{z})$ passes the verification w.h.p.

We leverage Lemma 14 to show that the simulated $(\tilde{a}, \text{ch}, \tilde{z})$ is statistically close an honest proof (a, ch, z) . It is easy to see that $\tilde{\mathbf{s}}_{mix}$ in $\tilde{z}$ and $\mathbf{s}_{mix}$ in z are both uniformly random in $\mathbb{Z}_q^n$. Note that $\|\text{ch} \cdot \mathbf{e}\| \leq \sqrt{m}B$ and $\sigma = \sqrt{m}B \cdot \omega(\sqrt{\log m})$, Lemma 14 shows that $\mathbf{e}_{mix} = \mathbf{e}_0 + \text{ch} \cdot \mathbf{e}$ in z – when output with probability $\min(\frac{D_{\mathbb{Z}^m, \sigma}(\mathbf{e}_{mix})}{M \cdot D_{\mathbb{Z}^m, \sigma, \text{ch} \cdot \mathbf{e}}(\mathbf{e}_{mix})}, 1)$ – and $\tilde{\mathbf{e}}_{mix}$ in $\tilde{z}$ – when output with probability $1/M$ – have statistical distance at most $2^{-100}/M$.

Overall, the simulated proof $\tilde{z} = (\tilde{\mathbf{s}}_{mix}, \tilde{\mathbf{e}}_{mix})$ and the honest $z = (\mathbf{k}_{mix}, \mathbf{e}_{mix})$ have statistical distance at most $2^{-100}/M$. This completes the proof of special zero-knowledge. $\square$

To construct a simulation sound QA-NIZK, we require the output of $\Sigma.\text{Prove}_2$ is non-malleable to avoid the adversary breaking soundness by manipulating the simulated proof of a false statement. Below we introduce the notion of *quasi unique response* introduced by Fischlin in [24].

Definition 13 (Quasi Unique Response). A Σ -protocol has quasi unique responses if for any language parameter ρ , $\text{crs} \leftarrow \Sigma.\text{CRSGen}(\rho)$ and $b = 0, 1$, it hold that,

$$\Pr \left[\exists (x, a, \text{ch}, z^{(0)}, z^{(1)}) \text{ s.t. } (z^{(0)} \neq z^{(1)}) \wedge \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z^{(b)} = 1) \right] \leq \text{negl}(\lambda).$$

Below we show that, as long as $\mathbf{A} \leftarrow \mathbb{Z}_q^{m \times n}$, $m > n \log q + \lambda$ and $q > 4\tilde{B}$, then the LWE map $\mathbb{Z}_q^n \times [-\tilde{B}, \tilde{B}]^m \mapsto \mathbb{Z}_q : (\mathbf{s}, \mathbf{e}) \rightarrow \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ is injection with overwhelming probability, which implies the quasi unique responses of our Σ -protocol.

Lemma 15 (Unique Response of Σ). *If $m > n \log q + \lambda$ and $q > 2\tilde{B}$, then the Σ -protocol described above has quasi unique responses. More precisely,*

$$\Pr \left[\exists (x, a, \text{ch}, z^{(0)}, z^{(1)}) \text{ s.t. } \begin{matrix} (z^{(0)} \neq z^{(1)}) \wedge \\ \Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z^{(b)} = 1) \end{matrix} \right] \leq 2^{-\lambda}.$$

Proof. Suppose there exists $(x, a, \text{ch}, z^{(0)}, z^{(1)})$ s.t. $(z^{(0)} \neq z^{(1)})$ and $\Sigma.\text{Vrfy}(\text{crs}, x, a, \text{ch}, z^{(b)}) = 1$ for both $b = 0, 1$. By our construction, we can parse $x = \mathbf{c}, a = \mathbf{c}_0$ and $z^{(0)} = (\mathbf{s}_{\text{mix}}^{(0)}, \mathbf{e}_{\text{mix}}^{(0)}), z^{(1)} = (\mathbf{s}_{\text{mix}}^{(1)}, \mathbf{e}_{\text{mix}}^{(1)})$. Since the verification passes, it must hold that:

$$\begin{aligned} \|\mathbf{e}_{\text{mix}}^{(0)}\|_{\infty}, \|\mathbf{e}_{\text{mix}}^{(1)}\|_{\infty} &\leq \tilde{B}/2, \\ \mathbf{c}_0^{\top} + \text{ch} \cdot \mathbf{c}^{\top} &= \mathbf{s}_{\text{mix}}^{(0)\top} \mathbf{A} + \mathbf{e}_{\text{mix}}^{(0)\top} \pmod{q}, \\ \mathbf{c}_0^{\top} + \text{ch} \cdot \mathbf{c}^{\top} &= \mathbf{s}_{\text{mix}}^{(1)\top} \mathbf{A} + \mathbf{e}_{\text{mix}}^{(1)\top} \pmod{q}. \end{aligned}$$

This implies that $(\mathbf{s}_{\text{mix}}^{(0)}, \mathbf{e}_{\text{mix}}^{(0)}) \neq (\mathbf{s}_{\text{mix}}^{(1)}, \mathbf{e}_{\text{mix}}^{(1)})$ but $\mathbf{s}_{\text{mix}}^{(0)\top} \mathbf{A} + \mathbf{e}_{\text{mix}}^{(0)\top} = \mathbf{s}_{\text{mix}}^{(1)\top} \mathbf{A} + \mathbf{e}_{\text{mix}}^{(1)\top} \pmod{q}$. Note that if $\mathbf{s}_{\text{mix}}^{(0)} = \mathbf{s}_{\text{mix}}^{(1)}$ implies $\mathbf{e}_{\text{mix}}^{(0)} = \mathbf{e}_{\text{mix}}^{(1)}$. Thus it must hold that $\mathbf{s}_{\text{mix}}^{(0)} \neq \mathbf{s}_{\text{mix}}^{(1)}$. As $\mathbf{A}$ is uniformly random in $\mathbb{Z}_q^{n \times m}$, then for any fix $\mathbf{s}$, $\mathbf{s}^{\top}$ is uniformly random in $\mathbb{Z}_q^n$ w.h.p. Consequently, we have the probability

$$\Pr[(\mathbf{s}_{\text{mix}}^{(0)} - \mathbf{s}_{\text{mix}}^{(1)})^{\top} \mathbf{A} = \mathbf{e}_{\text{mix}}^{(1)\top} - \mathbf{e}_{\text{mix}}^{(0)\top}] \approx \frac{q^n \cdot \tilde{B}^m}{q^m} \leq q^n \cdot 2^{-m} \leq 2^{-\lambda},$$

and Lemma 15 follows. $\square$

Similarly, the soundness error of above Σ -protocol presented in this section can be made negligibly small via $O(\lambda)$ parallel repetitions. With such the above Σ -protocol for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})}$, we leverage FS-transformation in random oracle to obtain a tag-based QA-NIZK for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})}$.

Tag-based OTSS QA-NIZK for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})}$. The syntax of tag-based QA-NIZK is shown in Def. 7 and 8. Note that in the random oracle, we can simulate a proof without the trapdoor td_{crs} , so in the following description we will ignore SimGen and always set $\text{td}_{\rho} = \perp$. Let $\mathcal{T}$ denote that tag space. Below we construct a tag-based OTSS QA-NIZK for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ specified by (31) and (32) from a Σ -protocol for $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})}$ and a hash function $\mathbf{H} : \{0, 1\}^* \rightarrow \{0, 1\}^{\lambda}$ modeled as a random oracle.

- $\text{crs} \leftarrow \text{CRSGen}(\rho)$: Invoke $\text{crs}_{\Sigma} \leftarrow \Sigma.\text{CRSGen}(\rho)$ and return $\text{crs} = \text{crs}_{\Sigma}$.
- $\pi \leftarrow \text{Prove}(\text{crs}, \tau, x, w)$: For $i \in [\lambda]$, invoke $(a_i, \text{st}_i) \leftarrow \Sigma.\text{Prove}_1(\text{crs}, x, w)$ and compute $(\text{ch}_1, \dots, \text{ch}_{\lambda}) := \mathbf{H}(x, \tau, \mathbf{a} := (a_1, \dots, a_{\lambda}))$. For $i \in [\lambda]$, invoke $z_i \leftarrow \Sigma.\text{Prove}_2(\text{crs}, x, w, a_i, \text{st}_i, \text{ch}_i)$ and set $\mathbf{z} = (z_1, \dots, z_{\lambda})$. Return the proof $\pi := (\mathbf{a}, \mathbf{z})$.
- $0/1 \leftarrow \text{Vrfy}(\text{crs}, \tau, x, \pi)$: Compute $(\text{ch}_1, \dots, \text{ch}_{\lambda}) := \mathbf{H}(\mathbf{a}, x, \tau)$. Parse $\mathbf{a} = (a_1, \dots, a_{\lambda})$ and $\mathbf{z} = (z_1, \dots, z_{\lambda})$. If for all $i \in [\lambda]$, $\Sigma.\text{Vrfy}(\text{crs}, x, a_i, \text{ch}_i, z_i) = 1$, return 1. Otherwise, return 0.

- $\pi \leftarrow_s \text{Sim}(\text{crs}, \tau, x)$: Sample $(\text{ch}_1, \dots, \text{ch}_\lambda) \leftarrow_s \{0, 1\}^\lambda$. For $i \in [\lambda]$, invoke $(\tilde{a}_i, \tilde{z}_i) \leftarrow_s \Sigma.\text{Sim}(\text{crs}, x, \text{ch}_i)$. Let $\tilde{\mathbf{a}} := (\tilde{a}_1, \dots, \tilde{a}_\lambda)$ and $\tilde{\mathbf{z}} := (\tilde{z}_1, \dots, \tilde{z}_\lambda)$. Program $\text{H}(x, \tau, \tilde{\mathbf{a}}) = (\text{ch}_1, \dots, \text{ch}_\lambda)$ and return $\pi = (\tilde{\mathbf{a}}, \tilde{\mathbf{z}})$.

Theorem 6. *If Σ is a Σ -protocol for $\mathcal{GL}_A^{(\text{LWE})}$ that has quasi unique responses and H is a random oracle maps $\{0, 1\}^* \rightarrow \{0, 1\}^\lambda$, then the above construction is a tag-based OTSS QA-NIZK.*

Proof of Theorem 6. The completeness and zero-knowledge can be easily reduced to the completeness and zero-knowledge property of Σ -protocol, respectively. We below focus on the one-time simulation soundness.

When the adversary $\mathcal{A}$ issues a query of $\mathcal{O}_{\text{SIM}}(\tau, x)$, the challenger invokes $\pi \leftarrow_s \text{Sim}(\text{crs}, \tau, x)$ and returns π to $\mathcal{A}$ (note that the Sim algorithm can always generate valid proof π for any x) and add (τ^*, x^*, π^*) to $\mathcal{Q}_{\text{SIM}}$.

Assume that at some moment, $\mathcal{A}$ outputs $(\tau^*, x^*, \pi^* = (\mathbf{a}^*, \mathbf{z}^*))$ such that

$$(x^* \notin \tilde{\mathcal{L}}_\rho) \wedge ((\tau^*, x^*, \pi^*) \notin \mathcal{Q}_{\text{SIM}}) \wedge (\text{Vrfy}(\text{crs}, \tau^*, x^*, \pi^*) = 1).$$

Let $\mathbf{a}^* = (a_1^*, \dots, a_\lambda^*)$ and $\mathbf{z}^* = (z_1^*, \dots, z_\lambda^*)$. We consider the following two cases:

- $(\tau^*, x^*, (\mathbf{a}^*, \cdot)) \notin \mathcal{Q}_{\text{SIM}}$. In this case, since $x^* \notin \tilde{\mathcal{L}}_A^{(\text{LWE})}$, by the special soundness of Σ , for each $i \in [\lambda]$, (x^*, a_i^*) uniquely determines the challenge ch_i^* s.t. $\Sigma.\text{Vrfy}(x^*, a_i^*, \text{ch}_i^*, z_i) = 1$ for some third message z_i . Thus, there is only λ -bit string $(\text{ch}_1^*, \dots, \text{ch}_\lambda^*)$ s.t. all λ verifications of $\Sigma.\text{Vrfy}(x^*, a_i, \text{ch}_i, z_i) = 1$ pass. Since the output of $\text{H}(\tau^*, x^*, \mathbf{a}^*)$ is unpredictable, the probability $\Pr[\text{H}(\tau^*, x^*, \mathbf{a}^*) = (\text{ch}_1^*, \dots, \text{ch}_\lambda^*)] = 2^{-\lambda}$ is negligible.
- $(\tau^*, x^*, (\mathbf{a}^*, \cdot)) \in \mathcal{Q}_{\text{SIM}}$. In this case, since $x^* \notin \tilde{\mathcal{L}}_A^{(\text{LWE})}$, similar to the analysis above, by the special soundness of Σ , $(x^*, \mathbf{a}^*)$ uniquely determines the λ -bit challenge string $(\text{ch}_1^*, \dots, \text{ch}_\lambda^*)$ all λ verifications of $\Sigma.\text{Vrfy}(x^*, a_i^*, \text{ch}_i^*, z_i) = 1$ pass for some z_i . So the only method of $\mathcal{A}$ to make $(\tau^*, x^*, (\mathbf{a}^*, \mathbf{z}^*)) \notin \mathcal{Q}_{\text{SIM}}$ is to find at least one different $\tilde{z}_i^* \neq z_i^*$ for some $i \in [\lambda]$ and $\Sigma.\text{Vrfy}(a_i^*, \text{ch}_i^*, \tilde{z}_i^*) = 1$. Since the Σ -protocol has quasi unique responses, the probability of $\mathcal{A}$ can find such $\tilde{z}_i^*$ is negligible.

In conclusion, in both cases, $\mathcal{A}$ breaks the simulation soundness game with only negligible advantage and Theorem 6 follows. $\square$

We note that in the above proof, we do not require the $\mathcal{A}$ can only queried $\mathcal{O}_{\text{SIM}}$ once. Thus, the same proof remains valid in the case of unbounded simulation soundness.

F CPA-Secure Threshold Decryption Scheme for Small N

In this section, we propose a CPA-secure threshold decryption scheme TD_0 for small N with polynomial modulus based on the LWE assumption in the standard model.

Our construction will utilize the ReRand algorithm introduced in [31], as recalled below.

Lemma 16 (Noise Rerandomization [31, Lemma 1]). *Let q, ℓ, m be positive integers and let r a positive real satisfying $r > \max\{\omega(\sqrt{\log m}), \omega(\sqrt{\log \ell})\}$. Let σ be a positive real number. There exists a PPT algorithm **ReRand** satisfying the following property: for any $\mathbf{b} \in \mathbb{Z}_q^m$, any $\mathbf{V} \in \mathbb{Z}_q^{m \times \ell}$, if $\sigma > s_1(\mathbf{V})$, then*

$$\{\mathbf{b}'^\top \mid \mathbf{x} \leftarrow D_{\mathbb{Z}^m, r}, \mathbf{b}'^\top \leftarrow \text{ReRand}(\mathbf{V}, \mathbf{b}^\top + \mathbf{x}^\top, r, \sigma)\} \approx_s \{\tilde{\mathbf{b}}'^\top = \mathbf{b}^\top \mathbf{V} + \mathbf{x}'^\top \mid \mathbf{x}' \leftarrow D_{\mathbb{Z}^\ell, 2r\sigma}\}.$$

For completeness, we recall the details of the **ReRand** algorithm in Fig. 11.

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 $\mathbf{b}'^\top \leftarrow \text{ReRand}(\mathbf{V}, \mathbf{b}^\top + \mathbf{x}^\top, r, \sigma) :$ 
 $\mathbf{c} \leftarrow D_r^m, \mathbf{d} \leftarrow D_{\sqrt{2r}(\sigma^2 \mathbf{I}_\ell - \mathbf{V}^\top \mathbf{V})^{1/2}}$  //these are continuous Gaussian distributions
 $\mathbf{f}^\top \leftarrow D_{\mathbb{Z}^{1 \times \ell} - (\mathbf{c}^\top \mathbf{V} + \mathbf{d}^\top), \sqrt{2r}\sigma}$ 
 $\mathbf{b}'^\top := ((\mathbf{b}^\top + \mathbf{x}^\top) + \mathbf{c}^\top) \mathbf{V} + \mathbf{d}^\top + \mathbf{f}^\top$ 
Return  $\mathbf{b}'^\top$ 

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Fig. 11. The **ReRand** algorithm in [31, Lemma 1].

F.1 Parameters and Building Blocks

Let t, N be integers, where $t \leq N$. The construction of TD_0 uses the following parameters, and we give a suggestion for asymptotic parameter setting in Table 4, so that all requirements in this section are met. (We refer to Remark 3 for more efficient choices of the modulus q when only CPA security is required and robustness is not considered.)

- Let $n, m, q, p, M \in \mathbb{N}$, where $q = M \cdot p$ for primes M, p .
- Let $\sigma_k, \sigma_0, \sigma_1, \sigma_y \in \mathbb{R}$ and $B_k, B_0, B_1, B_y \in \mathbb{R}$, s.t. $B_k \geq \alpha \sigma_k, B_0 \geq \alpha \sigma_0, B_1 \geq \alpha \sigma_1, B_y \geq \alpha \cdot 2\sigma_0 \sigma_y$ for $\alpha = \omega(\sqrt{\log \lambda})$. According to Lemma 5, B_k, B_0, B_1, B_y are tail bounds of the distributions $D_{\mathbb{Z}, \sigma_k}, D_{\mathbb{Z}, \sigma_0}, D_{\mathbb{Z}, \sigma_1}, D_{\mathbb{Z}, 2\sigma_0 \sigma_y}$, respectively.
- Let $\tilde{B}_k, \tilde{B}_0, \tilde{B}_y \in \mathbb{R}$, s.t. $\tilde{B}_k \geq \gamma B_k, \tilde{B}_0 \geq \gamma B_0$ for $\gamma = \omega(\log \lambda \sqrt{\log m}) \cdot \sqrt{m}$ and $\tilde{B}_y \geq \omega(\log \lambda) \cdot B_y$.

Table 4. Asymptotic parameter setting for TD_0 , where the $O(\cdot)$ notation is omitted.

Parameters	n	m	q	p	M	σ_k	σ_0	σ_1	σ_y
Setting	$\lambda \log \lambda$	$\lambda^2 \log \lambda$	$N^t \cdot \lambda^7 \log^8 \lambda$	$N^t \cdot \lambda^6 \log^8 \lambda$	λ	λ	λ	$N^{t/2} \cdot \lambda^3 \log^{1.5} \lambda$	$\lambda^2 \log^{1.5} \lambda$

Para.	B_k	B_0	B_1	B_y	$\tilde{B}_k$	$\tilde{B}_0$	$\tilde{B}_y$
Setting	$\lambda \log \lambda$	$\lambda \log \lambda$	$N^{t/2} \cdot \lambda^3 \log^{2.5} \lambda$	$\lambda^3 \log^{2.5} \lambda$	$\lambda^2 \log^{3.5} \lambda$	$\lambda^2 \log^{3.5} \lambda$	$\lambda^3 \log^{3.5} \lambda$

Our scheme TD_0 also employs the following building blocks.

- The RSS scheme $\text{RSS} = (\text{Share}, \text{Select}, \text{Recover})$ (cf. Fig. 1 in Subsect. 2.2).
- An adaptively sound $\Pi = (\text{CRSGen}, \text{Prove}, \text{Vrfy}, \text{SimGen}, \text{Sim})$ ¹¹ (cf. Def. 8) for gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{LWE})} = (\mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})})$ defined as follows, with language parameter $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$:

$$\begin{aligned}\mathcal{L}_{\mathbf{A}}^{(\text{LWE})} &:= \{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-B_0, B_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \}, \\ \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{LWE})} &:= \{ \mathbf{c} \mid \exists \mathbf{s} \in \mathbb{Z}^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m \text{ s.t. } \mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \}.\end{aligned}$$

- An adaptively sound NIZK scheme $\overline{\Pi} = (\overline{\text{CRSGen}}, \overline{\text{Prove}}, \overline{\text{Vrfy}}, \overline{\text{SimGen}}, \overline{\text{Sim}})$ (cf. Def. 7) for gap language $\mathcal{GL}_{\mathbf{A}}^{(\text{lin})} = (\mathcal{L}_{\mathbf{A}}^{(\text{lin})}, \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})})$ defined as follows, with language parameter $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and $B^* = mB_0B_k + B_y, \tilde{B}^* := m\tilde{B}_0\tilde{B}_k + \tilde{B}_y$:

$$\begin{aligned}\mathcal{L}_{\mathbf{A}}^{(\text{lin})} &:= \left\{ (\mathbf{pk}, h, \mathbf{c}) \mid \begin{array}{l} \exists (\mathbf{k} \in [-B_k, B_k]^m, \\ e' \in [-B^*, B^*]) \text{ s.t. } \mathbf{pk} = \mathbf{A}\mathbf{k} \wedge \\ h = \mathbf{c}^\top \mathbf{k} + e' \end{array} \right\}, \\ \tilde{\mathcal{L}}_{\mathbf{A}}^{(\text{lin})} &:= \left\{ (\mathbf{pk}, h, \mathbf{c}) \mid \begin{array}{l} \exists (\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} \in [-\tilde{B}_0, \tilde{B}_0]^m, \\ y \in [-\tilde{B}^*, \tilde{B}^*]) \text{ s.t. } \mathbf{c} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \wedge \\ h = \mathbf{s}^\top \mathbf{pk} + y \end{array} \right\}.\end{aligned}$$

- The re-randomization algorithm ReRand as defined in Lemma 16.

The NIZKs used in this section are the same as those in Sect. 4, and we refer to Appendix E.2 and Appendix E.3 for details.

F.2 Construction of TD_0

Our (t, N) -threshold decryption scheme $\text{TD}_0 = (\text{Setup}, \text{KGen}, \text{Enc}, \text{PDec}, \text{PVer}, \text{Comb})$ for small N is shown in Fig. 12.

Correctness. The correctness of our scheme TD_0 follows from the completeness of Π and $\overline{\Pi}$, the correctness of RSS (cf. Subsect. 2.2), the property of ReRand (cf. Lemma 16), and our parameter selection. Firstly, it is clear that the Vrfy and $\overline{\text{Vrfy}}$ algorithms in PDec , PVer and Comb will output 1 except with negligible probability, by the completeness of Π and $\overline{\Pi}$.

Next, we show that the Comb algorithm can recover the message μ correctly except with negligible probability. For any message $\mu \in \mathbb{Z}_p$, the encryption algorithm $\text{Enc}(\mathbf{pk}, \mu)$ generates ciphertext $\text{ct} = (\mathbf{c}, d, \pi)$ by computing $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, $d := \mathbf{s}^\top \mathbf{A}\mathbf{k} + e' + p \cdot \mu$ with $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$, $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$. Then for $(h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}, \pi_i)$ generated by the partial decryption algorithm $\text{PDec}(\text{sk}_i, \text{ct})$, each $h_{i,I}$ is computed by $h_{i,I} \leftarrow \text{ReRand}(\mathbf{k}_I, \mathbf{c}^\top, \sigma_0, \sigma_y)$. Since $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ and $\sigma_y > s_1(\mathbf{k}_I)$, by the property of ReRand (cf. Lemma 16), we know that $h_{i,I} \leftarrow \text{ReRand}(\mathbf{k}_I, \mathbf{c}^\top, \sigma_0, \sigma_y)$ is statistically close to $h_{i,I} = \mathbf{s}^\top \mathbf{A}\mathbf{k}_I + \tilde{e}_{i,I}$ where $\tilde{e}_{i,I} \leftarrow D_{\mathbb{Z}, 2\sigma_0\sigma_y}$. Thus for any $S \subseteq [N]$ with $|S| \geq t$, in

¹¹ In the case of CPA security, we employ NIZK only for robustness and do not require that the ciphertext is non-malleable. Thus, we do not need that Π is gap-based.

<u>pp</u> $\leftarrow$ Setup: $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$ $\text{crs} \leftarrow \text{CRSGen}(\mathbf{A}), \overline{\text{crs}} \leftarrow \overline{\text{CRSGen}}(\mathbf{A})$ Return $\text{pp} := (\mathbf{A}, \text{crs}, \overline{\text{crs}})$ <u>(pk, {(pk_i, sk_i)}_{i ∈ [N]})</u> $\leftarrow$ KGen(pp): $(\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N)) \leftarrow \text{Share}(N, t, D_{\mathbb{Z}^m, \sigma_k})$ $\text{pk} := \mathbf{A}\mathbf{k} \in \mathbb{Z}_q^n, \text{sk} := \mathbf{k} \in \mathbb{Z}_q^m$ $\mathcal{I} := \{I \mid I \subseteq [N], I = N - t + 1\}$ For $i \in [N]$: Parse $\text{sk}_i = \{\mathbf{k}_I \mid I \in \mathcal{I}, i \in I\}$ $\text{pk}_i := \{\text{pk}_{i,I} := \mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_i}$ Return $(\text{pk}, \{(\text{pk}_i, \text{sk}_i)\}_{i \in [N]})$ <u>ct</u> $\leftarrow$ Enc(pk, $\mu \in \mathbb{Z}_p$): $\mathbf{s} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}, \mathbf{e}' \leftarrow D_{\mathbb{Z}, \sigma_1}$ $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ $d := \mathbf{s}^\top \text{pk} + \mathbf{e}' + p \cdot \mu \in \mathbb{Z}_q$ $\pi \leftarrow \text{Prove}(\text{crs}, \mathbf{c} \in \mathcal{L}_{\mathbf{A}}^{(\text{LWE})}, (\mathbf{s}, \mathbf{e}))$ Return $\text{ct} := (\mathbf{c}, d, \pi)$	<u>(h_i, π_i)</u> $\leftarrow$ PDec(sk_i, ct = (c, d, π)): If Vrfy(crs, c, π) = 0: Return $\perp$ For each $\mathbf{k}_I \in \text{sk}_i$: $h_{i,I} \leftarrow \text{ReRand}(\mathbf{k}_I, \mathbf{c}^\top, \sigma_0, \sigma_y)$ $\mathbf{e}'_{i,I} := h_{i,I} - \mathbf{c}^\top \mathbf{k}_I$ $\pi_{i,I} \leftarrow \overline{\text{Prove}}(\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}) \in \mathcal{L}_{\mathbf{A}}^{(\text{lin})}, (\mathbf{k}_I, \mathbf{e}'_{i,I}))$ Return $(h_i := \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}, \pi_i := \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i})$ <u>0/1</u> $\leftarrow$ PVer(pk_i, ct = (c, d, π), h_i, π_i): Parse $h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ and $\pi_i = \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ For each $\text{pk}_{i,I} \in \text{pk}_i$: If Vrfy($\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}), \pi_{i,I}) = 0$: Return 0 Return 1 <u>μ</u> $\leftarrow$ Comb({pk_i}_{i ∈ [N]}, ct = (c, d, π), S, {(h_i, π_i)}_{i ∈ S}): If Vrfy(crs, c, π) = 0: Return $\perp$ $T := \emptyset$ For each $i \in S$: If PVer(pk_i, ct, h_i, π_i) = 1: $T := T \cup \{i\}$ If $T < t$: Return $\perp$ $\mathbf{w} = (w_i)_{i \in T} \leftarrow \text{Select}(T)$ For $i \in T$: Parse $h_i = \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ $h := \sum_{i \in T} \sum_{I \in w_i} h_{i,I}$ Return $\mu := \lceil (d - h)/p \rceil$
--	--

Fig. 12. Our (t, N) -threshold decryption scheme TD_0 for small N .

the combination algorithm $\text{Comb}(\{\text{pk}_i\}_{i \in [N]}, \text{ct}, S, \{(h_i, \pi_i)\}_{i \in S})$, we have $T = S$ and

$$\begin{aligned}
 h &:= \sum_{i \in S} \sum_{I \in w_i} h_{i,I} = \sum_{i \in S} \sum_{I \in w_i} (\mathbf{s}^\top \mathbf{A}\mathbf{k}_I + \tilde{e}_{i,I}) \\
 &= \mathbf{s}^\top \mathbf{A} (\sum_{i \in S} \sum_{I \in w_i} \mathbf{k}_I) + \sum_{i \in S} \sum_{I \in w_i} \tilde{e}_{i,I} = \mathbf{s}^\top \mathbf{A}\mathbf{k} + \sum_{i \in S} \sum_{I \in w_i} \tilde{e}_{i,I},
 \end{aligned}$$

where the last equality holds by the correctness of RSS. Therefore, we have $d - h = p \cdot \mu + \mathbf{e}' - \sum_{i \in S} \sum_{I \in w_i} \tilde{e}_{i,I}$. By our parameter setting in Table 4 and the tail bound (Lemma 5), the noise term $\mathbf{e}' - \sum_{i \in S} \sum_{I \in w_i} \tilde{e}_{i,I}$ is bounded by $B_1 + CB_y \leq p/2$ with $C := \binom{N}{N-t+1}$ except with negligible probability. Consequently, $\lceil (d - h)/p \rceil$ correctly recovers μ .

F.3 Security Proof of TD_0

Theorem 7 (IND-TCPA Security of TD_0). Assume that (i) Π has zero-knowledge, (ii) the matrix hint-LWE assumption holds for the parameters chosen above, then the proposed scheme TD_0 in Fig. 12 is IND-TCPA secure.

More precisely, for any adversary $\mathcal{A}$, there exists PPT $\mathcal{B}_1$ and $\mathcal{B}_2$ s.t.

$$\text{Adv}_{\text{TD}_0, \mathcal{A}}^{\text{CPA}}(\lambda) \leq \text{Adv}_{\Pi, \mathcal{B}_1}^{\text{zk}}(\lambda) + \text{Adv}_{q, m, n, D_{\mathbb{Z}^m, \sigma_0/2}}^{\text{LWE}}(\lambda) + \text{negl}(\lambda).$$

Proof of Theorem 7. We use a series of games G_0 - G_5 to show the advantage $\text{Adv}_{\text{TD}_0, \mathcal{A}}^{\text{CCA}}(\lambda)$ is negligible, where G_0 is the original IND-TCCA experiment and $\mathcal{A}$ has no advantage in G_5 .

Game G_0 : This is the original IND-TCPA experiment $\text{Exp}_{\text{TD}_0, \mathcal{A}}^{\text{CPA}}$ (cf. Fig. 2 without $\mathcal{O}_{\text{DEC}}$). By definition, we have $\text{Adv}_{\text{TD}_0, \mathcal{A}}^{\text{CPA}}(\lambda) = |\Pr[G_0 = 1] - \frac{1}{2}|$.

Let $\text{pp} = (\mathbf{A}, \text{crs}, \overline{\text{crs}})$. The challenger invokes $(\mathbf{k}, (\text{sk}_1, \dots, \text{sk}_N)) \leftarrow_{\$} \text{RSS}(N, t, D_{\mathbb{Z}^m, \sigma_k})$, sets $\text{pk} := \mathbf{A}\mathbf{k}$, $\text{pk}_i := \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \text{sk}_i}$ for each $i \in [N]$, and returns $(\text{pp}, \text{pk}, \{\text{pk}_i\}_{i \in [N]})$ to $\mathcal{A}$. The challenger initializes $\text{Cor} := \emptyset$, $\text{Hon} := [N]$. Then it answers $\mathcal{A}$'s $\mathcal{O}_{\text{COR}}$, $\mathcal{O}_{\text{ENC}}$ queries adaptively and generates the challenge ciphertext ct^* as follows.

- For corruption query $\mathcal{O}_{\text{COR}}(i)$, the challenger adds i to Cor and returns sk_i .
- For encryption query $\mathcal{O}_{\text{ENC}}(\mu)$, the challenger first generates $\text{ct} = (\mathbf{c}, d, \pi)$ as follows:
 - Sample $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d := \mathbf{s}^\top \text{pk} + e' + p \cdot \mu \in \mathbb{Z}_q$.
 - Generate a proof $\pi \leftarrow_{\$} \text{Prove}(\text{crs}, \mathbf{c}, (\mathbf{s}, \mathbf{e}))$.

Then, for each $i \in [N]$, the challenger invokes $\text{PDec}(\text{sk}_i, \text{ct})$ to obtain (h_i, π_i) as follows.

- For each $\mathbf{k}_I \in \text{sk}_i$, compute $h_{i,I} \leftarrow_{\$} \text{ReRand}(\mathbf{k}_I, \mathbf{c}, \sigma_0, \sigma_y)$, $e'_{i,I} := h_{i,I} - \mathbf{c}^\top \mathbf{k}_I$, and generate a proof $\pi_{i,I} \leftarrow \overline{\text{Prove}}(\overline{\text{crs}}, (\text{pk}_{i,I}, h_{i,I}, \mathbf{c}), (\mathbf{k}_I, e'_{i,I}))$.
- Set $h_i := \{h_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$ and $\pi_i := \{\pi_{i,I}\}_{\mathbf{k}_I \in \text{sk}_i}$.

Finally, the challenger returns $(\text{ct}, \{h_i, \pi_i\}_{i \in [N]})$ to $\mathcal{A}$.

- On receiving (μ_0, μ_1) , the challenger chooses $\beta \leftarrow_{\$} \{0, 1\}$ and generates the challenge ciphertext $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as an encryption of μ_β as follows:
 - Sample $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^{*\top} := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.
 - Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d^* := \mathbf{s}^\top \text{pk} + e' + p \cdot \mu_\beta \in \mathbb{Z}_q$.
 - Generate a proof $\pi^* \leftarrow_{\$} \text{Prove}(\text{crs}, \mathbf{c}^*, (\mathbf{s}, \mathbf{e}))$.

Game G_1 : It is the same as G_0 , except for the reply to encryption query $\mathcal{O}_{\text{ENC}}(\mu)$.

- Generate $\text{ct} = (\mathbf{c}, d, \pi)$, where $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ with $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$.
- For each $\text{pk}_{i,I} = \mathbf{A}\mathbf{k}_I \in \text{pk}_i$, compute $h_{i,I} := \mathbf{s}^\top \text{pk}_{i,I} + e'$ with $e' \leftarrow D_{\mathbb{Z}, 2\sigma_0\sigma_y}$, instead of $h_{i,I} \leftarrow_{\$} \text{ReRand}(\mathbf{k}_I, \mathbf{c}, \sigma_0, \sigma_y)$.

According to Lemma 16, the output of $\text{ReRand}(\mathbf{k}_I, \mathbf{c}, \sigma_0, \sigma_y)$ is statistically indistinguishable from $h_{i,I} := \mathbf{s}^\top \mathbf{A}\mathbf{k}_I + e' = \mathbf{s}^\top \text{pk}_{i,I} + e'$ with $e' \leftarrow D_{\mathbb{Z}, 2r\sigma}$. Thus, we have $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{negl}(\lambda)$.

Game G_2 : It is the same as G_1 except for the generation of challenge ciphertext ct^* . More precisely, the challenger generates $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as follows.

Step 1: Sample $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^{*\top} := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.

Step 2: Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d^* := \mathbf{s}^\top \text{pk} + e' + p \cdot \mu_\beta \in \mathbb{Z}_q$.

Step 3: Invoke $\pi^* \leftarrow_{\$} \text{Sim}(\text{crs}, \mathbf{c}^*)$.

By the zero-knowledge property of Π , $|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq \text{Adv}_{\Pi, \mathcal{B}_1}^{\text{zk}}(\lambda)$.

Game G_3 : It is the same as G_2 except for the generation of challenge ciphertext ct^* . More precisely, the challenger generates $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as follows.

Step 1: Sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$ and compute $\mathbf{c}^{*\top} := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$.

Step 2: Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and compute $d^* := \mathbf{c}^{*\top} \mathbf{k} + e' - \mathbf{e}^\top \mathbf{k} + p \cdot \mu_\beta \in \mathbb{Z}_q$.

Step 3: Invoke $\pi^* \leftarrow \text{Sim}(\text{crs}, \mathbf{c}^*)$.

Since $\mathbf{pk} = \mathbf{A}\mathbf{k}$ and $\mathbf{c}^{*\top} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$, the d^* in $\mathbf{G}_2$ equals $\mathbf{s}^\top \mathbf{A}\mathbf{k} + e' + p \cdot \mu_\beta = (\mathbf{c}^{*\top} - \mathbf{e}^\top) \mathbf{k} + e' + p \cdot \mu_\beta$, which is just the d^* in $\mathbf{G}_3$. So the change is conceptual and we have $\Pr[\mathbf{G}_2 = 1] = \Pr[\mathbf{G}_3 = 1]$.

Game $\mathbf{G}_4$: It is the same as $\mathbf{G}_3$ except for the generation of challenge ciphertext ct^* . More precisely, the challenger generates $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$ as follows.

Step 1: Sample $\mathbf{c}^{*\top} \leftarrow \mathbb{Z}_q^{1 \times m}$.

Step 2: Sample $e' \leftarrow D_{\mathbb{Z}, \sigma_1}$ and $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, \sigma_0}$. Compute $d^* := \mathbf{c}^{*\top} \cdot \mathbf{k} + e' - \mathbf{e}^\top \cdot \mathbf{k} + p \cdot \mu_\beta \in \mathbb{Z}_q$.

Step 3: Compute $\tau^* := H(\mathbf{c}^*, d^*)$ and invoke $\pi^* \leftarrow \text{Sim}(\text{crs}, \tau^*, \mathbf{c}^*)$.

By the Matrix-hint LWE assumption (cf. Def. 2), the $\mathbf{c}^{*\top} = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top$ in $\mathbf{G}_3$ is computationally indistinguishable from the $\mathbf{c}^{*\top} \leftarrow \mathbb{Z}_q^{1 \times m}$ in $\mathbf{G}_4$, even conditioned on the $e' - \mathbf{e}^\top \cdot \mathbf{k}$ involved in the computation of d^* . Formally, we have the following claim, whose proof is essentially the same as that for Claim 1 in the proof of Theorem 1.

Claim 8. $|\Pr[\mathbf{G}_3 = 1] - \Pr[\mathbf{G}_4 = 1]| \leq \text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda)$, where $\text{para} := ((q, m, n, D_{\mathbb{Z}^m, \sigma_0}), (D_{\mathbb{Z}, \sigma_1}, [-CB_k, CB_k]^m))$ with $C = \binom{N}{N-t+1}$. By Lemma 2 our parameter setting, there exist $\mathcal{B}_2'$ s.t. $\text{Adv}_{\text{para}, \mathcal{B}_2}^{\text{Mat-hLWE}}(\lambda) \leq \text{Adv}_{q, m, n, D_{\mathbb{Z}^m, \sigma_0/2}, \mathcal{B}_2'}^{\text{LWE}}(\lambda)$.

Game $\mathbf{G}_5$: It is the same as $\mathbf{G}_4$ except for the generation of $\text{ct}^* = (\mathbf{c}^*, d^*, \pi^*)$. More precisely, the challenger implements steps 1-3 like $\mathbf{G}_4$ but step 2 is changed as follows.

Step 2: Sample $d^* \leftarrow \mathbb{Z}_q$.

Claim 9. $|\Pr[\mathbf{G}_4 = 1] - \Pr[\mathbf{G}_5 = 1]| \leq 1/2^\lambda$.

Proof. Let $\mathcal{I} := \{I \mid I \subseteq [N], |I| = N - t + 1\}$. Below we first analyze the information about $\mathbf{sk} = \mathbf{k}$ and its shares $\mathbf{sk}_i = \{\mathbf{k}_I \mid I \in \mathcal{I}, i \in I\}$ ($i \in [N]$) that might be leaked to $\mathcal{A}$ in $\mathbf{G}_4$ and $\mathbf{G}_5$ (except for the challenge ciphertext ct^*).

- Firstly, $\mathcal{A}$ obtains $\mathbf{pk} = \mathbf{A}\mathbf{k}$ and $\mathbf{pk}_1 = \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \mathbf{sk}_1}, \dots, \mathbf{pk}_N = \{\mathbf{A}\mathbf{k}_I\}_{\mathbf{k}_I \in \mathbf{sk}_N}$. Since $\cup_{i \in [N]} \mathbf{sk}_i = \{\mathbf{k}_I\}_{I \in \mathcal{I}}$, $\mathcal{A}$ in fact gets the values of $\mathbf{A}\mathbf{k}$ and $\{\mathbf{A}\mathbf{k}_I\}_{I \in \mathcal{I}}$.
- Through $\mathcal{O}_{\text{COR}}$ queries, $\mathcal{A}$ obtains $\mathbf{sk}_i = \{\mathbf{k}_I \mid i \in I, I \in \mathcal{I}\}$ for $i \in \text{Cor}$. Given at most $t - 1$ corruptions, there must exist $\tilde{I} \in \mathcal{I}$ such that $\mathbf{k}_{\tilde{I}} \notin \{\mathbf{sk}_i\}_{i \in \text{Cor}}$.
- Through $\mathcal{O}_{\text{ENC}}$ queries, the challenger generates ct using public key $\mathbf{pk}$ and generates $\{h_i\}_{i \in [N]}$ using $\mathbf{pk}_i$ (due to the game change in $\mathbf{G}_1$), so $\mathcal{O}_{\text{ENC}}$ queries leaks nothing about $\mathbf{sk}$ beyond $\mathbf{pk}$ and $\{\mathbf{pk}_i\}_{i \in [N]}$.

Therefore, the information about $\mathbf{sk}$ and its shares $\{\mathbf{sk}_i\}_{i \in [N]}$ that $\mathcal{A}$ might learn is upper-bounded by $\text{View}_{\mathcal{A}} = (\mathbf{A}\mathbf{k}, \{\mathbf{A}\mathbf{k}_I\}_{I \in \mathcal{I}}, \{\mathbf{k}_I\}_{I \in \mathcal{I}, I \neq \bar{I}})$.

Next we will show that conditioned on $\mathcal{A}$'s view $\text{View}_{\mathcal{A}}$ in $\mathbf{G}_5/\mathbf{G}_4$, the min-entropy of $\mathbf{k}_{\bar{I}}$ is at least $2n \log q + 2\lambda$, i.e.,

$$\tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \mid \text{View}_{\mathcal{A}}) \geq 2n \log q + 2\lambda. \quad (36)$$

To see this, note that $\mathbf{A}\mathbf{k} = \sum_{I \in \mathcal{I}} \mathbf{A}\mathbf{k}_I$ and $\mathbf{k}_{\bar{I}}$ is independent of $\{\mathbf{k}_I\}_{I \in \mathcal{I}, I \neq \bar{I}}$, so $\tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \mid \text{View}_{\mathcal{A}}) = \tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \mid \mathbf{A}\mathbf{k}_{\bar{I}})$. By a similar analysis in the proof of Lemma 6 in Appendix B.2, in the case of $p/2 \geq \sqrt{m-n} \cdot \omega(\sqrt{n \log q}) + 2(m-n)\sigma_k \sqrt{\log \sigma_k}$ and $\sigma_k \geq \omega(\sqrt{n \log q}) \cdot \sqrt{\ln(2m(1+1\epsilon))}$, it holds that $\tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \bmod p \mid \mathbf{A}\mathbf{k}_{\bar{I}}) \geq (m-n) \log \sigma_k$. Note that $\mathbf{A}\mathbf{k}_{\bar{I}} \bmod M$ has at most M^n possible values, so $\tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \bmod M \mid \mathbf{A}\mathbf{k}_{\bar{I}}) \geq \mathbf{H}_{\infty}(\mathbf{k}_{\bar{I}} \bmod M) - n \log M$ by Lemma 11. Moreover, according to Corollary 1, it follows that $\mathbf{H}_{\infty}(\mathbf{k}_{\bar{I}} \bmod M) \geq m \log M - \log(1+4\epsilon)$. Therefore, $\tilde{\mathbf{H}}_{\infty}(\mathbf{k}_{\bar{I}} \mid \text{View}_{\mathcal{A}}) \geq (m-n) \log M - \log(1+4\epsilon)$, which further $\geq 2n \log q + 2\lambda$ by our parameter selection in Table 4. This yields (36).

Finally, based on (36), we will show that the d^* in $\mathbf{G}_4$ is statistically close to the $d^* \leftarrow_{\$} \mathbb{Z}_q$ in $\mathbf{G}_5$. Note that in $\mathbf{G}_4$, we have

$$\begin{aligned} d^* &= \mathbf{c}^{*\top} \cdot \mathbf{k} + e' - \mathbf{e}^{\top} \cdot \mathbf{k} + p \cdot \mu_{\beta} = (\mathbf{c}^* - \mathbf{e})^{\top} \mathbf{k} + e' + p \cdot \mu_{\beta} \\ &= (\mathbf{c}^* - \mathbf{e})^{\top} \mathbf{k}_{\bar{I}} + \sum_{I \in \mathcal{I}, I \neq \bar{I}} (\mathbf{c}^* - \mathbf{e})^{\top} \mathbf{k}_I + e' + p \cdot \mu_{\beta}. \end{aligned} \quad (37)$$

Recall that $\mathbf{c}^*$ is randomly chosen from $\mathbb{Z}_q^m$ and thus $\mathbf{c}^* - \mathbf{e}$ is also uniformly random over $\mathbb{Z}_q^m$. According to Leftover Hash Lemma (cf. Lemma 1), since (36) holds, it follows that $\Delta((\mathbf{c}^* - \mathbf{e}, (\mathbf{c}^* - \mathbf{e})^{\top} \mathbf{k}_{\bar{I}}), (\mathbf{c}^* - \mathbf{e}, u) \mid \text{View}_{\mathcal{A}}) \leq 1/2^{\lambda}$, where $u \leftarrow_{\$} \mathbb{Z}_q$. So the $(\mathbf{c}^* - \mathbf{e})^{\top} \mathbf{k}_{\bar{I}}$ in (37) is statistically close to uniform from the view of $\mathcal{A}$, and consequently, the d^* in $\mathbf{G}_4$ is statistically close to the $d^* \leftarrow_{\$} \mathbb{Z}_q$ in $\mathbf{G}_5$. This shows that $|\Pr[\mathbf{G}_4 = 1] - \Pr[\mathbf{G}_5 = 1]| \leq 1/2^{\lambda}$ and Claim 9 follows. $\blacksquare$

In $\mathbf{G}_5$, we have $d^* \leftarrow_{\$} \mathbb{Z}_q$, so the challenge bit β is totally hidden from $\mathcal{A}$ and it holds that $\Pr[\mathbf{G}_5 = 1] = \frac{1}{2}$. Putting all things together, Theorem 7 follows. $\square$

Theorem 8 (Robustness of TD_0). *If (i) Π has adaptive soundness, (ii) $\overline{\Pi}$ has adaptive soundness, (iii) $2C\tilde{B}^* \leq p/2$, then the proposed scheme TD_0 in Fig. 12 is RB secure.*

The proof is identical to that of Theorem 2, so we omit it here and refer to Appendix D.4 for a detailed proof.

Remark 3 (Improving Efficiency Without Robustness). The current parameter setting for TD_0 in Table 4 requires a relatively large polynomial modulus $q = 128N^t \cdot \lambda^7 \log^8 \lambda$, where the dominant constraint comes from the use of NIZK components $\Pi, \overline{\Pi}$ to ensure robustness and the requirement $2C\tilde{B}^* \leq p/2$ in Theorem 8.

If only CPA security is required while robustness is not considered, the scheme TD_0 can be made significantly more efficient by removing the NIZK components from the construction in Fig. 12. Concretely, the proof π is no longer needed

during encryption, and the proofs $\pi_{i,I}$ are no longer needed during partial decryption as well. This modification does not affect its CPA security, since the CPA experiment (cf. Fig. 2 without $\mathcal{O}_{\text{DEC}}$) only involves honestly generated (well-formed) ciphertexts. In this case, it only requires $CB_y \leq p/2$, which allows us to set $q = 8N^t \cdot \lambda^4 \log^{2.5} \lambda$.

Therefore, the CPA-secure TD_0 without robustness admits substantially better efficiency.

G Comparison with [20] and Concrete Efficiency of TD_2

In Table 5, we provide an asymptotic efficiency comparison between our TD_1 and the lattice-based TD scheme of [20] in the asynchronous setting. We focus on the LWE-based construction in [20] because it is the only existing adaptively CCA-secure non-interactive TD scheme prior to our work, but relies on an exponential modulus. All other existing non-interactive TD schemes achieve only selective security (cf. Table 1).

From Table 5 (a), we can see that the modulus of [20] is $q = O(2^\lambda)$, which is exponential in the security parameter λ , while the modulus of our TD_1 is $q = \sqrt{\ell} \cdot N^t \cdot \text{poly}(\lambda)$, which is polynomial for settings of small user number N .

Moreover, in Table 5 (a), we can see that both the ciphertext of our scheme and [20] includes a NIZK proof π , whose size $|\pi|$ is unspecified. This is because it is quite difficult to estimate the NIZK sizes, since the NIZKs of our scheme and [20] are both obtained from the LWE-based compiler of [36], which involves several cryptographic components whose parameters are unspecified. Nevertheless, as the NIZKs of our scheme and [20] are both obtained from the compiler of [36] for the same relation, so the two schemes would have comparable NIZK sizes $|\pi|$. Furthermore, the NIZK size will dominate the ciphertext size, so the two schemes would eventually have comparable ciphertext sizes. For practical deployment, one can replace the NIZKs by efficient RO-based NIZK schemes.

In Table 6, we provide the concrete parameters and efficiency for our TD_2 at the 128-bit security level. As reported by the lattice estimator (<https://github.com/malb/lattice-estimator>), the parameter setting in Table 6 indeed achieves ≥ 128 -bit security.

Finally, we stress that these efficiency estimates are only for reference. As stated in Sect. 1, the main contribution of our work is to investigating the feasibility of achieving non-interactive threshold decryption with adaptive CCA security and polynomial modulus, instead of efficiency optimization.

Table 5. Efficiency comparison with [20] in the asymptotic setting.

(a) Asymptotic efficiency comparison between TD_1 and DLN+21 [20]. Here $|\text{share}|$ denotes the number of secret shares each party holds and $|\pi|$ denotes the NIZK proof size. All quantities in the table are asymptotic, and the $O(\cdot)$ notation is omitted. Note that the message space in [20] is $\mathbb{Z}_M$ with $M = \Omega(N^{2.65} \cdot \lambda^3)$. To align with their setting, we set our modulus q also for the message space $\mathbb{Z}_M$ with $M = \Omega(N^{2.65} \cdot \lambda^3)$.

Scheme	n	m	q	$ \text{share} $
DLN+21 [20]	λ^2	λ^3	$2^\lambda \cdot N^{18.55} \cdot \lambda^{13.5}$	$N^{4.3}$
Our TD_1	$\lambda \log \lambda$	$\lambda^2 \log \lambda$	$\sqrt{\ell} \cdot N^{t+2.65} \cdot \lambda^{7.5}$	$\binom{N-1}{t-1}$

Scheme	Ciphertext size		Partial decryption size	
	$(\mathbf{c}, d) \in \mathbb{Z}_q^{m+1}$	π	$\{h_i \in \mathbb{Z}_q\}_{i \in [\text{share}]}$	$\{\pi_i\}_{i \in [\text{share}]}$
DLN+21 [20]	$\lambda^3(\lambda + \log N)$	$ \pi $	$ \text{share} \cdot (\lambda + \log N)$	$ \text{share} \cdot \pi $
Our TD_1	$\lambda^2 \log \lambda (\log \ell + t \log N + \log \lambda)$	$ \pi $	$ \text{share} \cdot (\log \ell + t \log N + \log \lambda)$	$ \text{share} \cdot \pi $

(b) The asymptotic parameter setting for [20] used in the comparison (a), which is mostly taken from [20], except that the modulus q is re-derived from their bounds. (We refer to Table 2 for the asymptotic parameter setting of our TD_1 used in (a), except that q is set for message space $\mathbb{Z}_M$ with $M = \Omega(N^{2.65} \cdot \lambda^3)$.)

Parameter	Asymptotic parameter setting in [20] used in (a)
n, m	$n = \Omega(\lambda^2), \quad m = \Omega(\lambda^2 \log q) = \Omega(\lambda^3)$
$e, d_S, d_{\max}$	$e = O(N^{5.3}), \quad d_S, d_{\max} \leq d = O(N^{5.3})$
σ	$\sigma = \Omega(\sqrt{e} m) = \Omega(N^{2.65} \lambda^{1.5})$
p	$p = \Omega(\sqrt{m} \sigma) = \Omega(N^{2.65} \lambda^3)$
δ, γ, r	$\delta = O(\log N), \quad \gamma = m = \Omega(\lambda^3), \quad r = \omega(\sqrt{\log m}) = \omega(\sqrt{\log \lambda})$
q	$q > 2^{\lambda+2} p d_S d_{\max} \delta \sigma^2 \gamma m \sqrt{m+1} r = \tilde{O}(2^\lambda N^{18.55} \lambda^{13.5})$

Table 6. Concrete parameters and efficiency of TD₂ for 128-bit security.

(a) Concrete parameter setting for TD₂ for 128-bit security.

Para.	N	ℓ	n	m	q	p	M	σ_{LWE}
Setting	1024	2^{64}	4096	2^{20}	2^{90}	2^{80}	1031	9

Para.	σ_k	σ_0	σ_1	σ_y	B_k	B_0	B_1	B_y	$\tilde{B}_0$
Setting	9	19	$2^{16.1}$	$2^{66.4}$	54	114	$2^{18.6}$	2^{69}	$2^{20.8}$

(b) Concrete sizes of the ciphertext and partial decryption for TD₂.

Ciphertext size (11.25MB + 1782MB(NIZK) in total)				Partial decryption size
$\mathbf{c} \in \mathbb{Z}_q^m$	$d \in \mathbb{Z}_q$	π		$h_{i,S} \in \mathbb{Z}_q$
		$\mathbf{a} \in \mathbb{Z}_q^{m\lambda}$	$\mathbf{z} \in \mathbb{Z}_q^{n\lambda} \times [\tilde{B}_0]^{m\lambda}$	
11.25 MB	12.5 B	1440 MB	341.625 MB	12.5 B

(c) Derivation process for the parameters in (a).

Parameter	Derivation process
N, M, ℓ	We set $N = 1024, M = 1031 > N, \ell = 2^{64}$
$n, m, q, \sigma_{\text{LWE}}$	$n = 4096, m = 2^{20} > 2n \log q, q \approx 2^{90}$ and $\sigma_{\text{LWE}} = 9$ ensures at least 128-bit LWE security, reported by LWE estimator
α, β	For $\mathbf{x} \leftarrow D_{\mathbb{Z}^m, \sigma}$, we choose $\alpha = 15/\sqrt{2\pi}$ and $\beta = (1.01/\sqrt{2\pi}) \cdot \sqrt{m}$ s.t. $\ \mathbf{x}\ _\infty \leq \alpha\sigma$ and $\ \mathbf{x}\ \leq \beta\sigma$ with probability $1 - 2^{-128}$ by Lemma 5.
ϵ	$2\epsilon = 2^{-128}$ so that $\sqrt{2} \cdot \eta_\epsilon(\mathbb{Z}^m) \leq \sqrt{2 \ln(2m(1+1/\epsilon))}/\pi \approx 8.12$ $\sigma_{\text{LWE}} = 9 \geq \sqrt{2} \cdot \eta_\epsilon(\mathbb{Z}^m)$ satisfies Lemma 2.
δ, σ_0, B_0	For Lemma 2, we set $\delta = 1$. $\sigma_0 = 19 \geq 2\sqrt{\sigma_{\text{LWE}}^2 + \delta^2}/2 \approx 18.06, B_0 := 114 > \alpha\sigma_0$.
$\tilde{B}_0$	For NIZK, $\tilde{\sigma}_0 := \ \mathbf{e}_0\ \cdot 15 \cdot \sqrt{2\pi} \approx 37\beta\sigma_0$ following Lemma 14, so that $\tilde{B}_0 := 2^{20.8} \geq \alpha \cdot \tilde{\sigma}_0$.
$\sigma_k, B_k, \sigma_y, B_y$	For Lemma 6, we set $c = 1$ and $\sigma_k = 9 \geq \sqrt{2} \cdot \eta_\epsilon(\mathbb{Z}^m)$, $B_k := 54 \geq \alpha \cdot \sigma_k, \sigma_y \approx 2^{66.4} \geq \sqrt{m\ell}\tilde{B}_0\sigma_k, B_y = \alpha\sigma_y \approx 2^{69}$. $\sigma_k \geq 8$ satisfies $m \log \sigma_k - \frac{m}{2} - n \log q \geq n \log q + 2\lambda$ (Lemma 1).
σ_1, B_1	Lemma 2 requires, $\sigma_1 \geq \sqrt{B}\sigma_0$, where $B = \ \mathbf{k}\ $ in our scheme. $\ \mathbf{k}\ \leq \beta\sigma_k$, we set $\sigma_1 := \beta\sigma_k\sigma_0 \approx 2^{16.1}$ and $B_1 = \alpha\sigma_1 \approx 2^{18.6}$.
p, q	$p \geq \max\{2\sqrt{m \log \sigma_k}(2\sqrt{m} + \sqrt{\ell}), 2(B_1 + mB_0B_k + NB_y)\}$, The latter dominates, so we set $p \approx 2^{80}$ and $q = Mp \approx 2^{90}$.

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